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Brennan et al.

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(45) **Date of Patent: Sep. 9, 2025**

(54) **IMPLANT FINDER**

(56) **References Cited**

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U.S. PATENT DOCUMENTS

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 114 days.

* cited by examiner

(21) Appl. No.: **18/130,673**

Primary Examiner — Anh Tuan T Nguyen

Assistant Examiner — Tommy T Ly

(22) Filed: **Apr. 4, 2023**

(74) *Attorney, Agent, or Firm* — Rothwell, Figg, Ernst & Manbeck, P.C.

(65) **Prior Publication Data**

US 2023/0309852 A1 Oct. 5, 2023

(57) **ABSTRACT**

Method and apparatus for locating an implant comprising magnetic material and/or a charge storage device in a living animal. The apparatus may include a magnetic field generator (e.g., an electromagnetic field generator) configured to generate a magnetic field (e.g., an electromagnetic field). The apparatus may include a sensor configured to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field. The magnetic material and/or the charge storage device of the implant may cause changes to the magnetic field as the sensor is moved over the implant. The apparatus may include a computer configured to use the sensor signal to detect a location of the implant.

Related U.S. Application Data

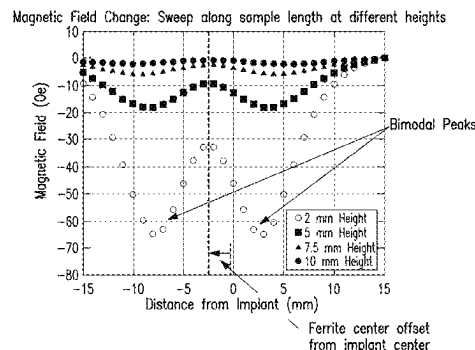
(60) Provisional application No. 63/327,566, filed on Apr. 5, 2022.

(51) **Int. Cl.**
A61B 5/06 (2006.01)

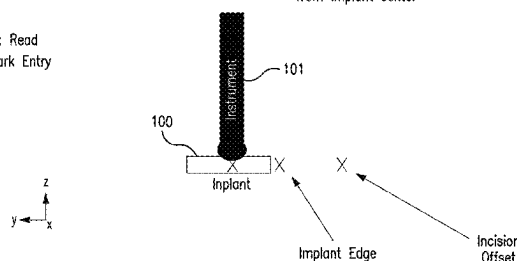
(52) **U.S. Cl.**
CPC **A61B 5/062** (2013.01)

(58) **Field of Classification Search**
CPC A61B 5/062; A61B 2034/2051
See application file for complete search history.

30 Claims, 53 Drawing Sheets



Step 3: Read
Depth, Mark Entry



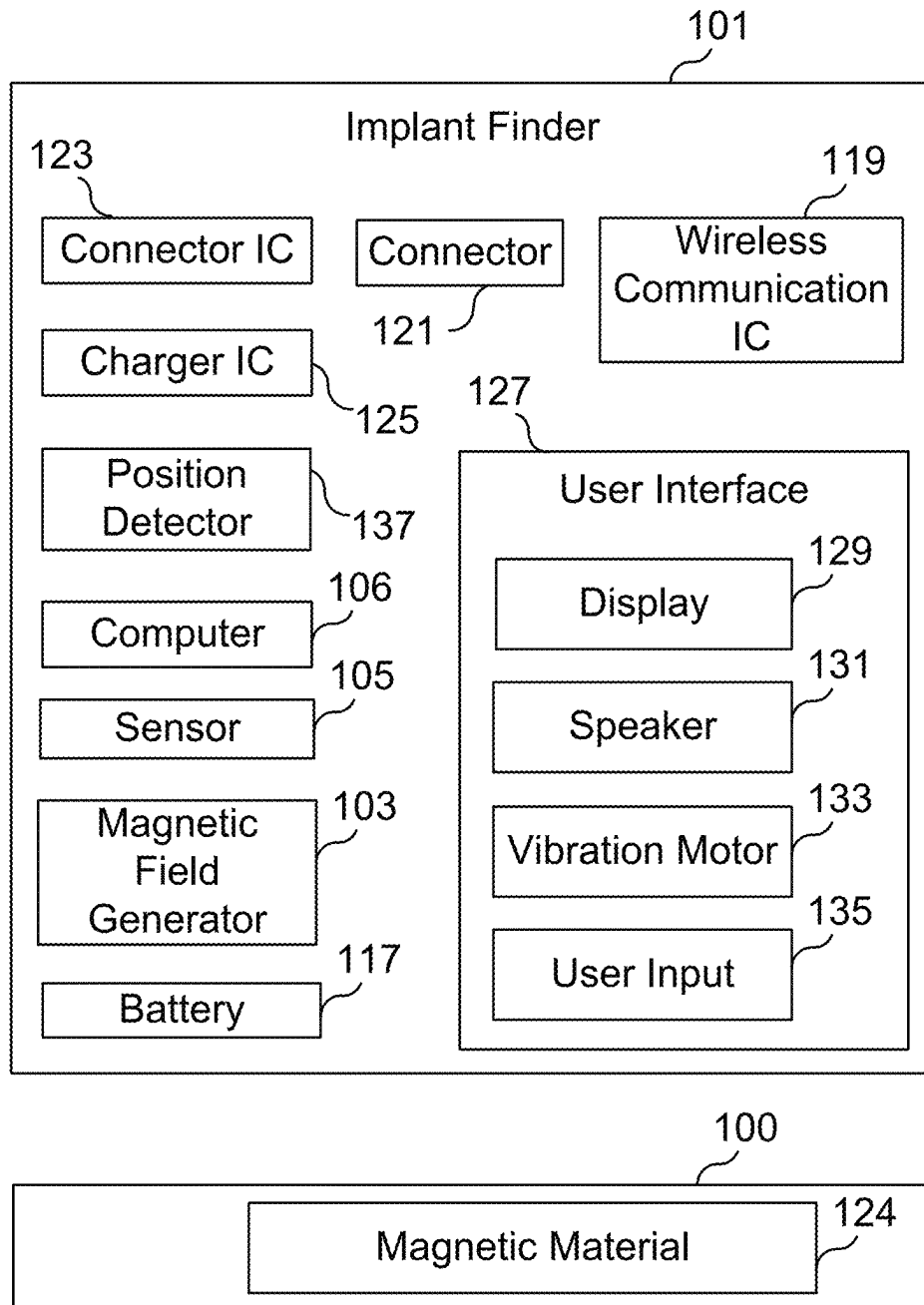


FIG. 1

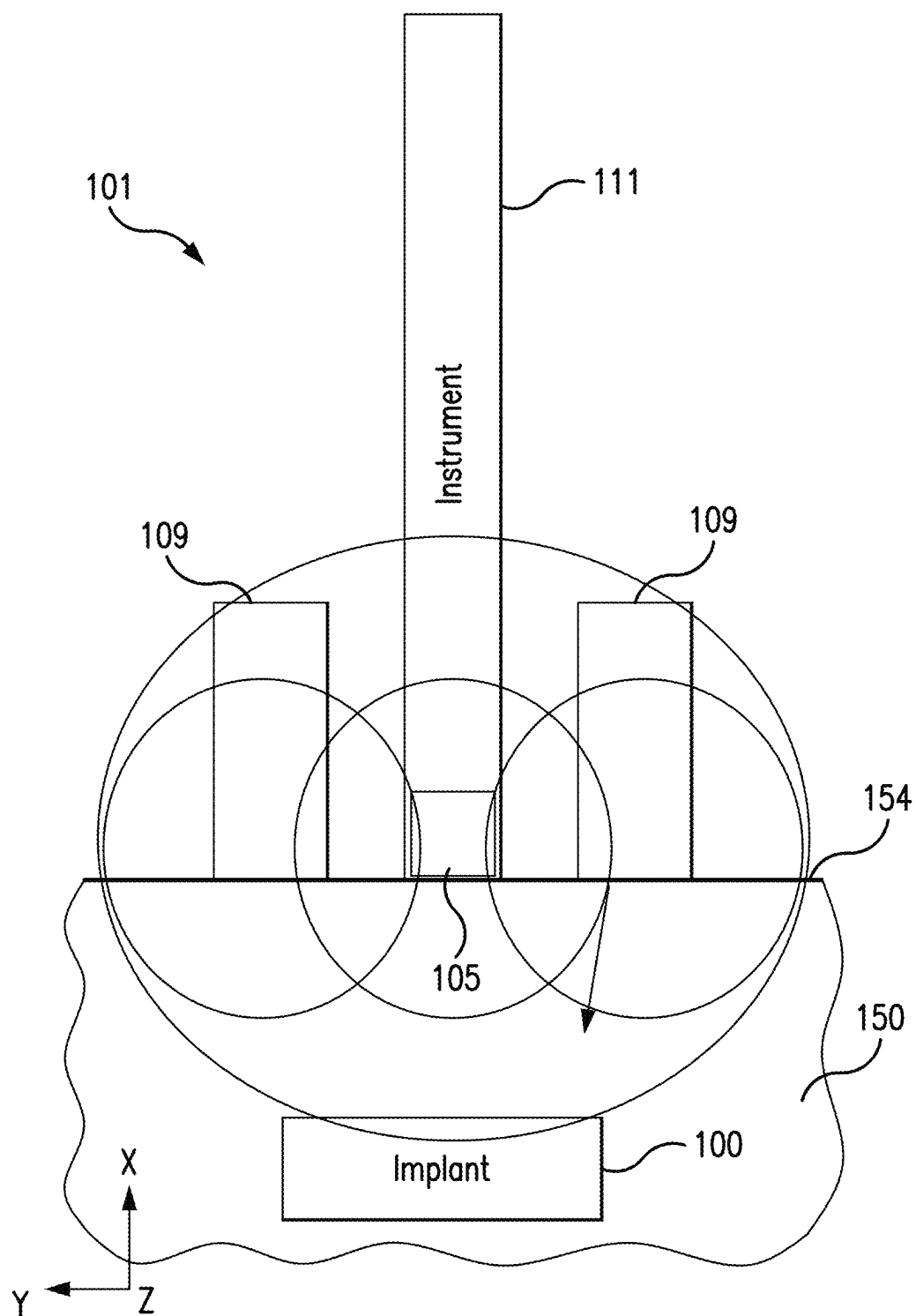


FIG. 2A

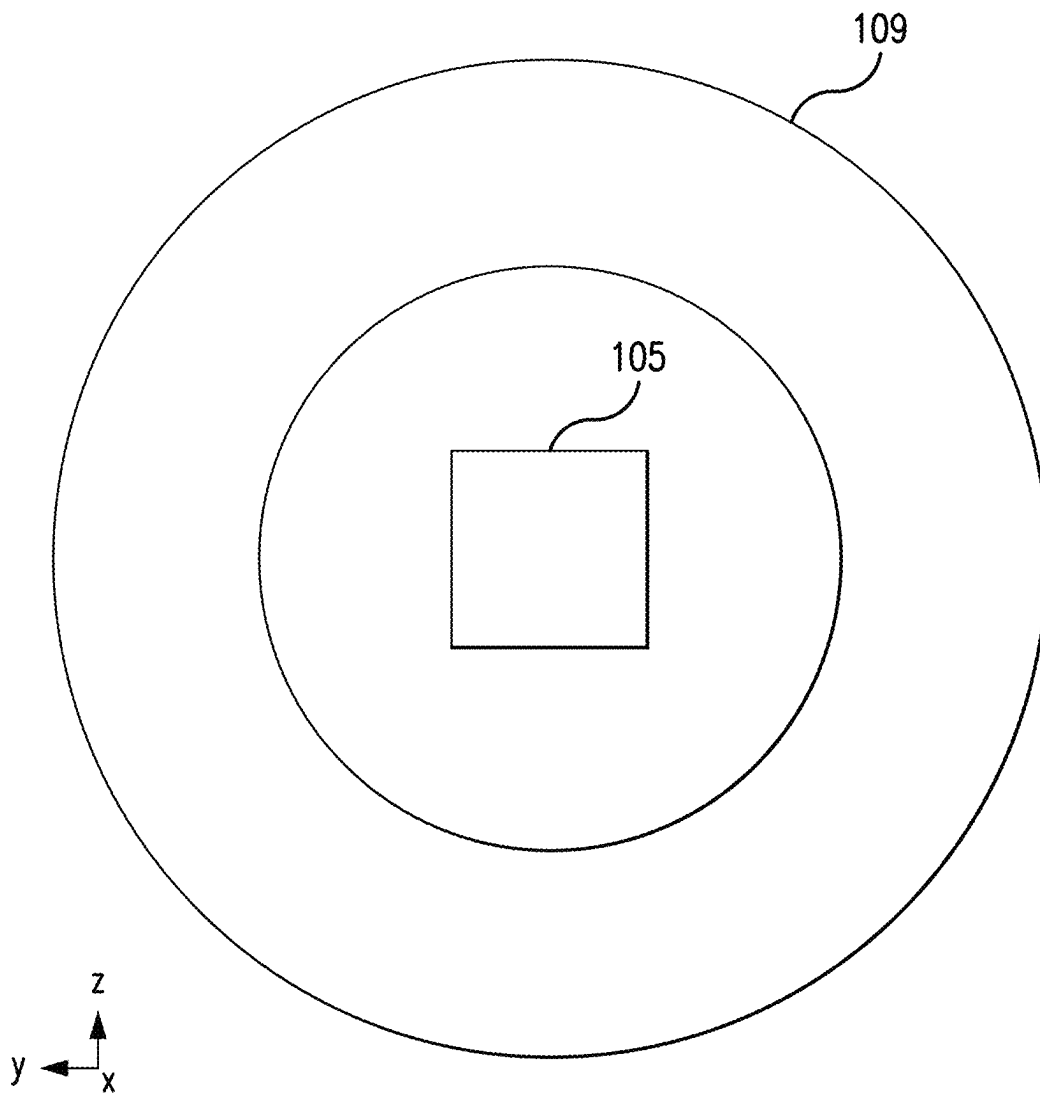


FIG. 2B

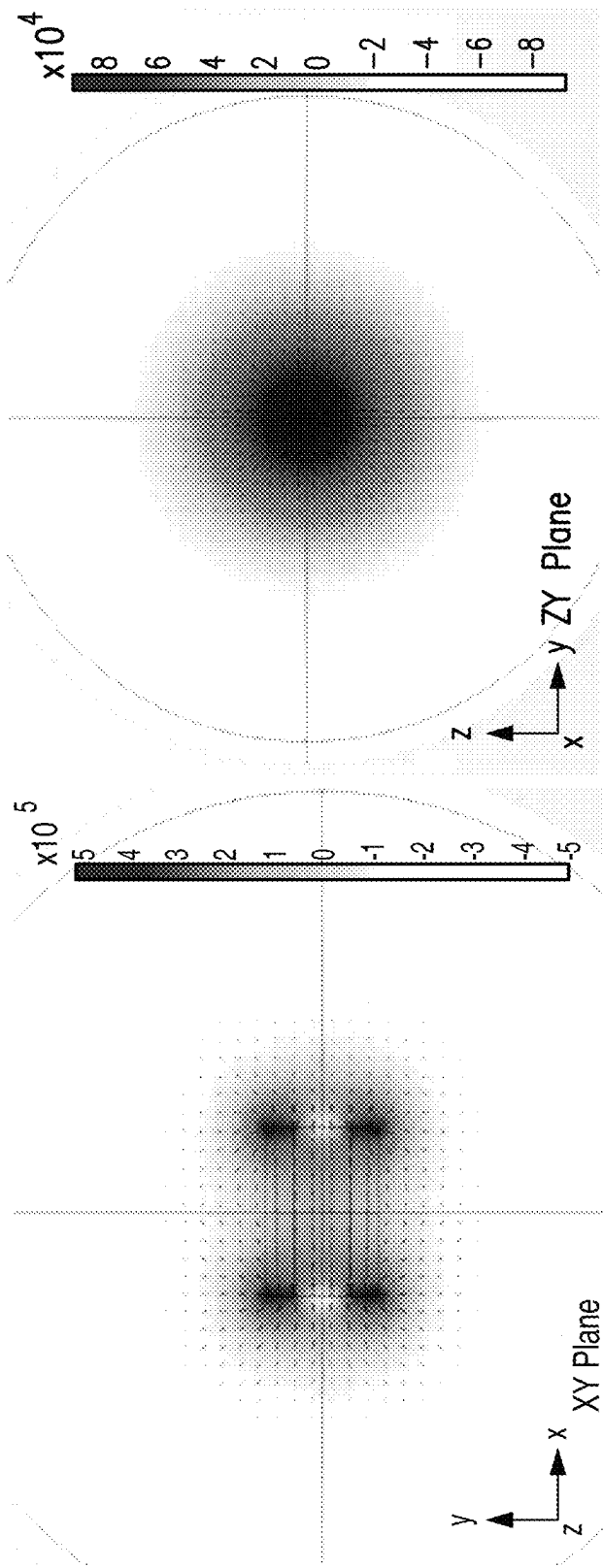


FIG. 2D

FIG. 2C

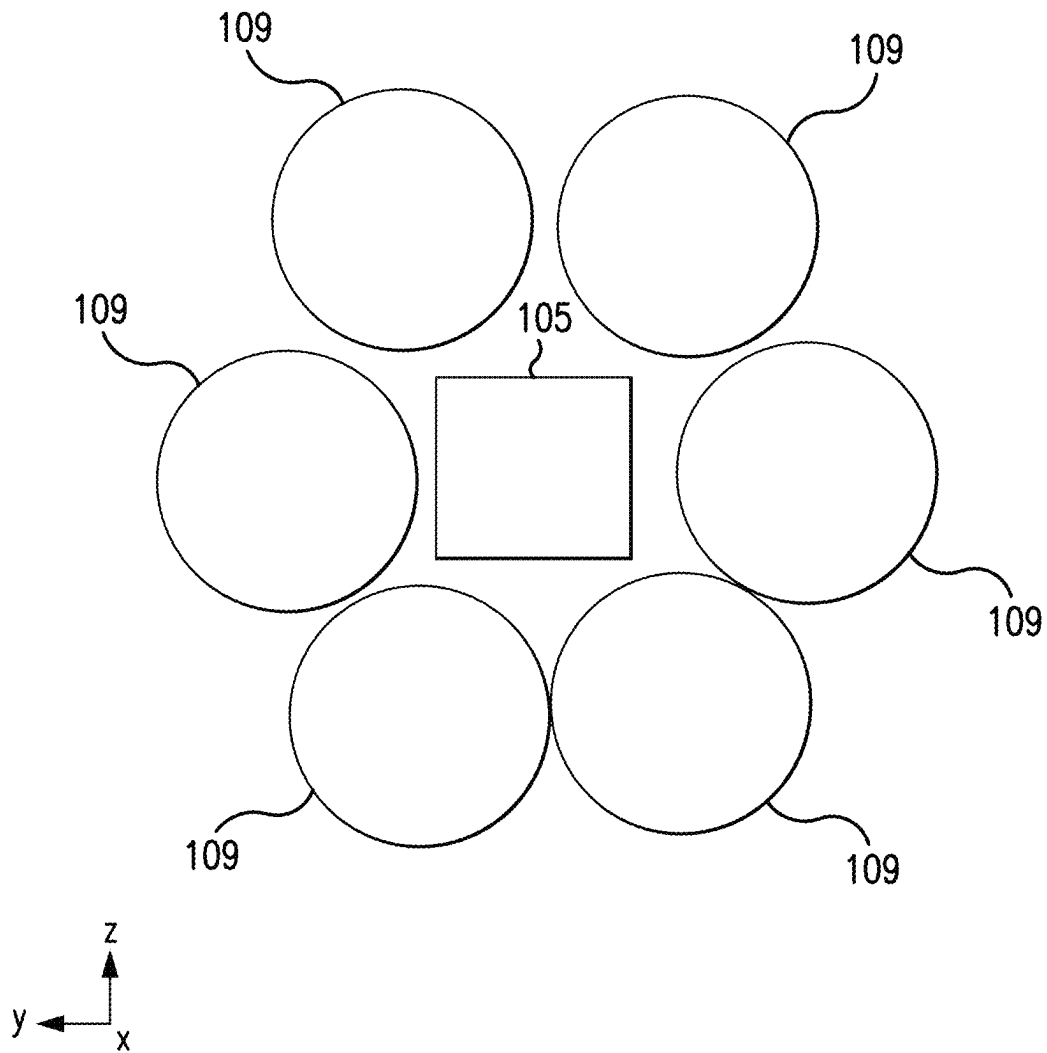


FIG. 2E

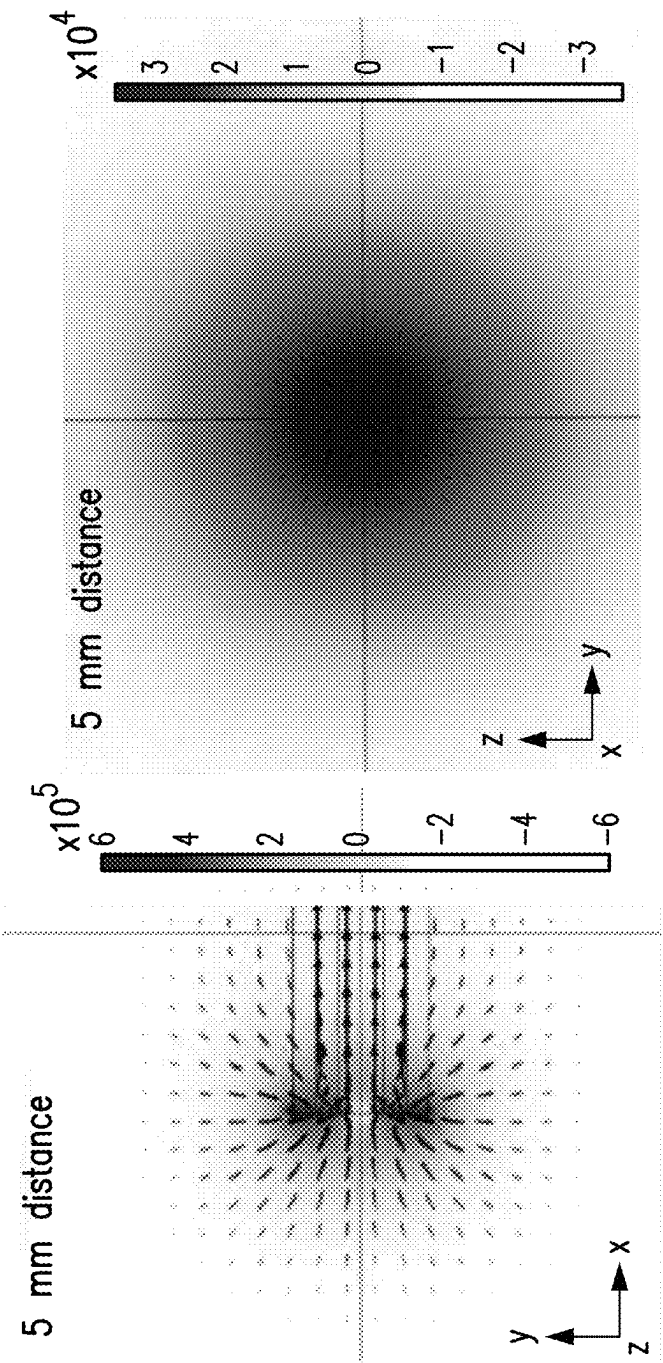


FIG. 2F

FIG. 2G

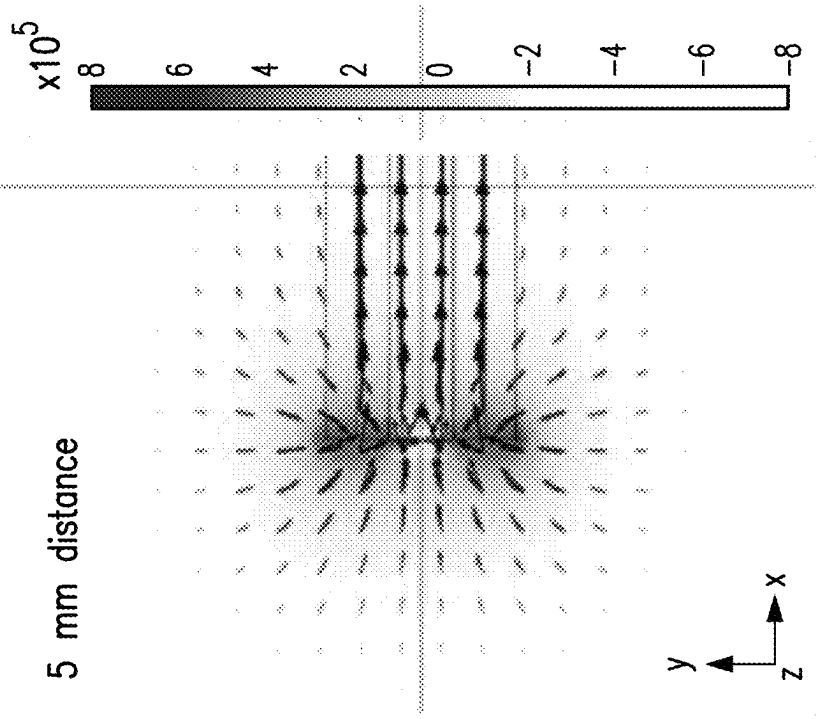


FIG. 2H

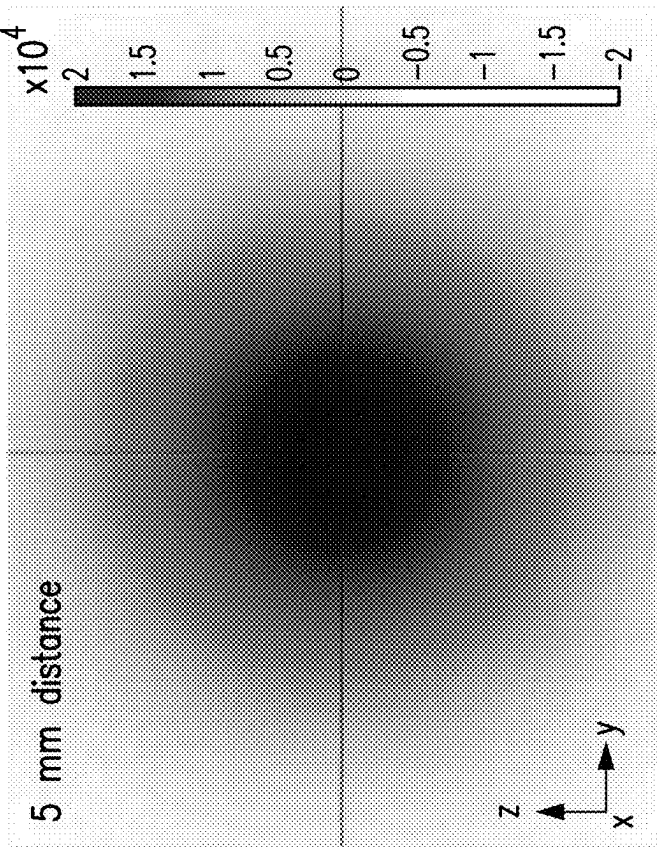


FIG. 2I

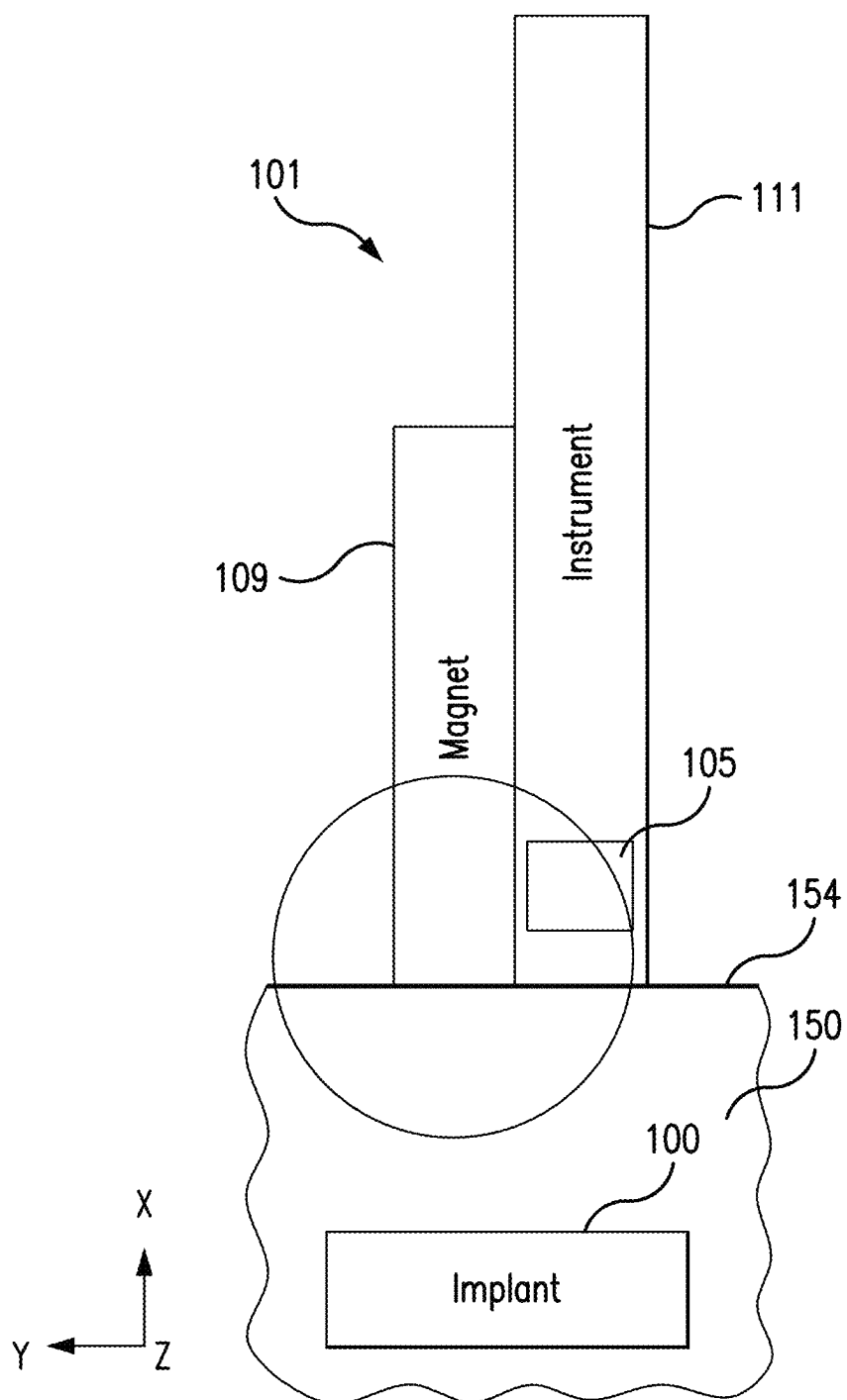


FIG. 3A

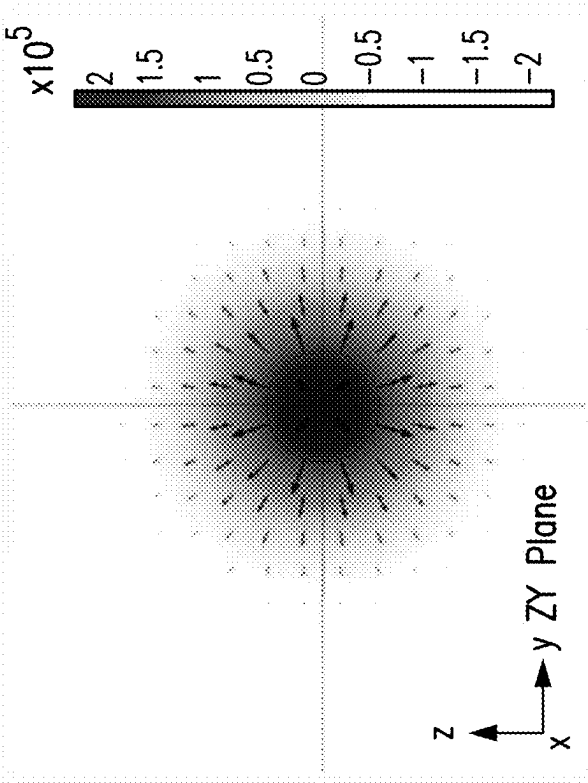


FIG. 3B

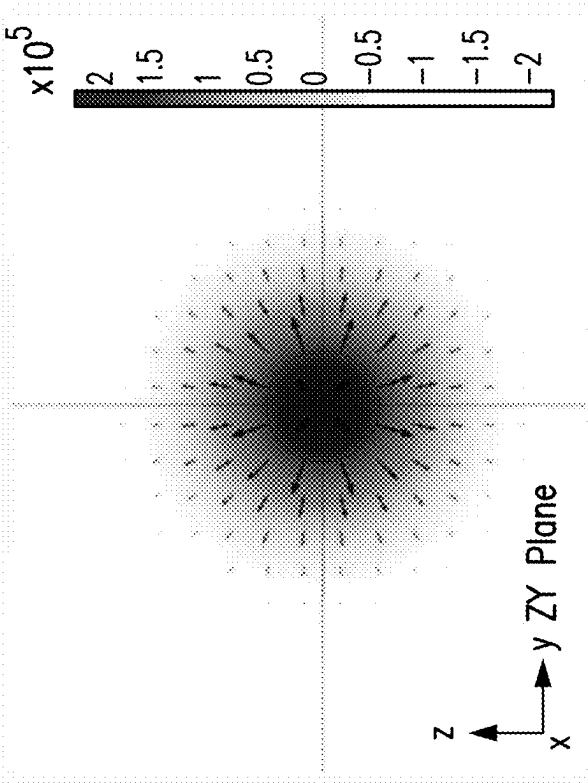


FIG. 3C

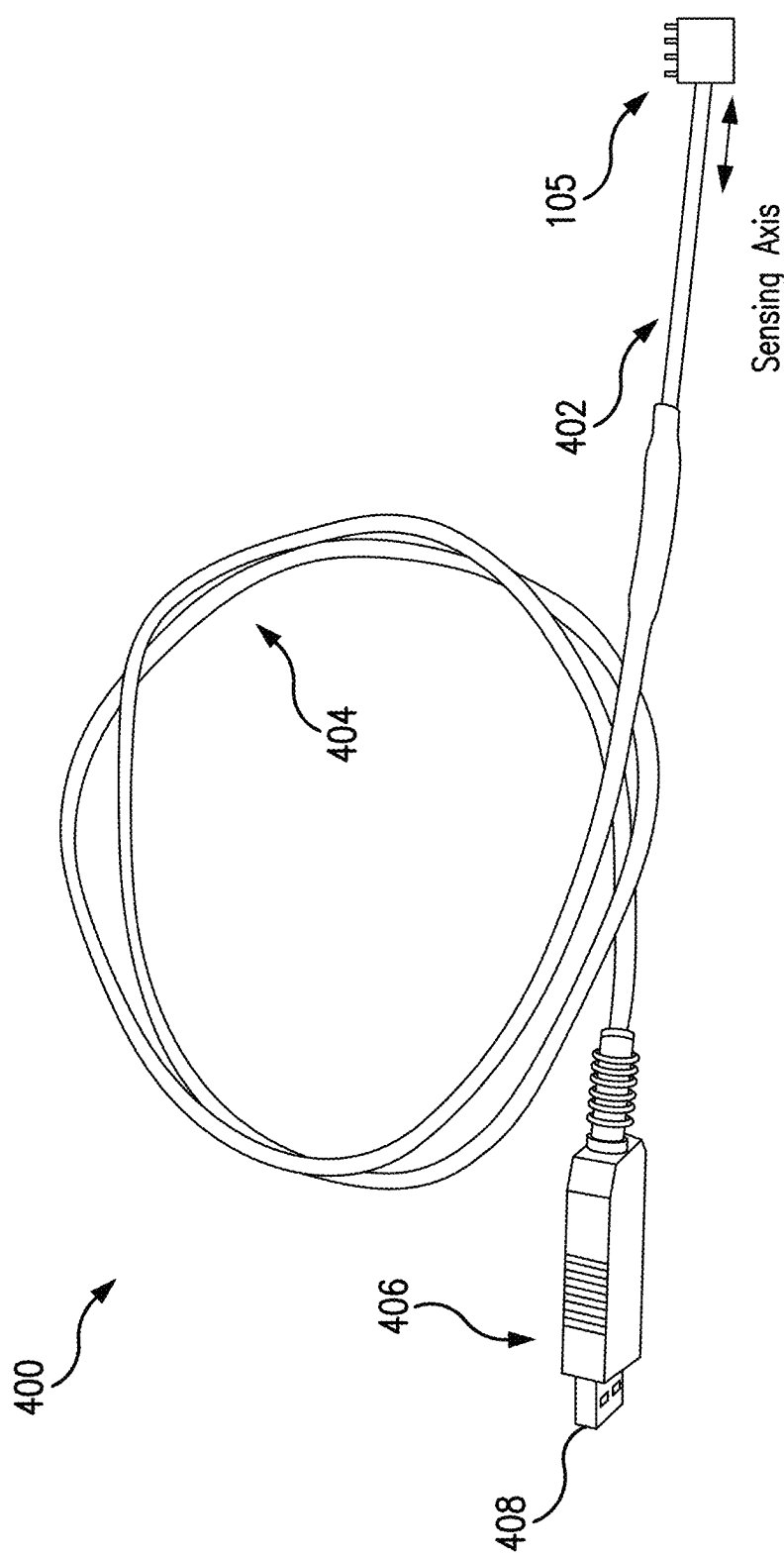


FIG. 4A

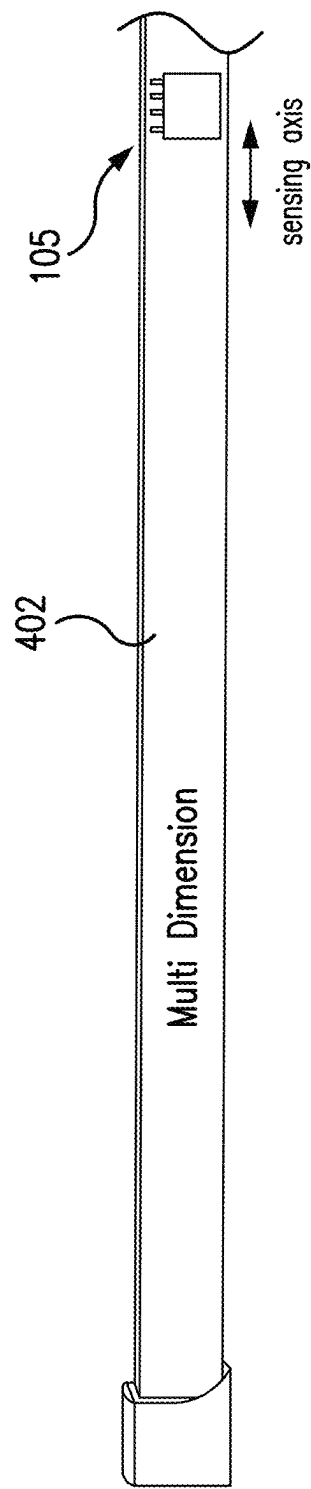


FIG. 4B

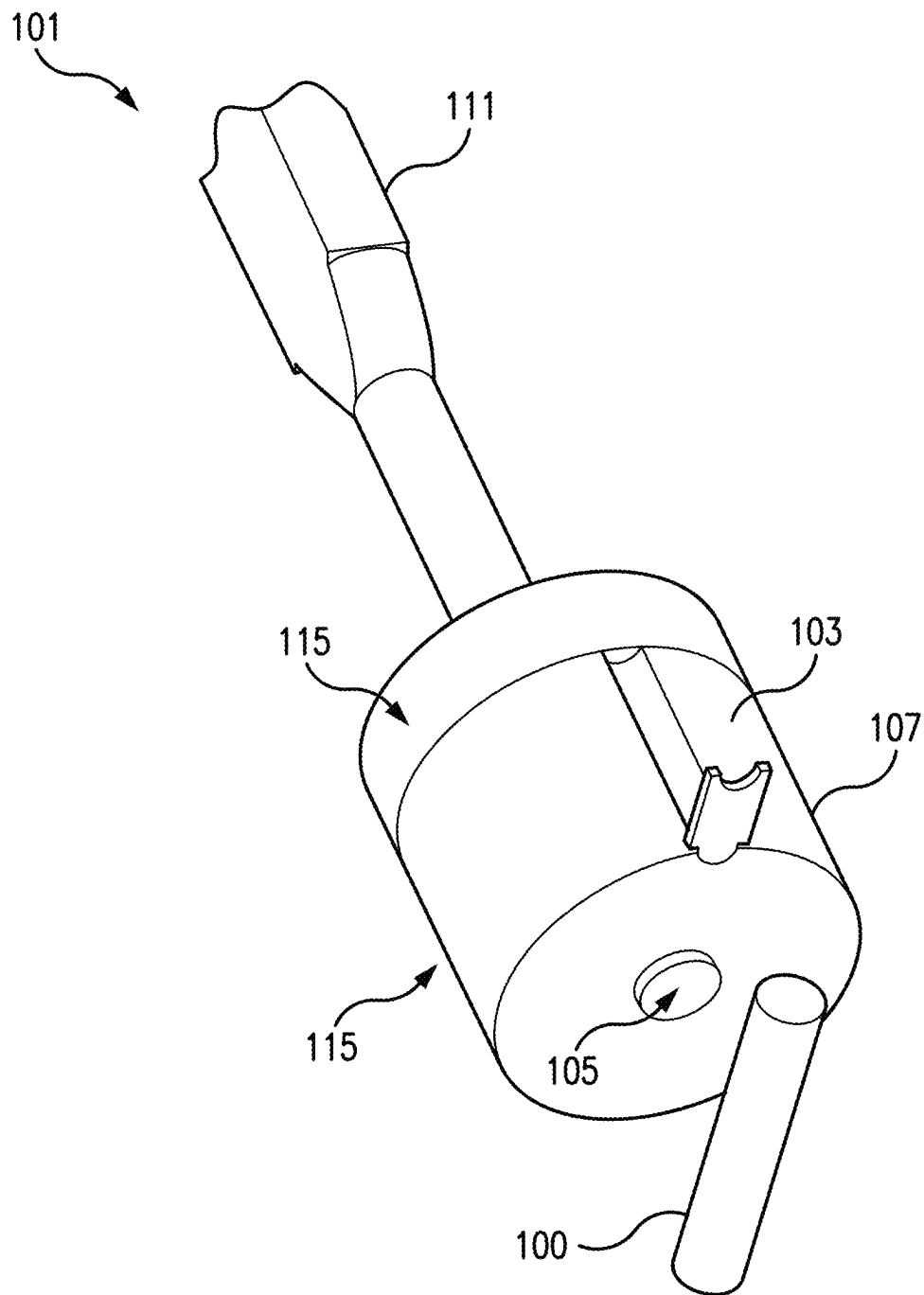


FIG. 5A

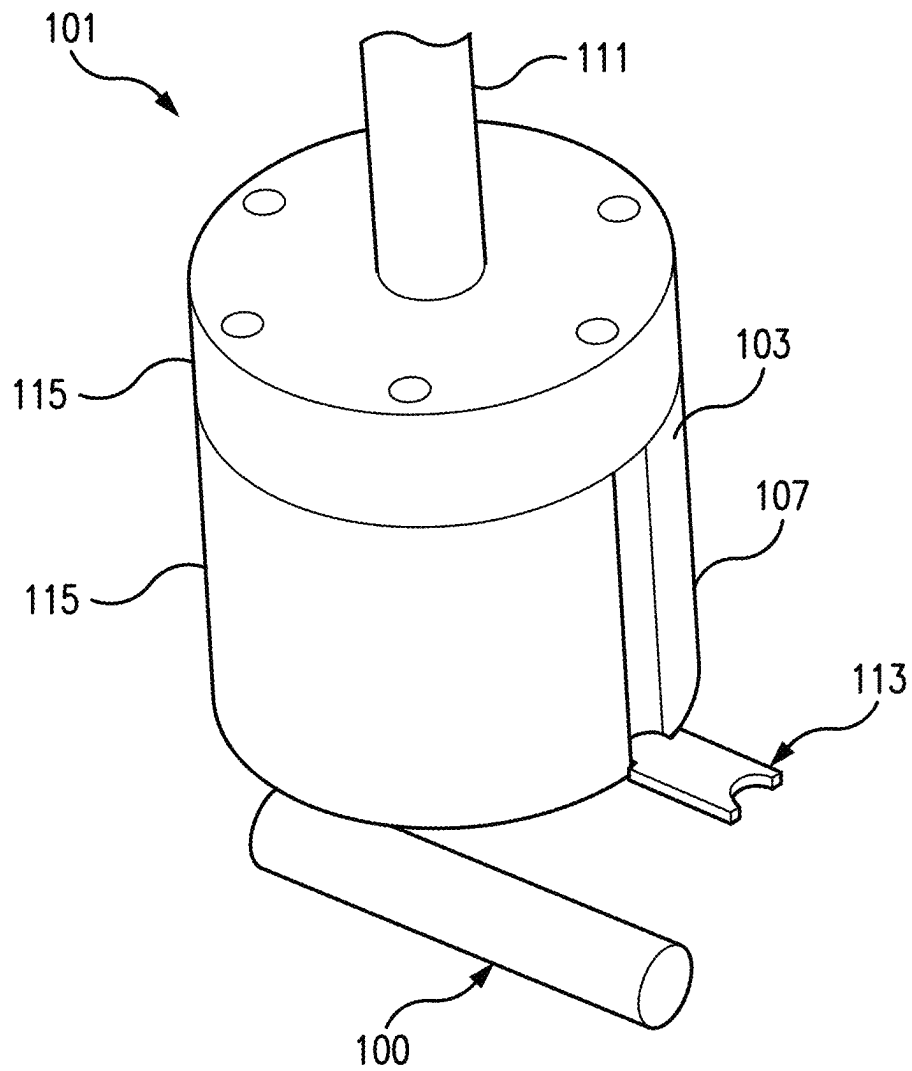


FIG. 5B

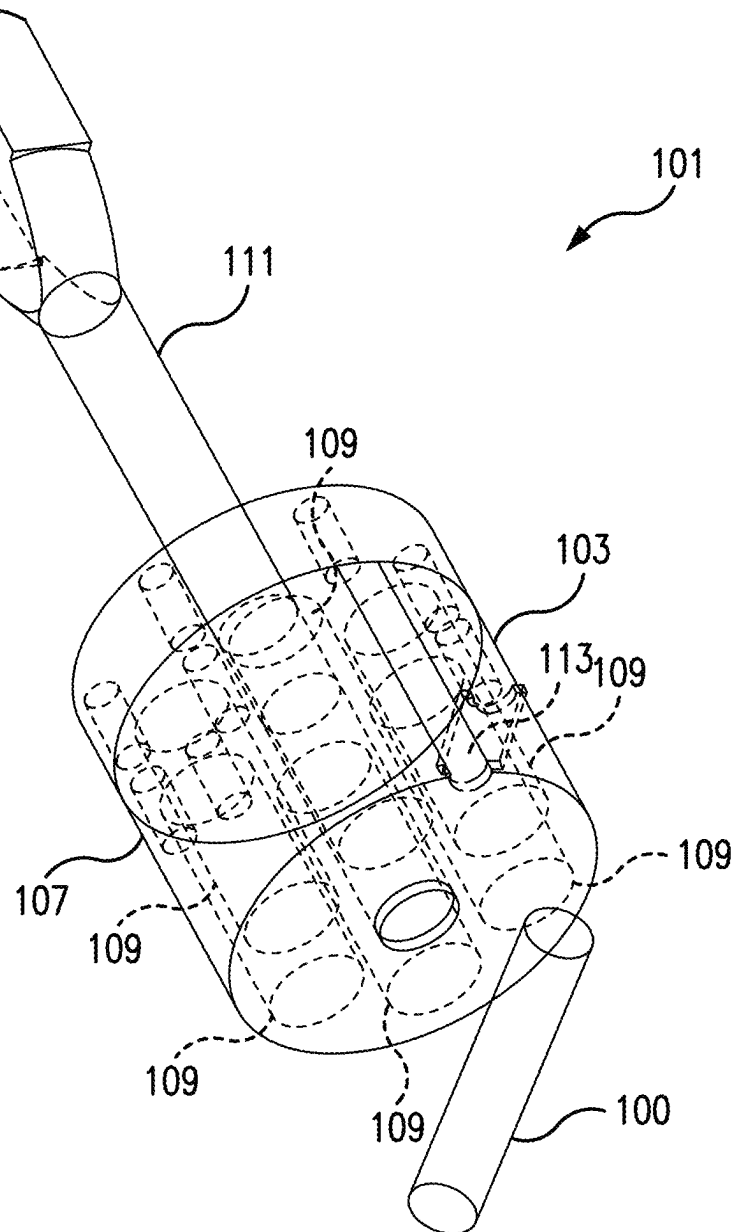


FIG. 5C

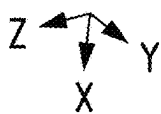
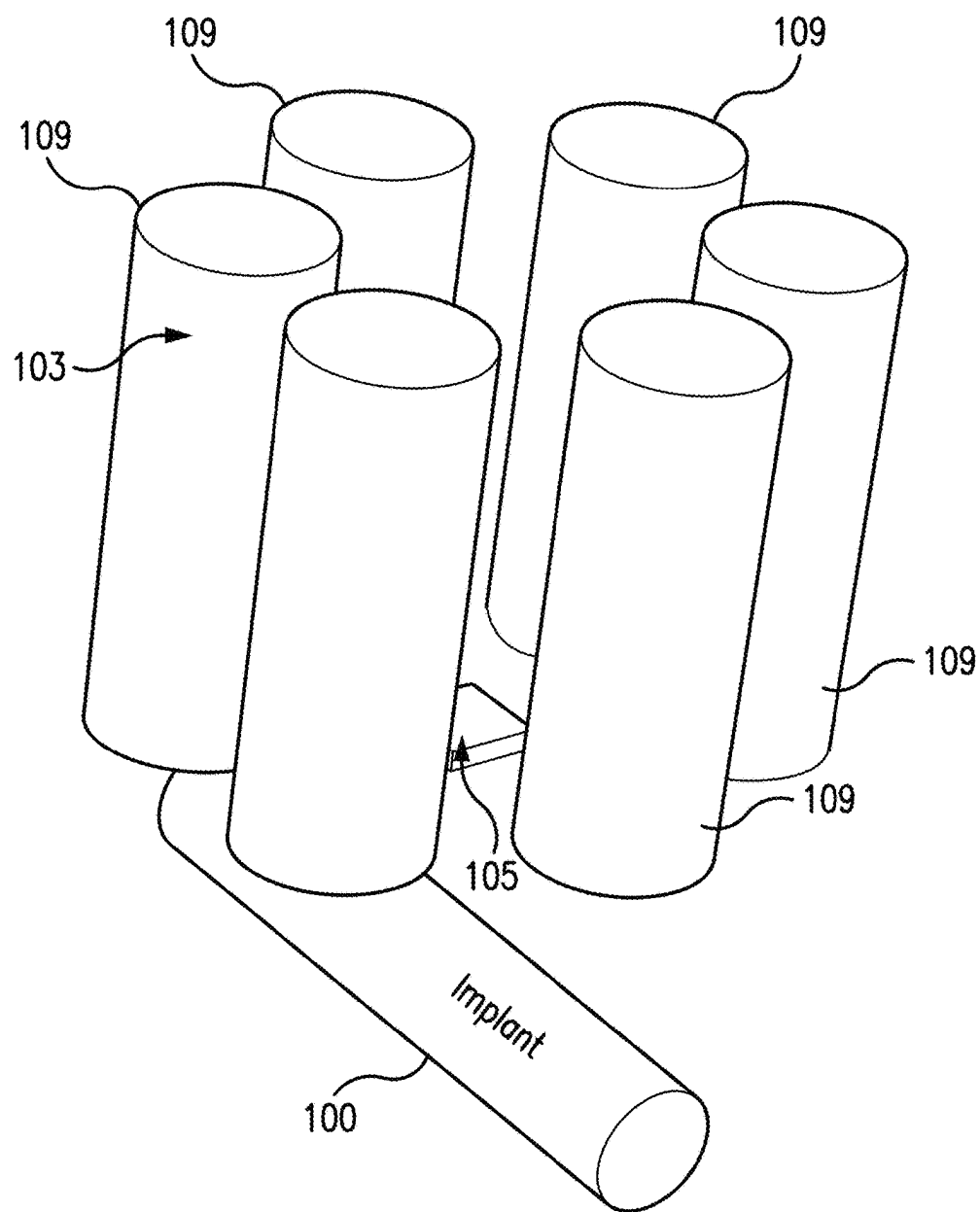


FIG. 5D

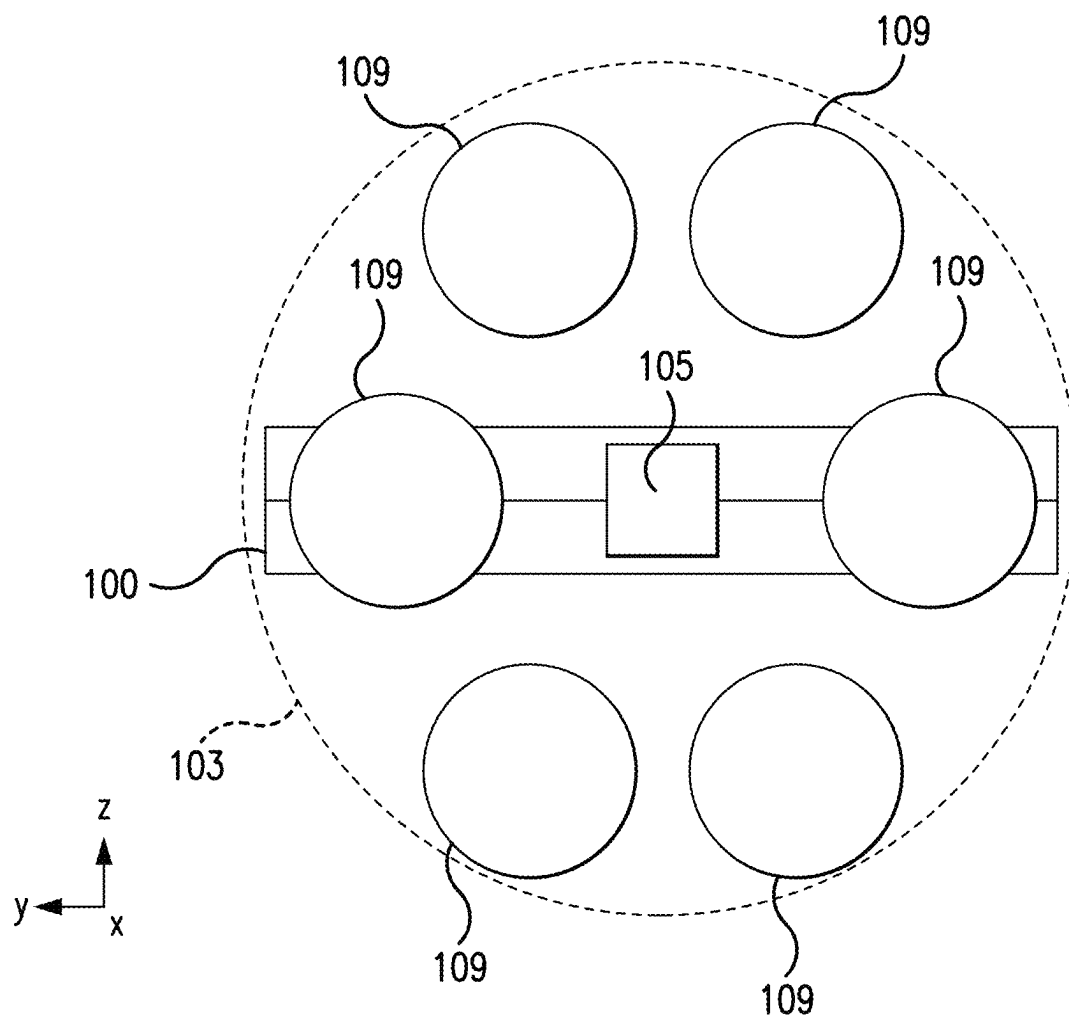


FIG. 5E

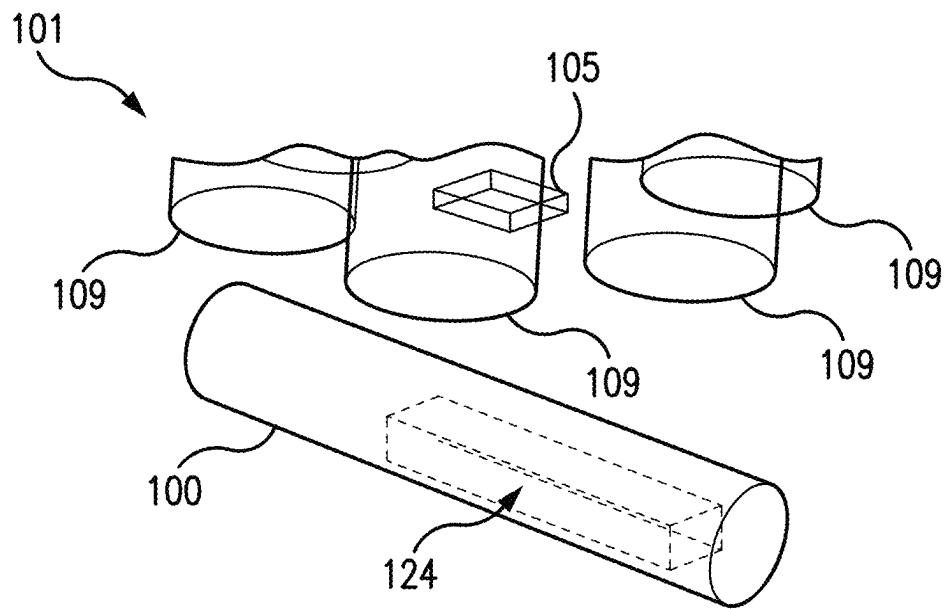


FIG. 5F

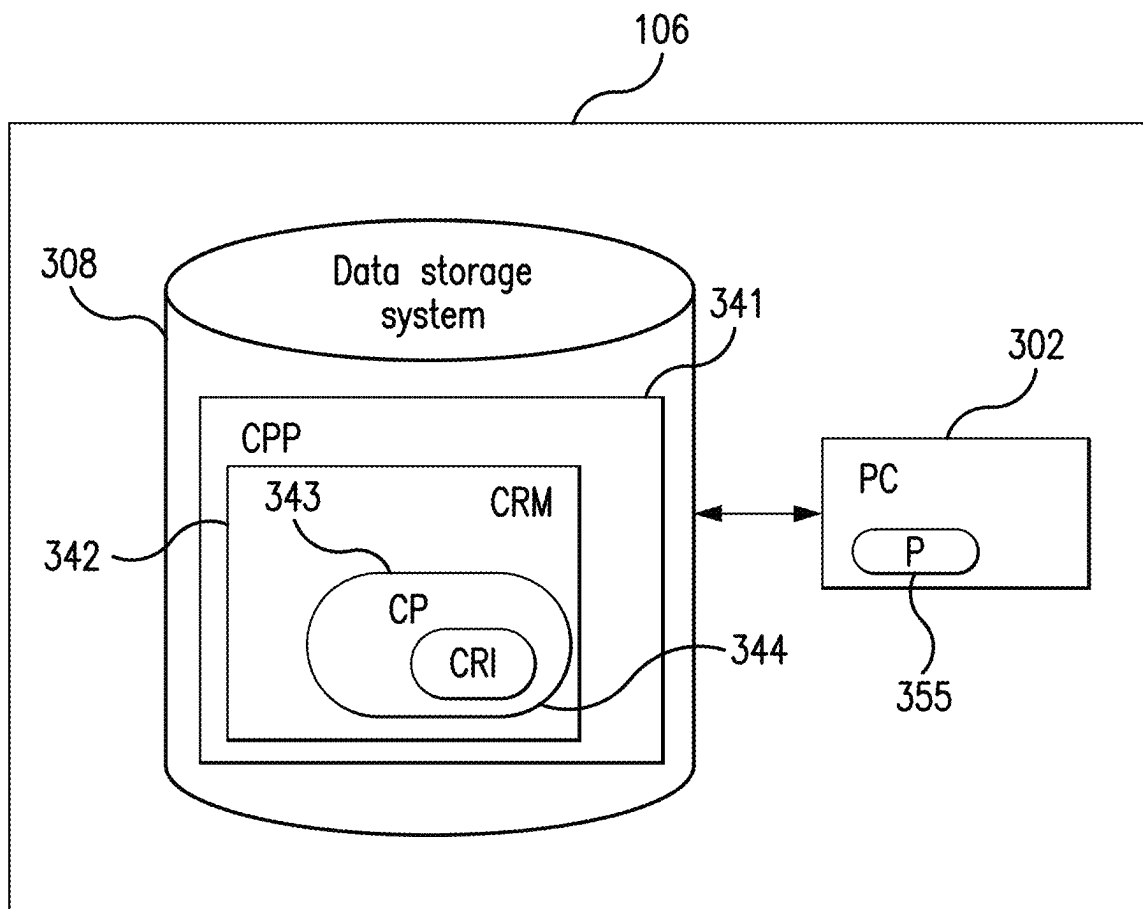


FIG. 6

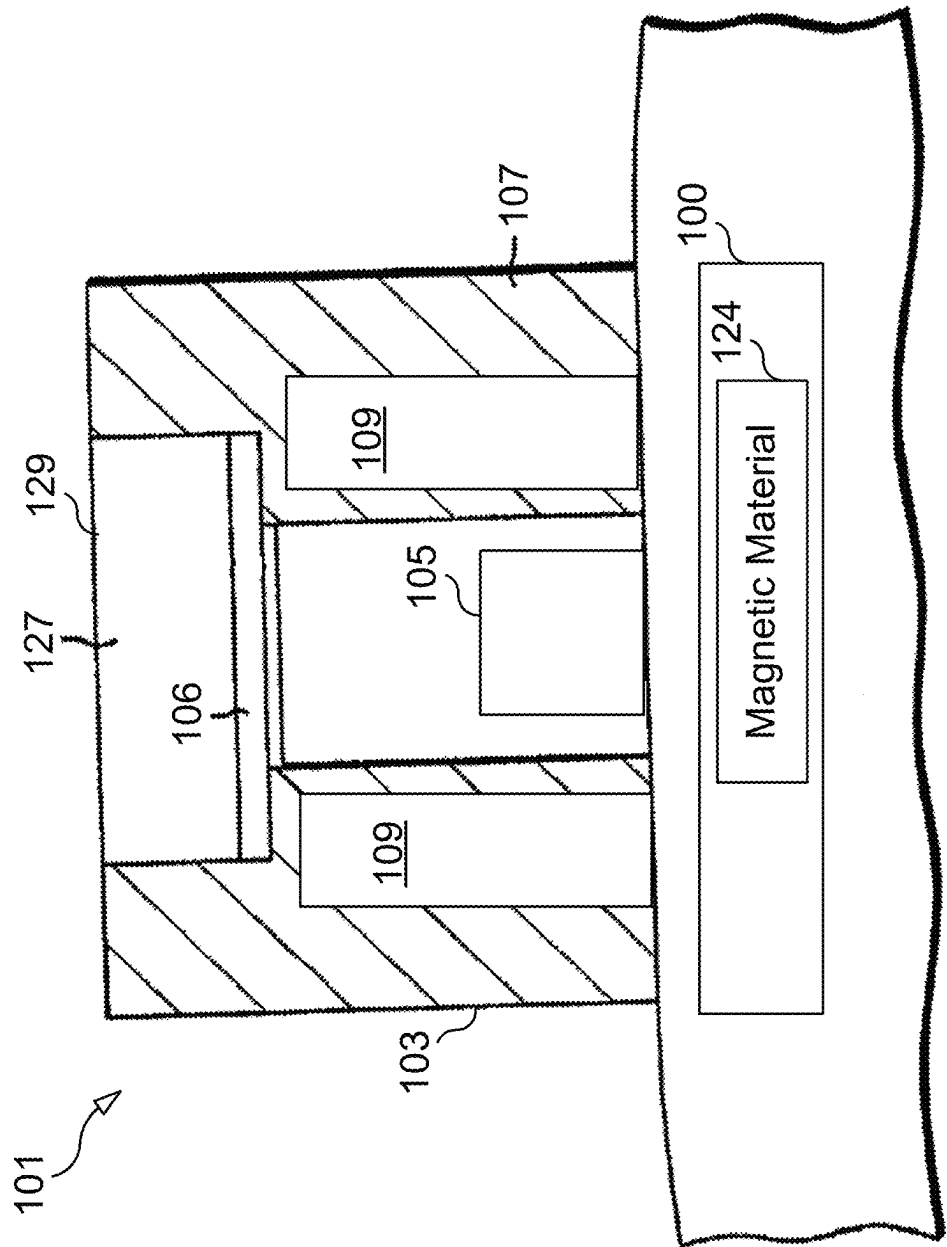


FIG. 7A

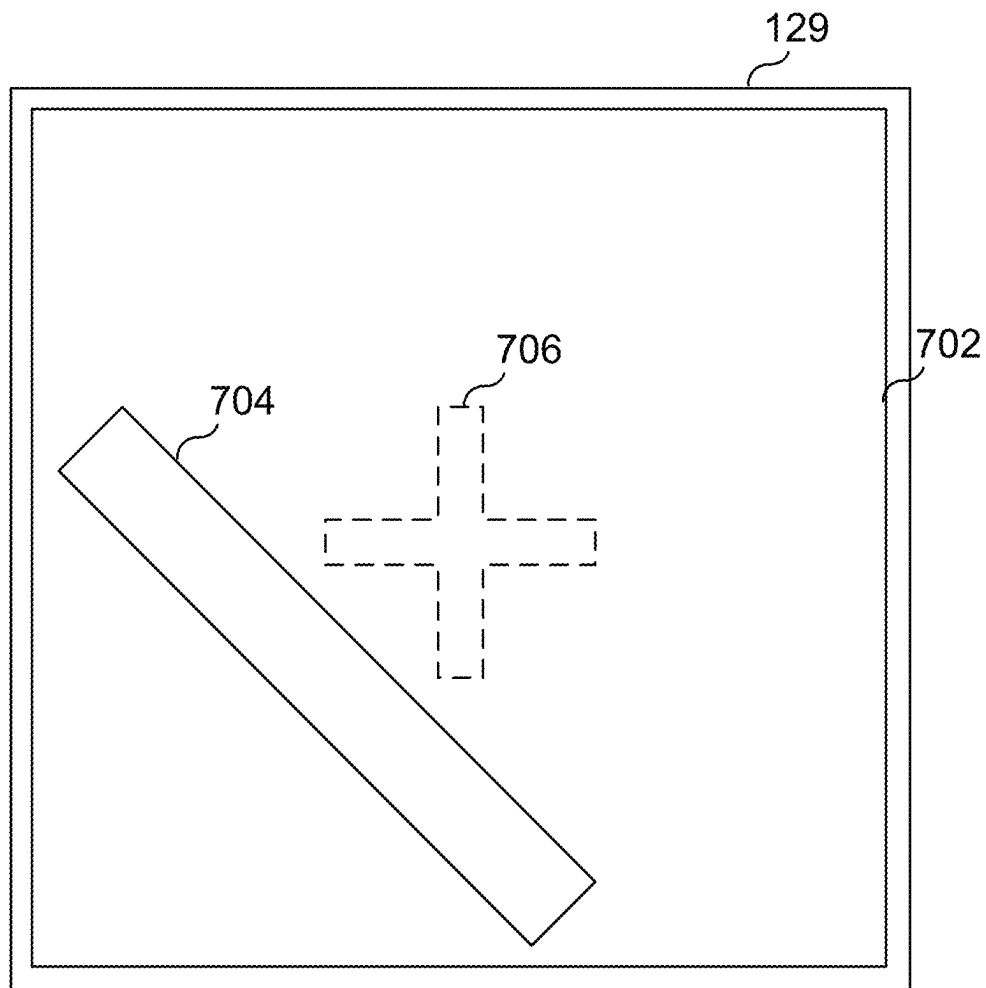


FIG. 7B

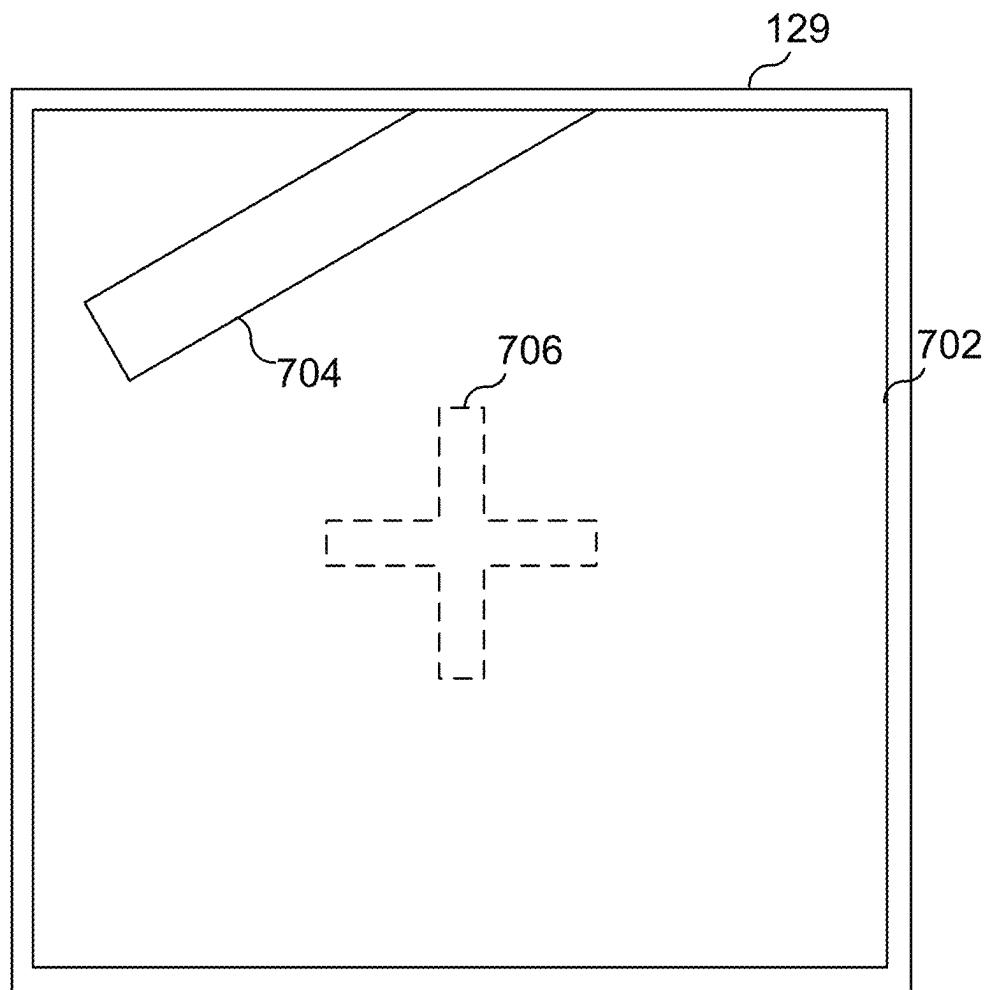


FIG. 7C

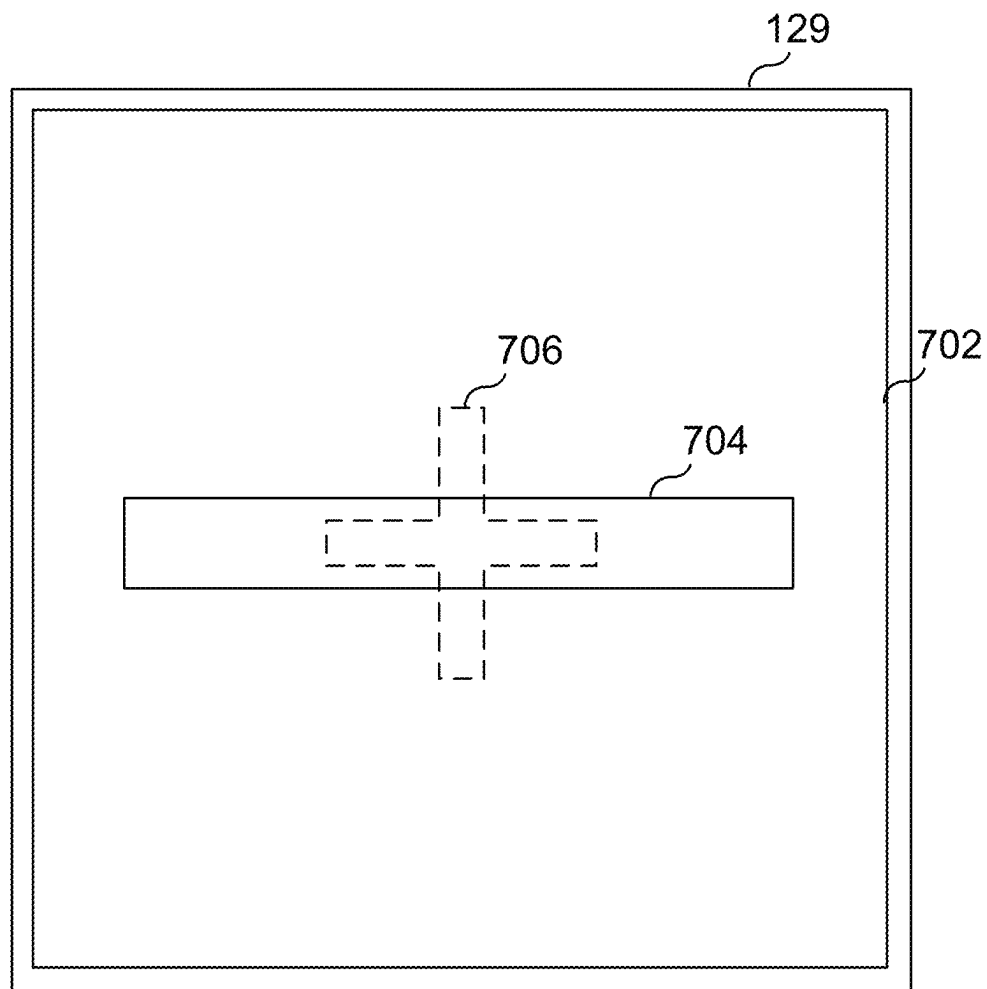


FIG. 7D

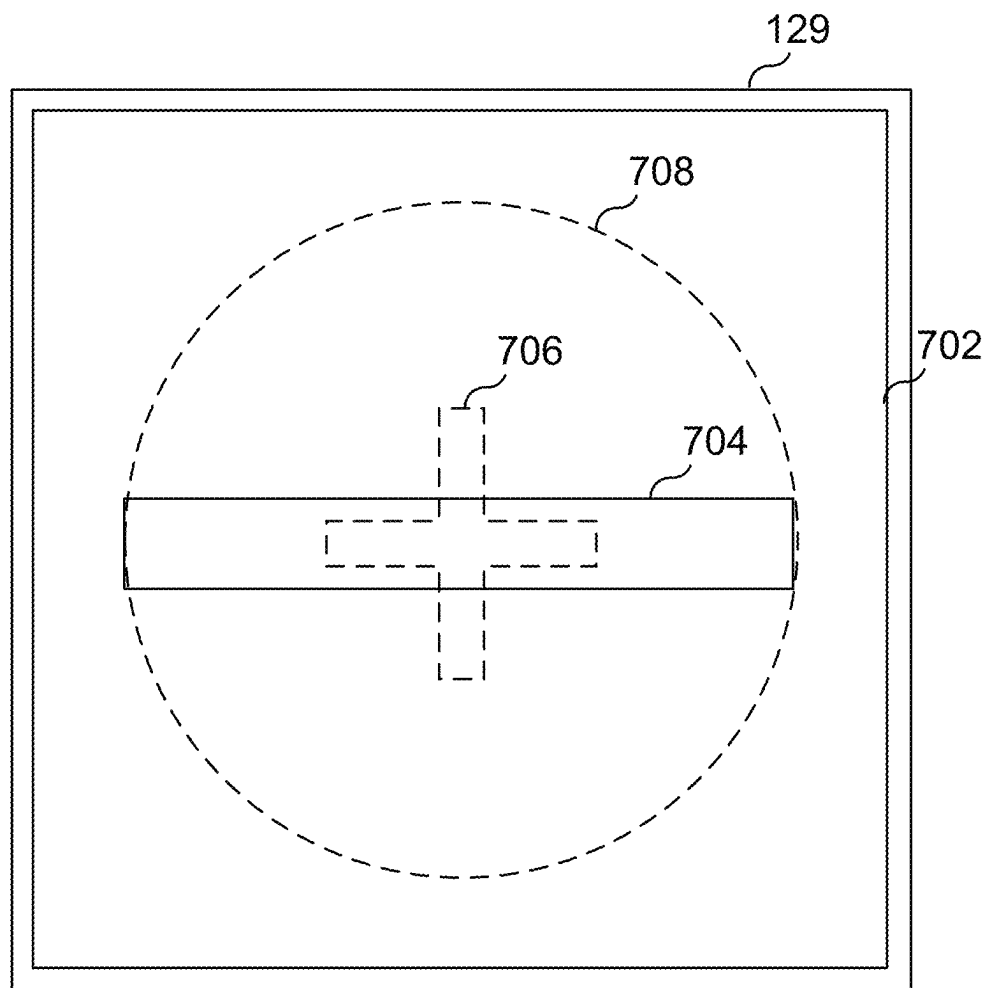


FIG. 7E

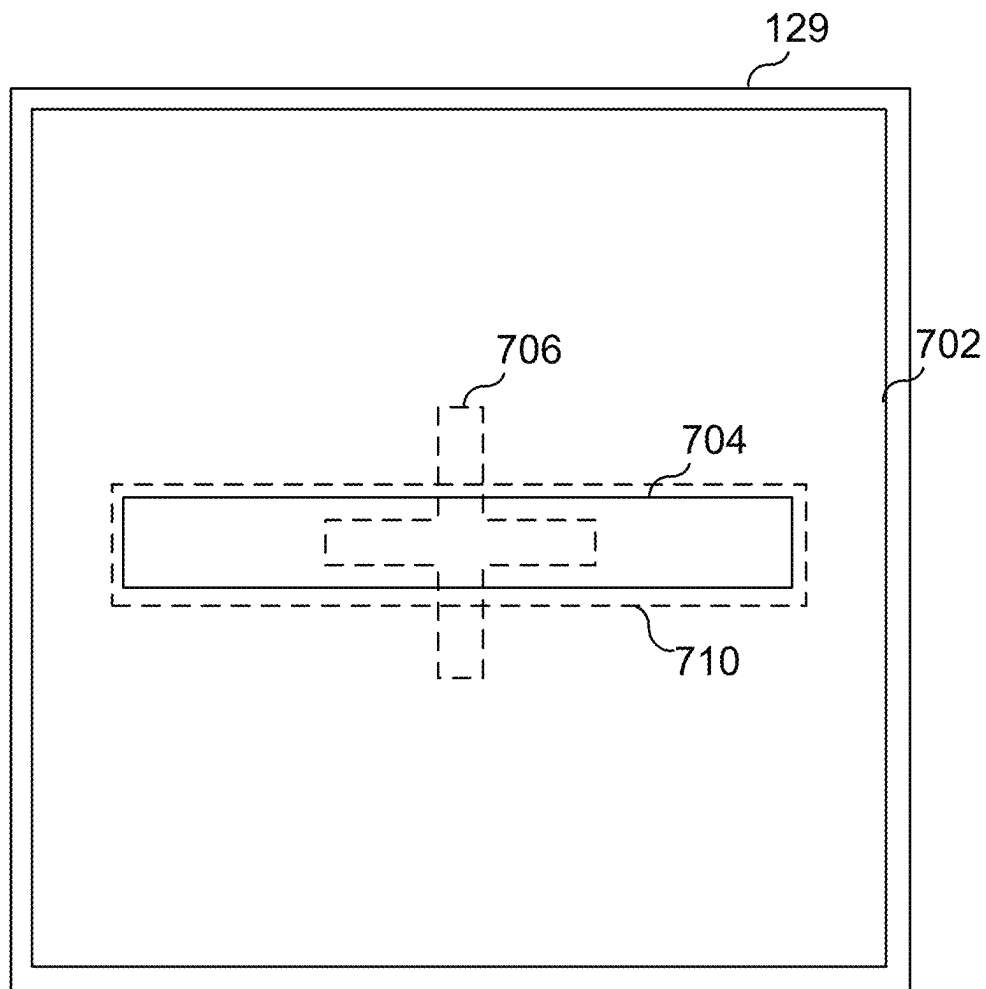


FIG. 7F

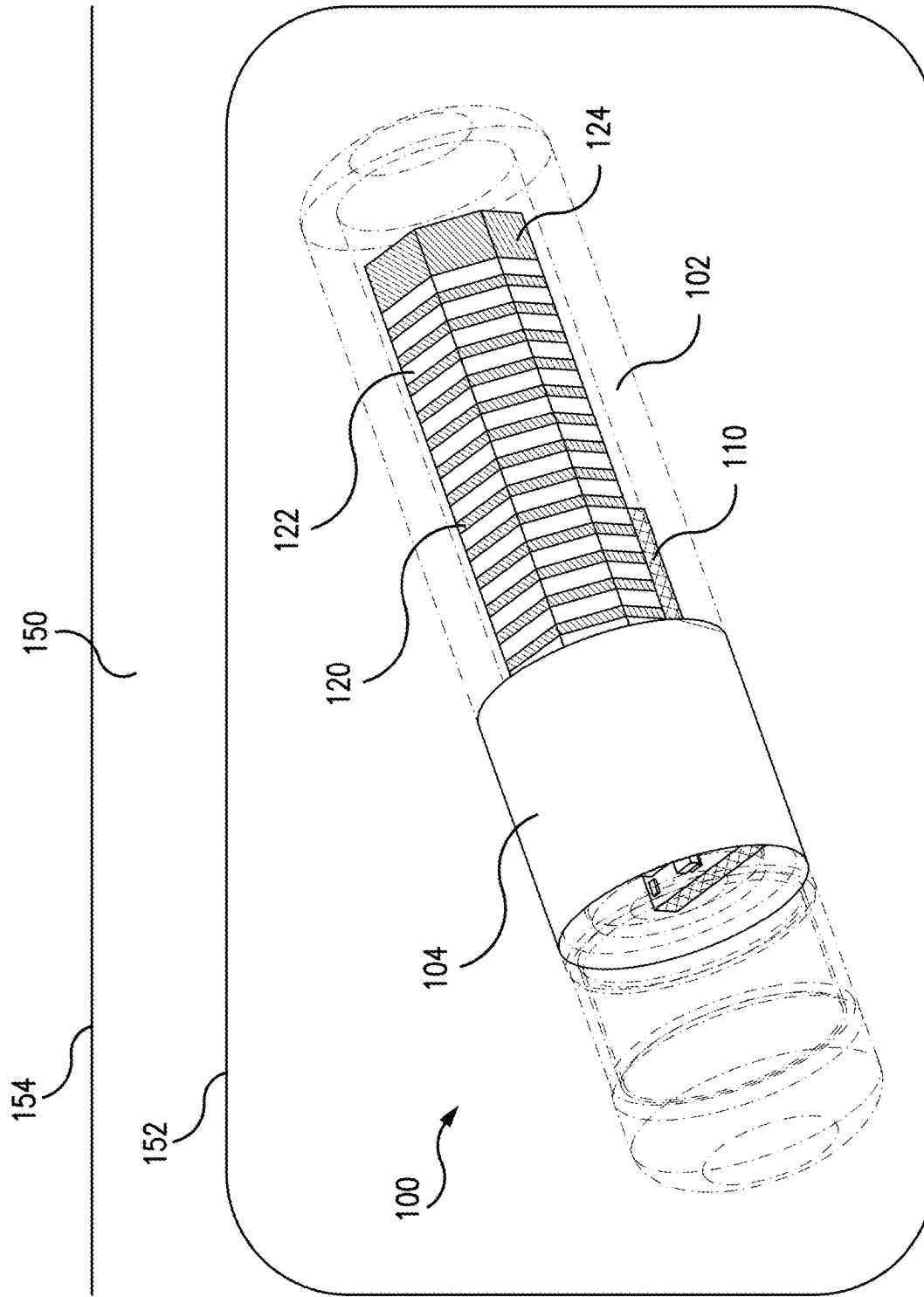


FIG. 8A

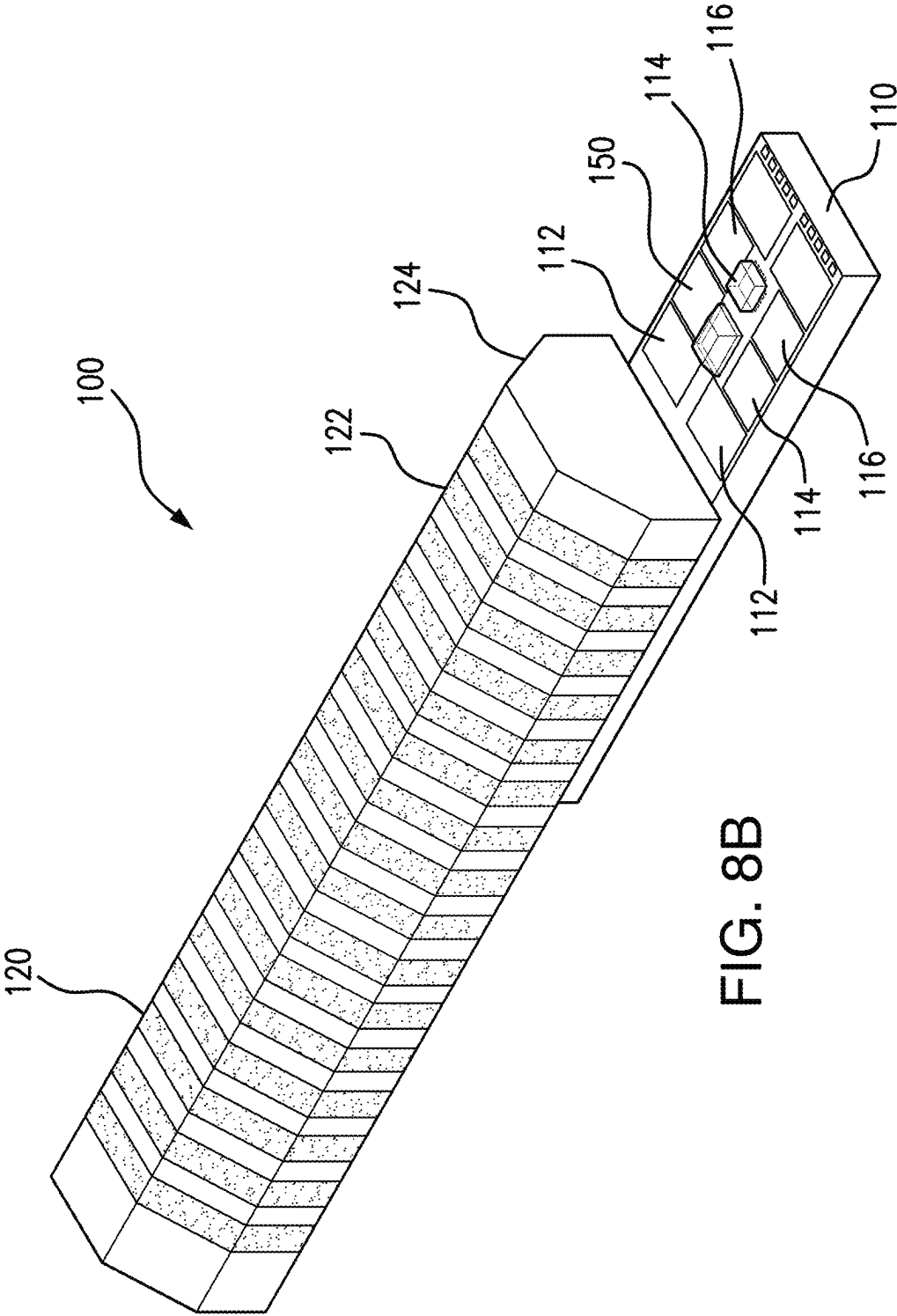


FIG. 8B

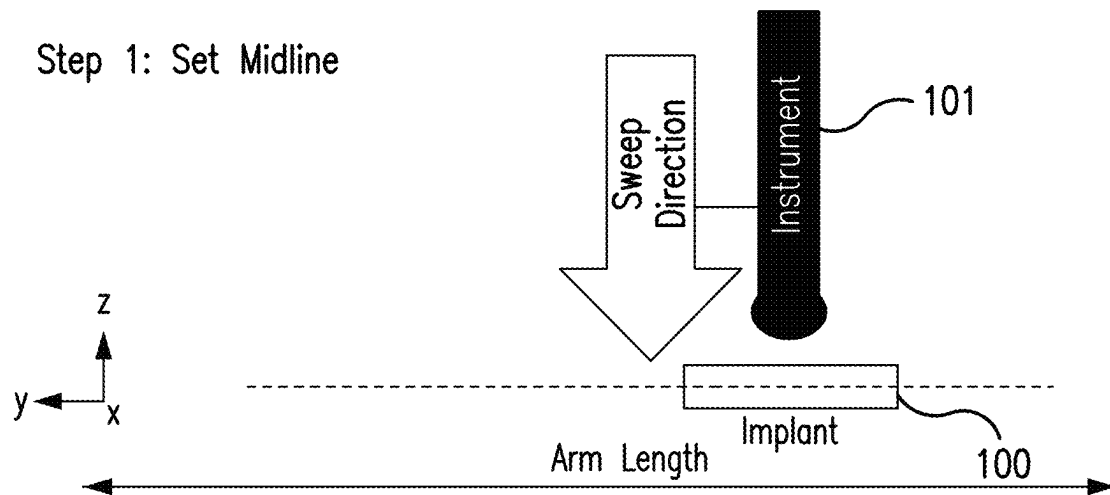


FIG. 9A

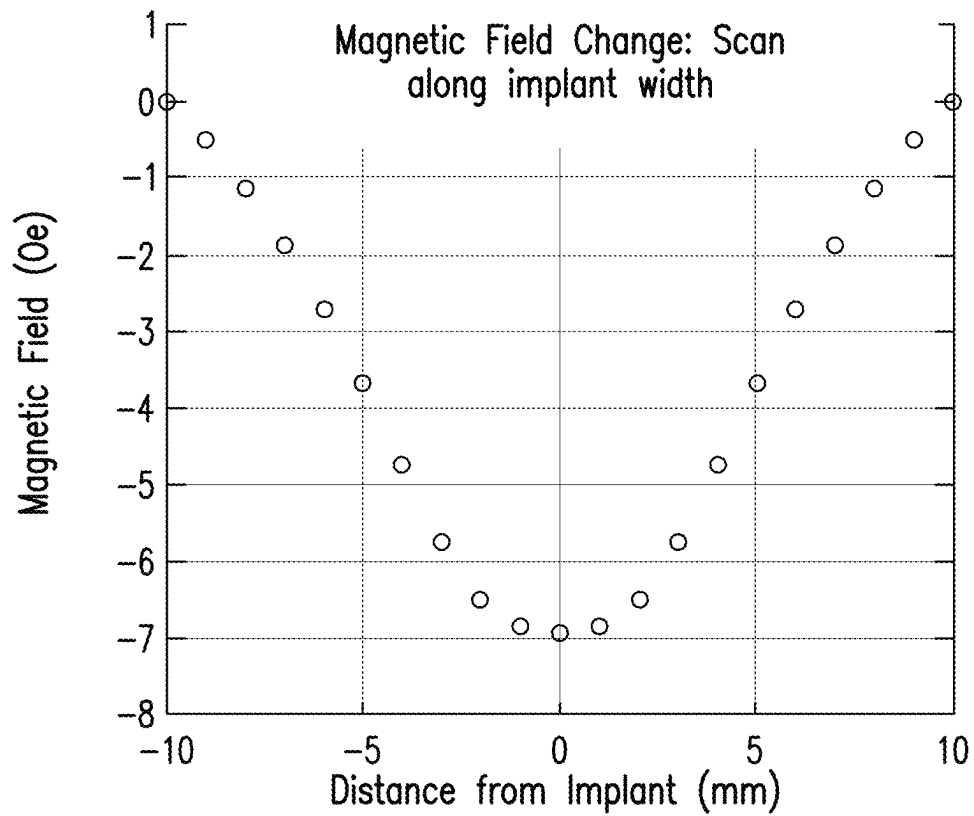


FIG. 9B

Step 2: Find center
and edge of implant

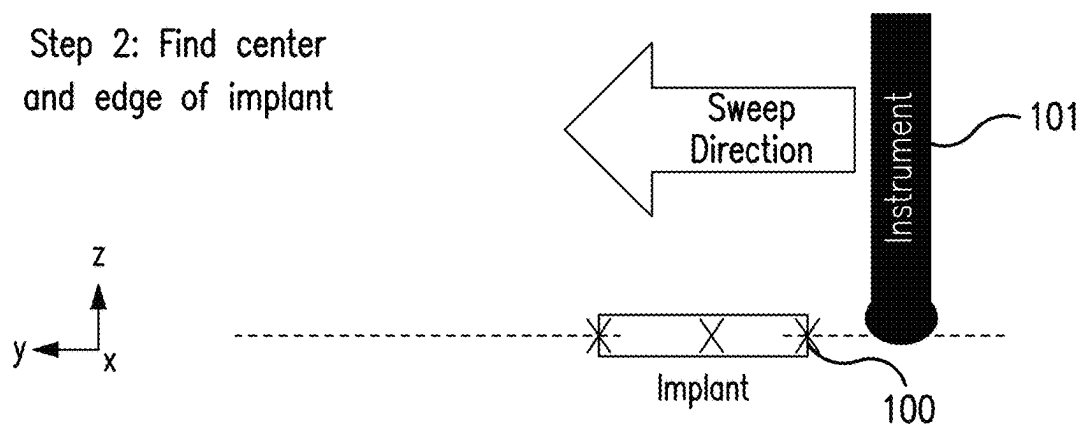


FIG. 9C

Magnetic Field Change: Sweep along sample length at different heights

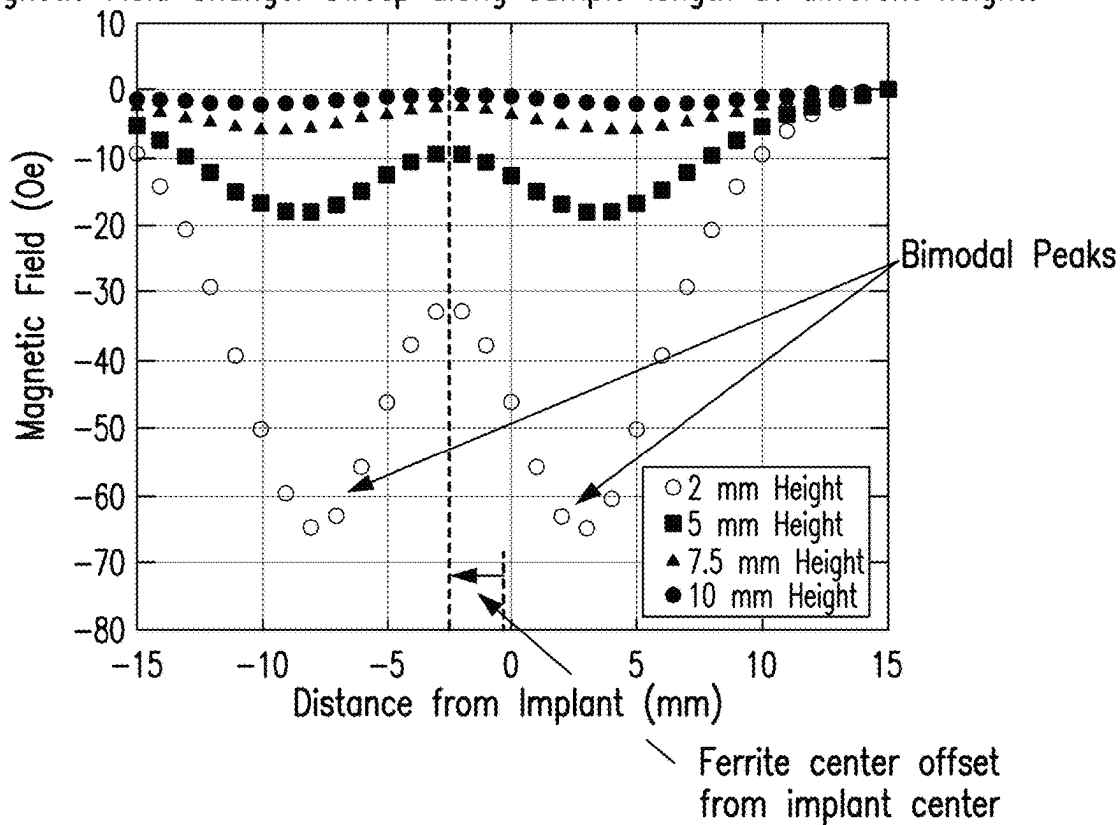


FIG. 9D

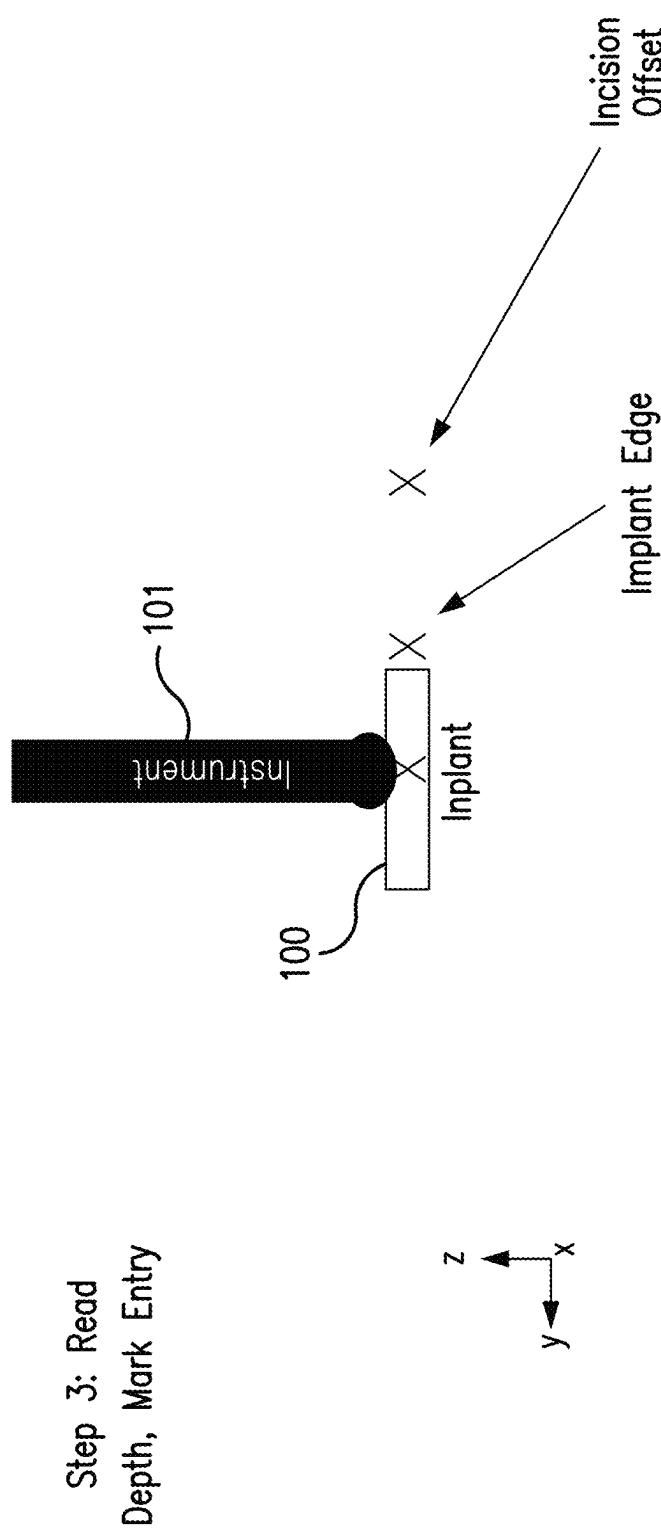


FIG. 9E

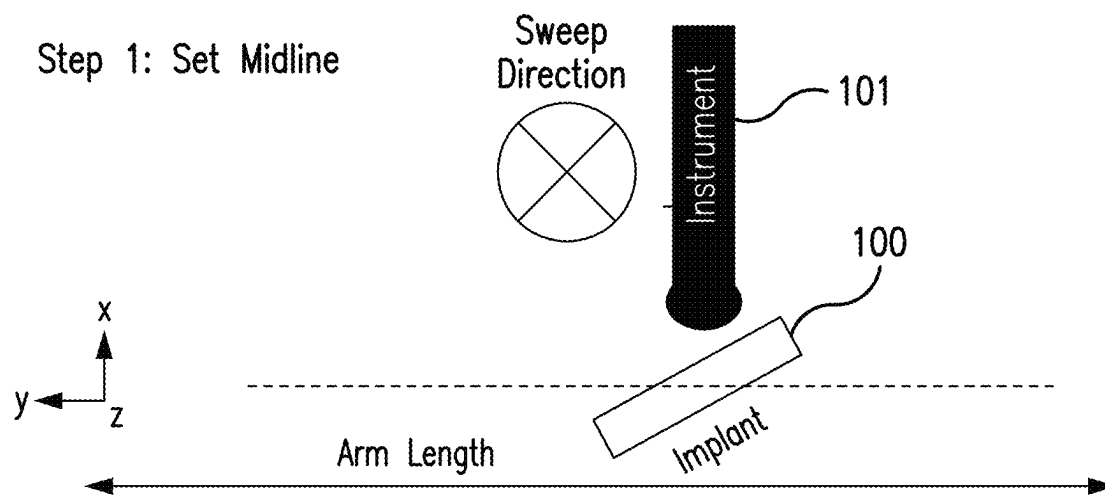


FIG. 10A

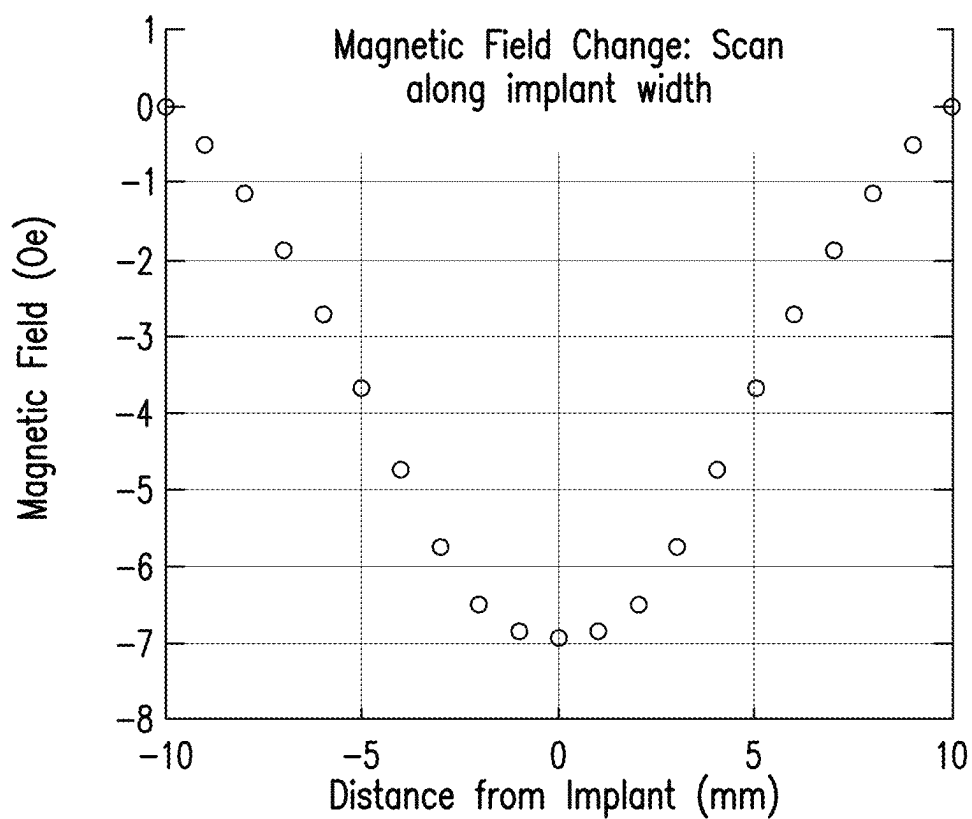


FIG. 10B

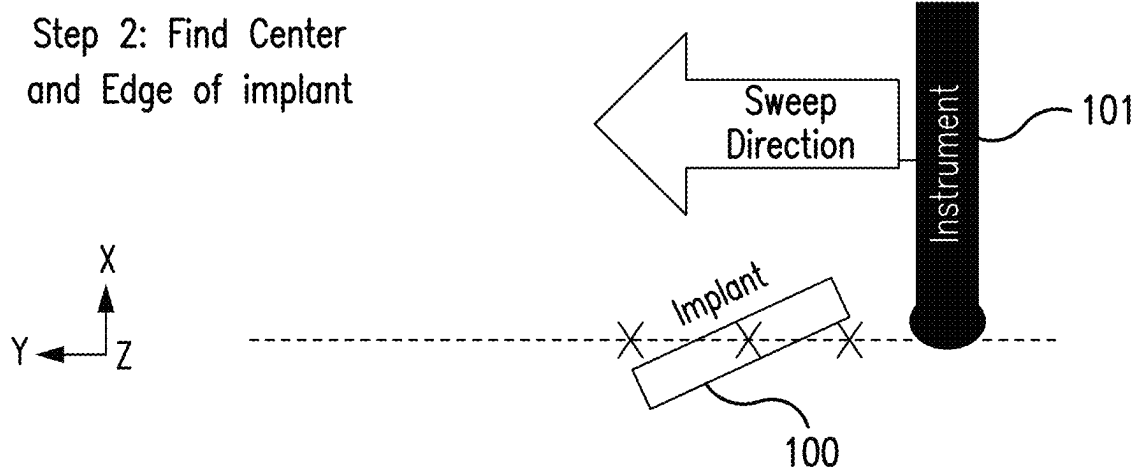
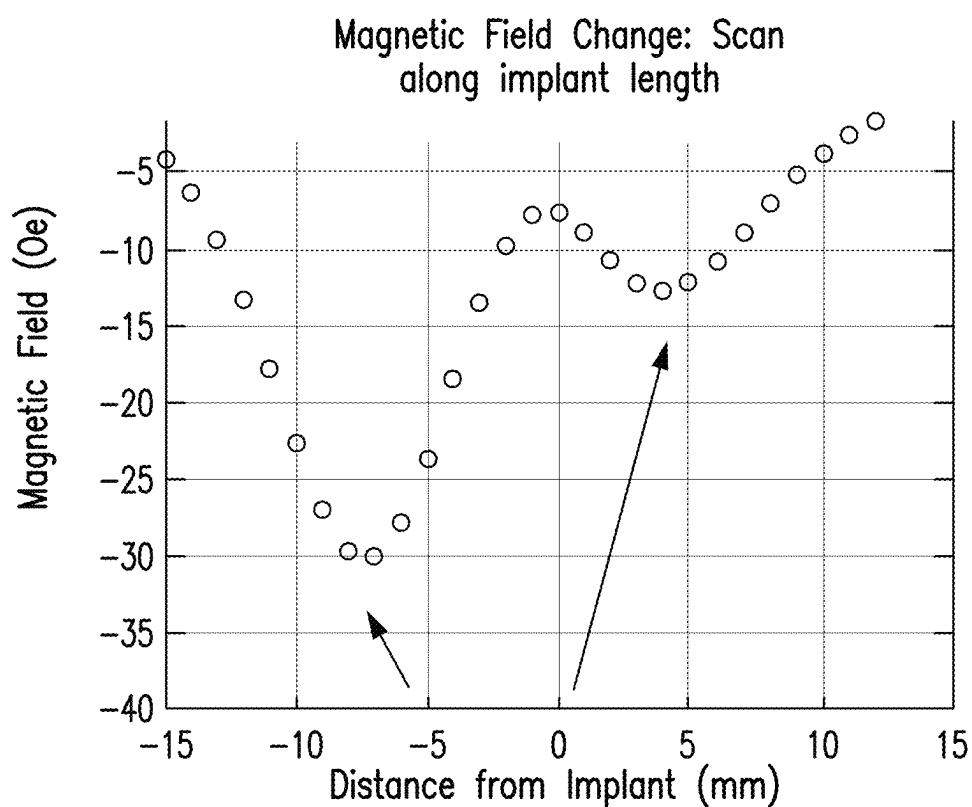


FIG. 10C



∨
Bimodal Peak
Differential

FIG. 10D

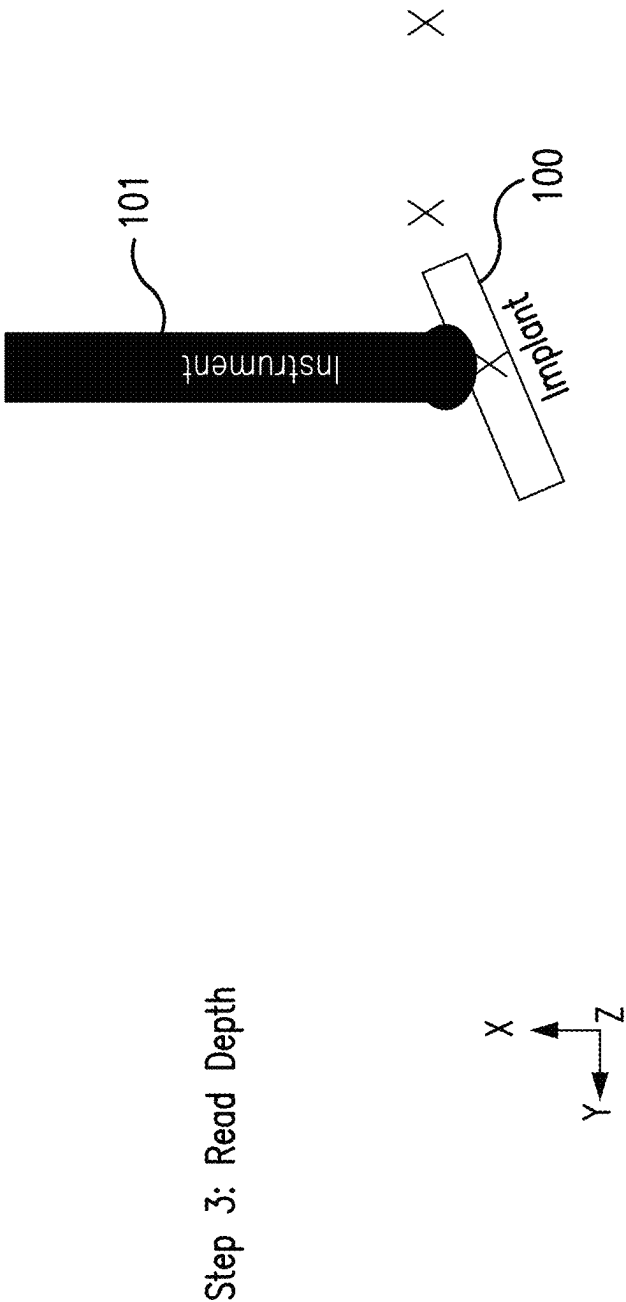


FIG. 10E

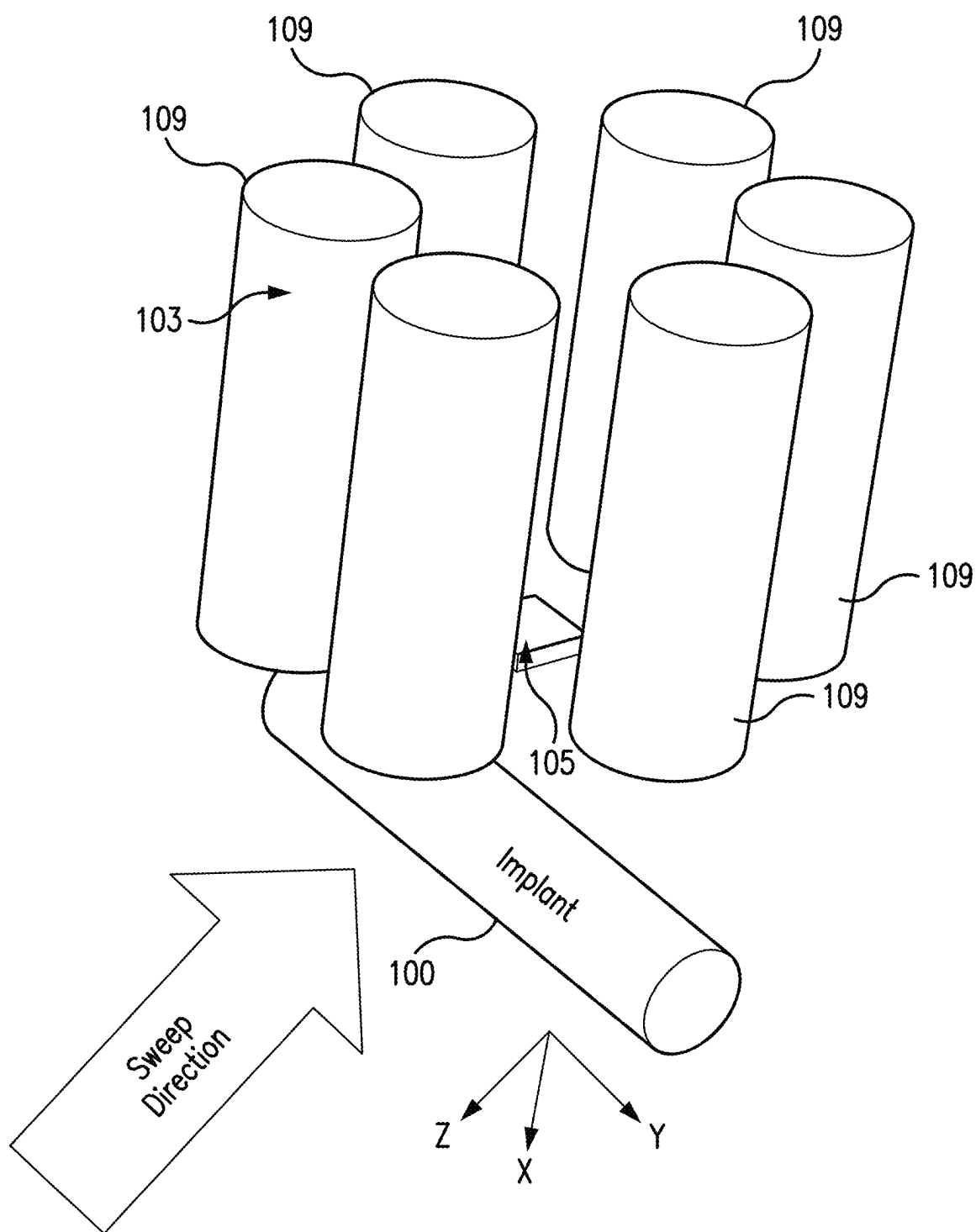


FIG. 11A

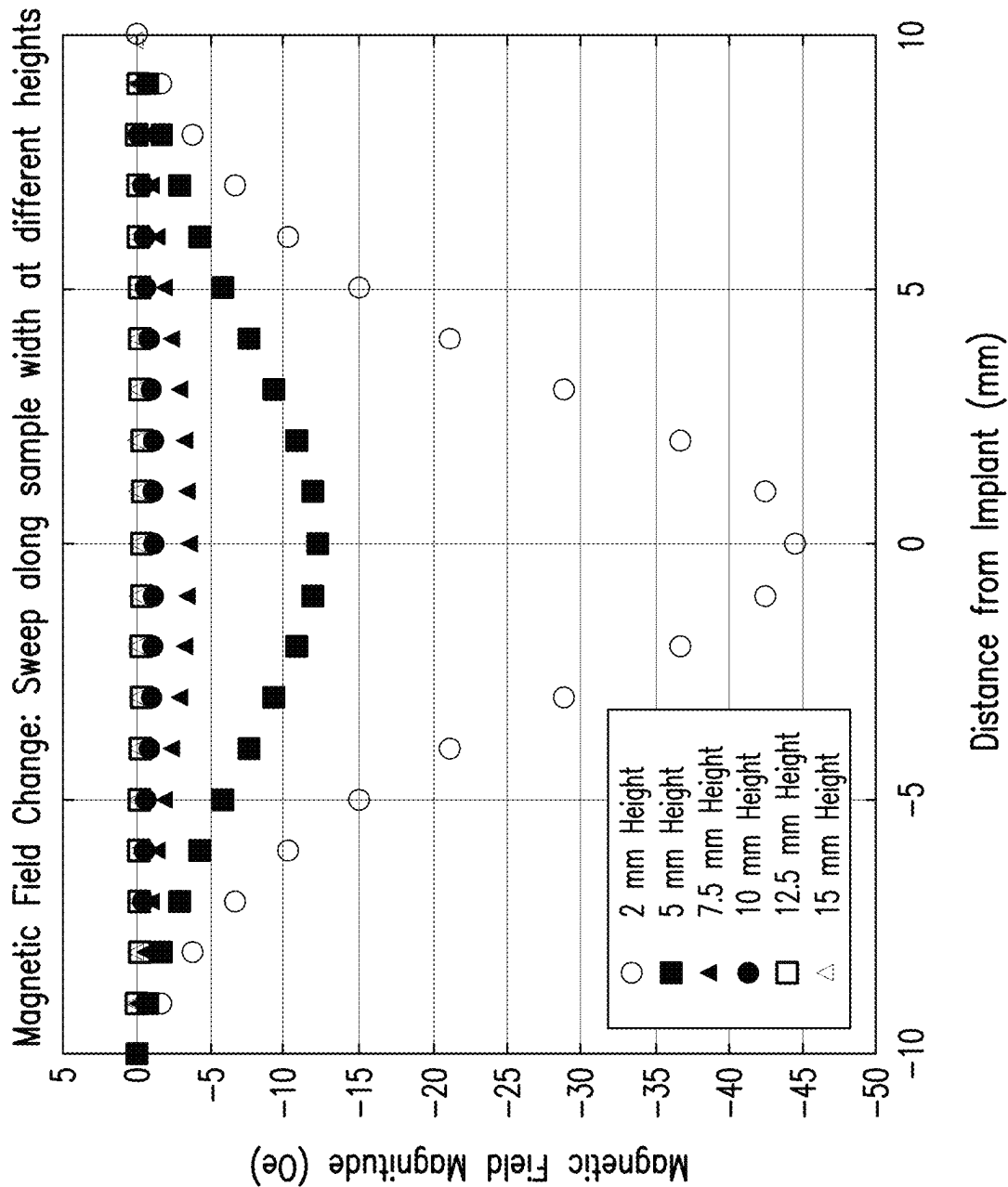


FIG. 11B

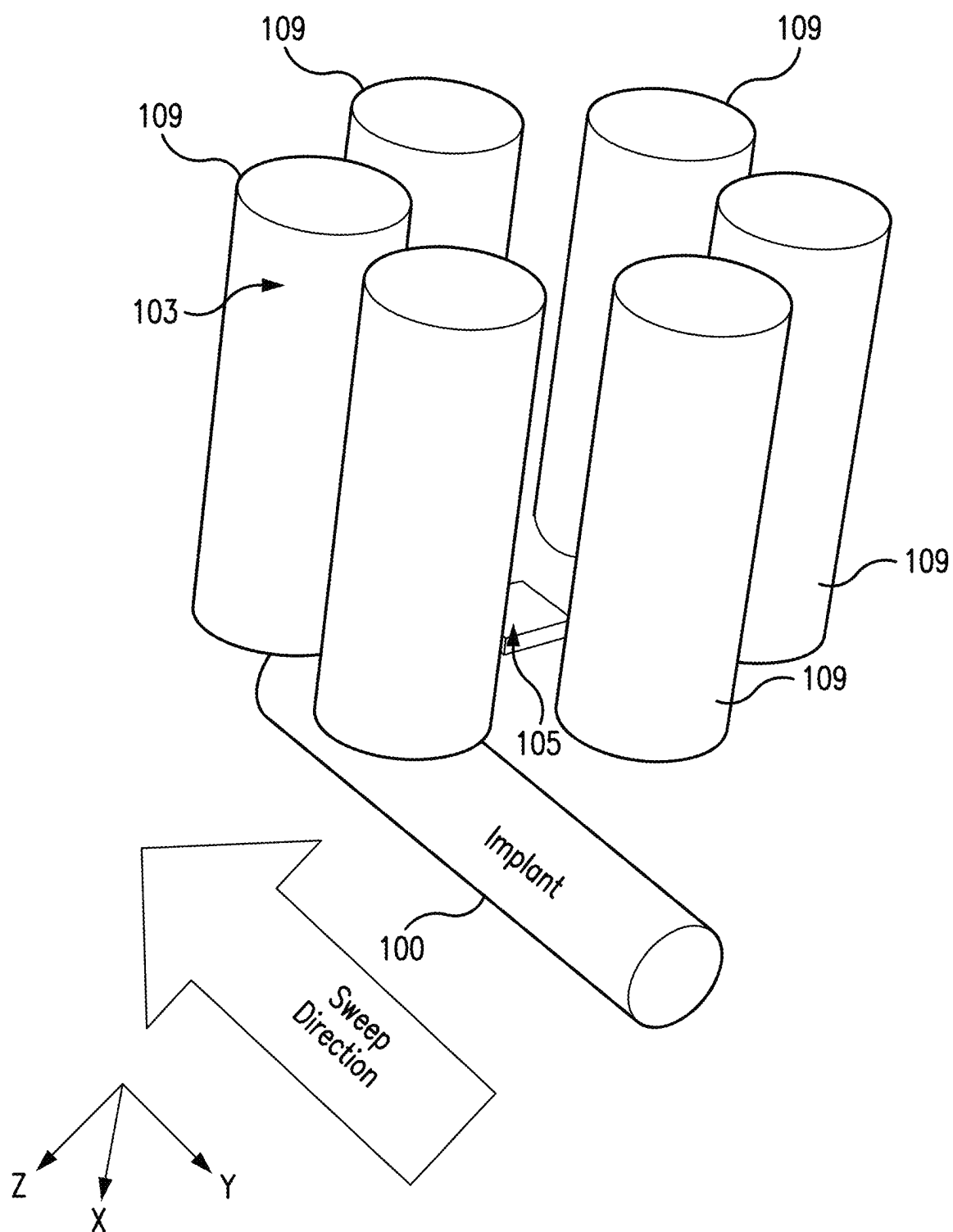


FIG. 12A

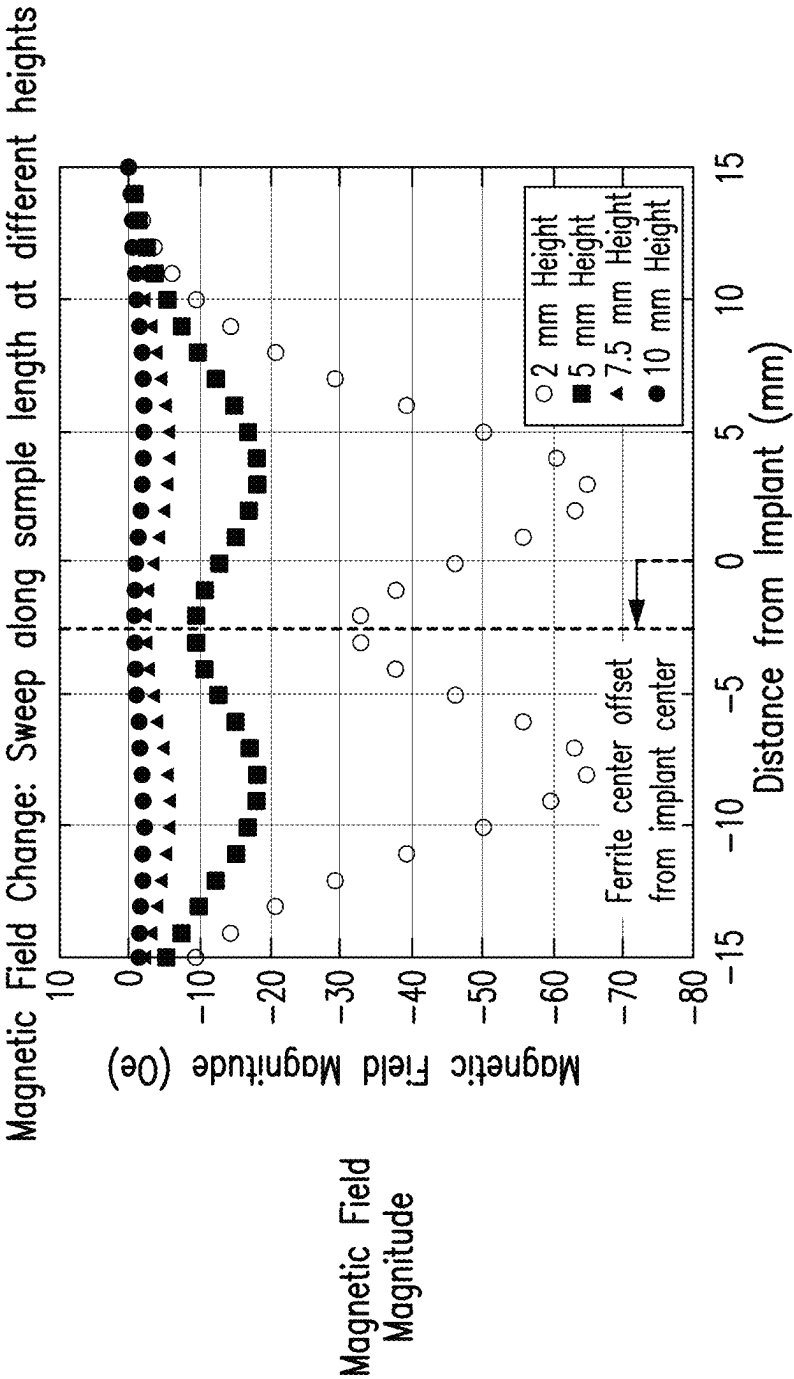


FIG. 12B

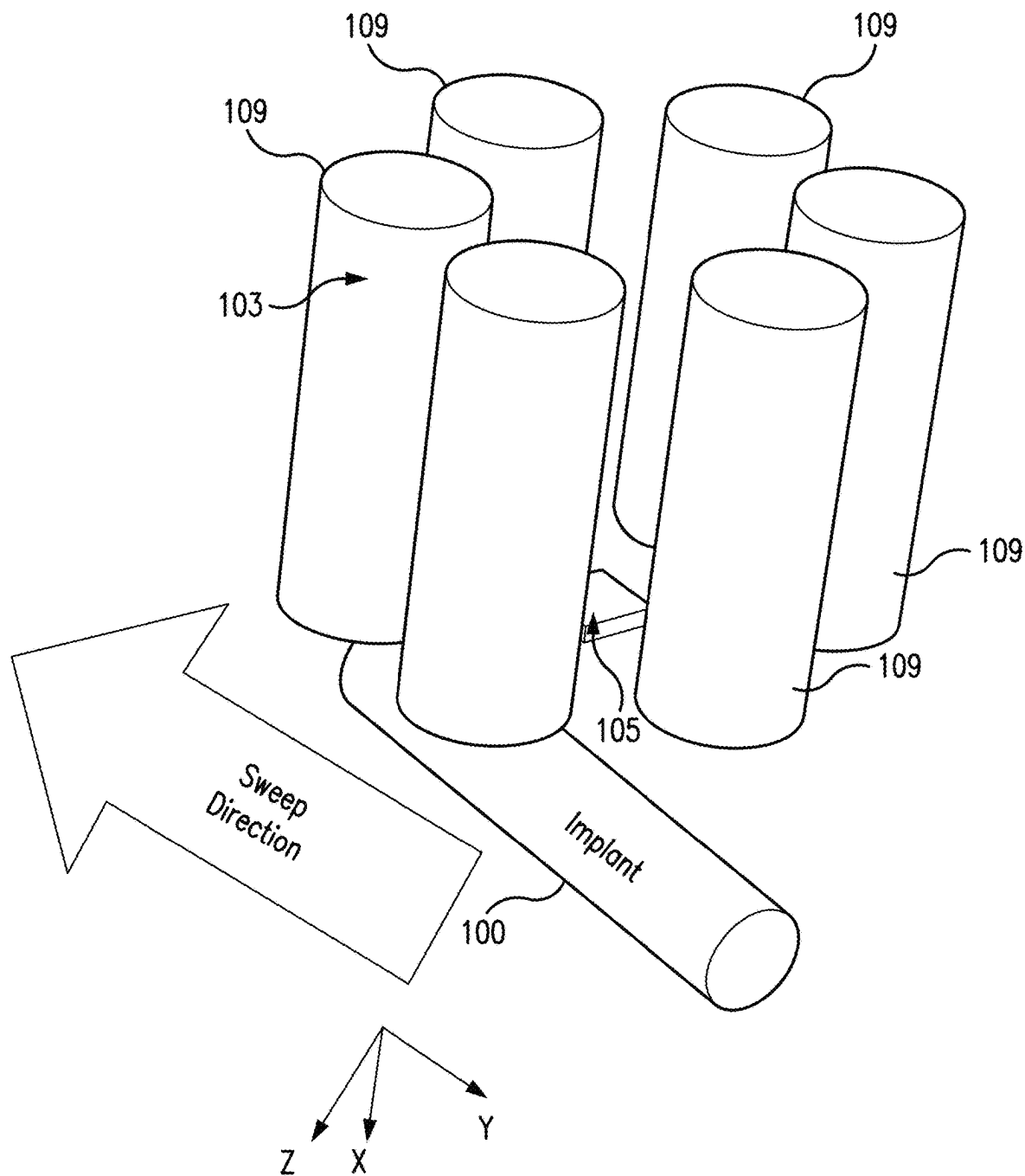


FIG. 13A

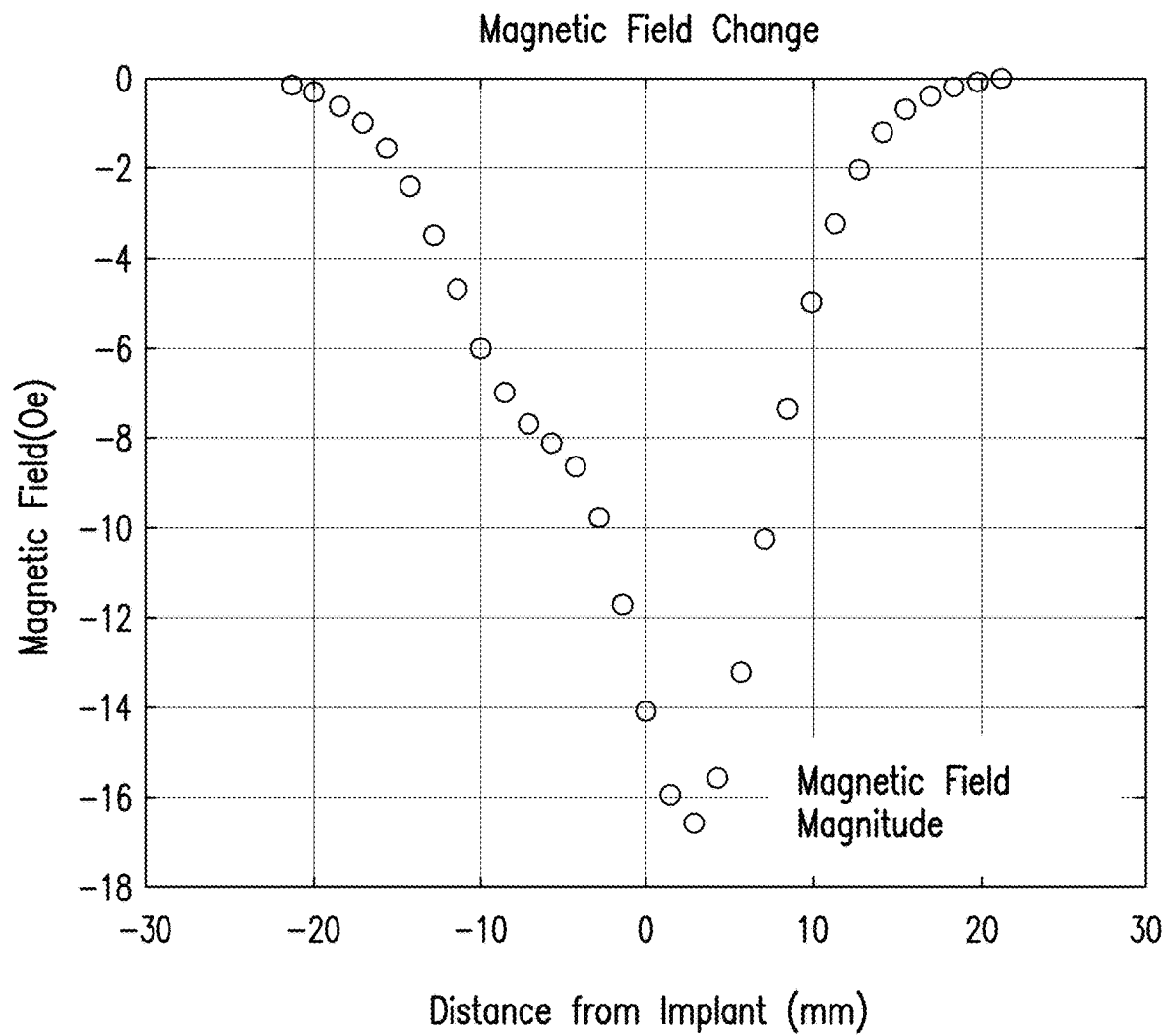


FIG. 13B

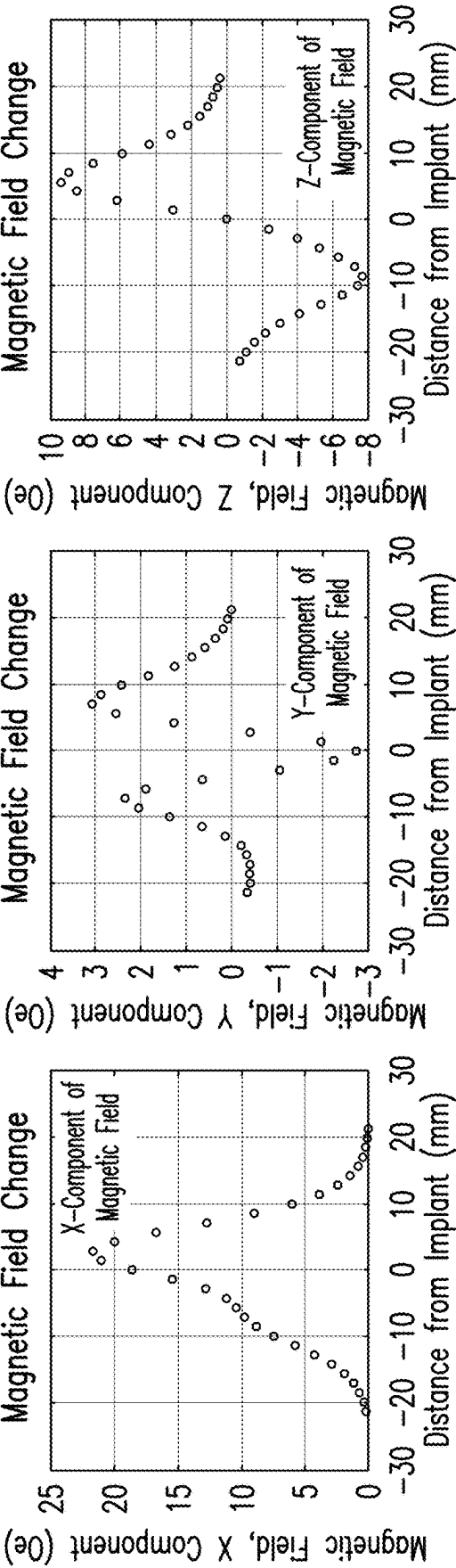


FIG. 13C

FIG. 13D

FIG. 13E

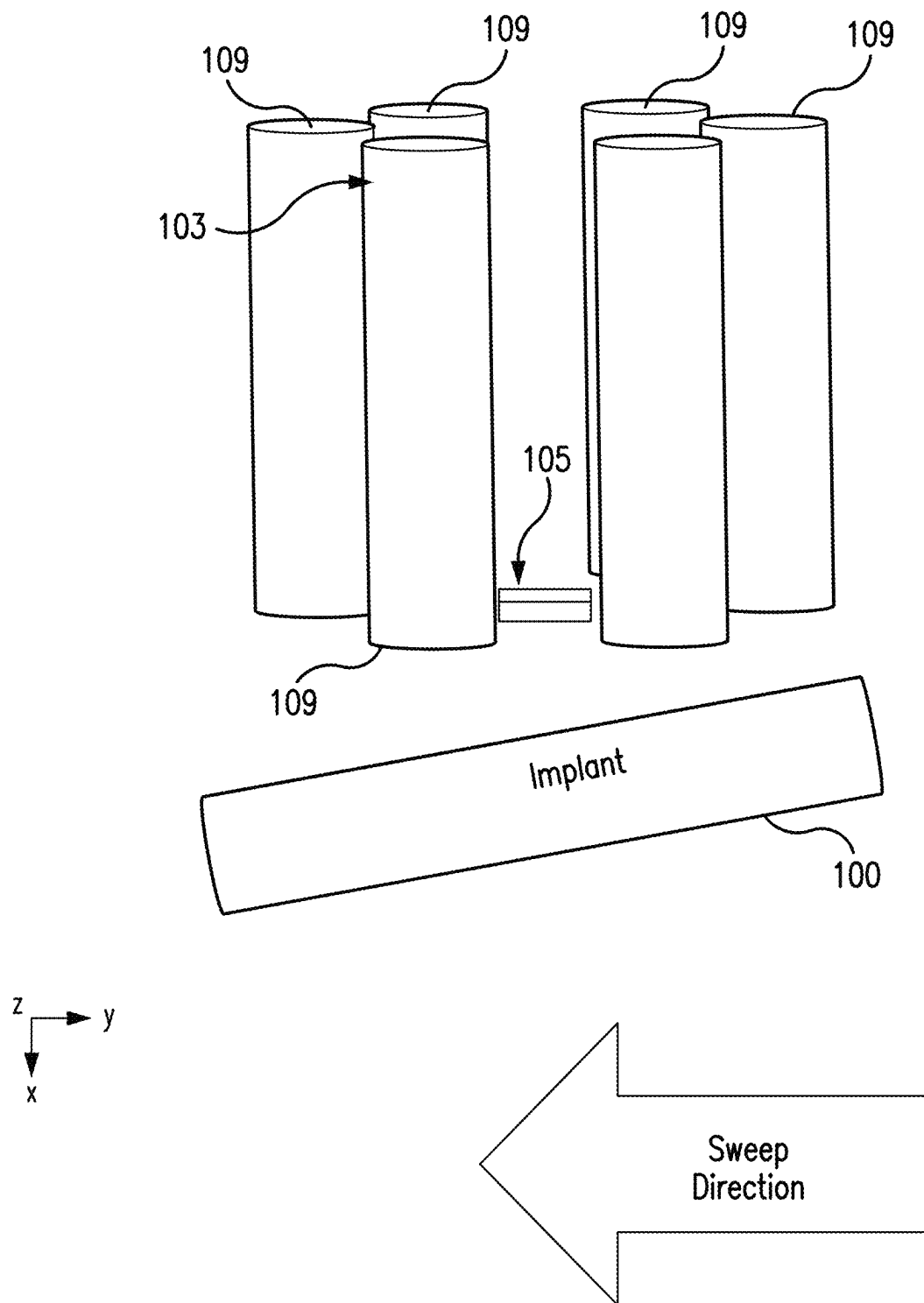


FIG. 14A

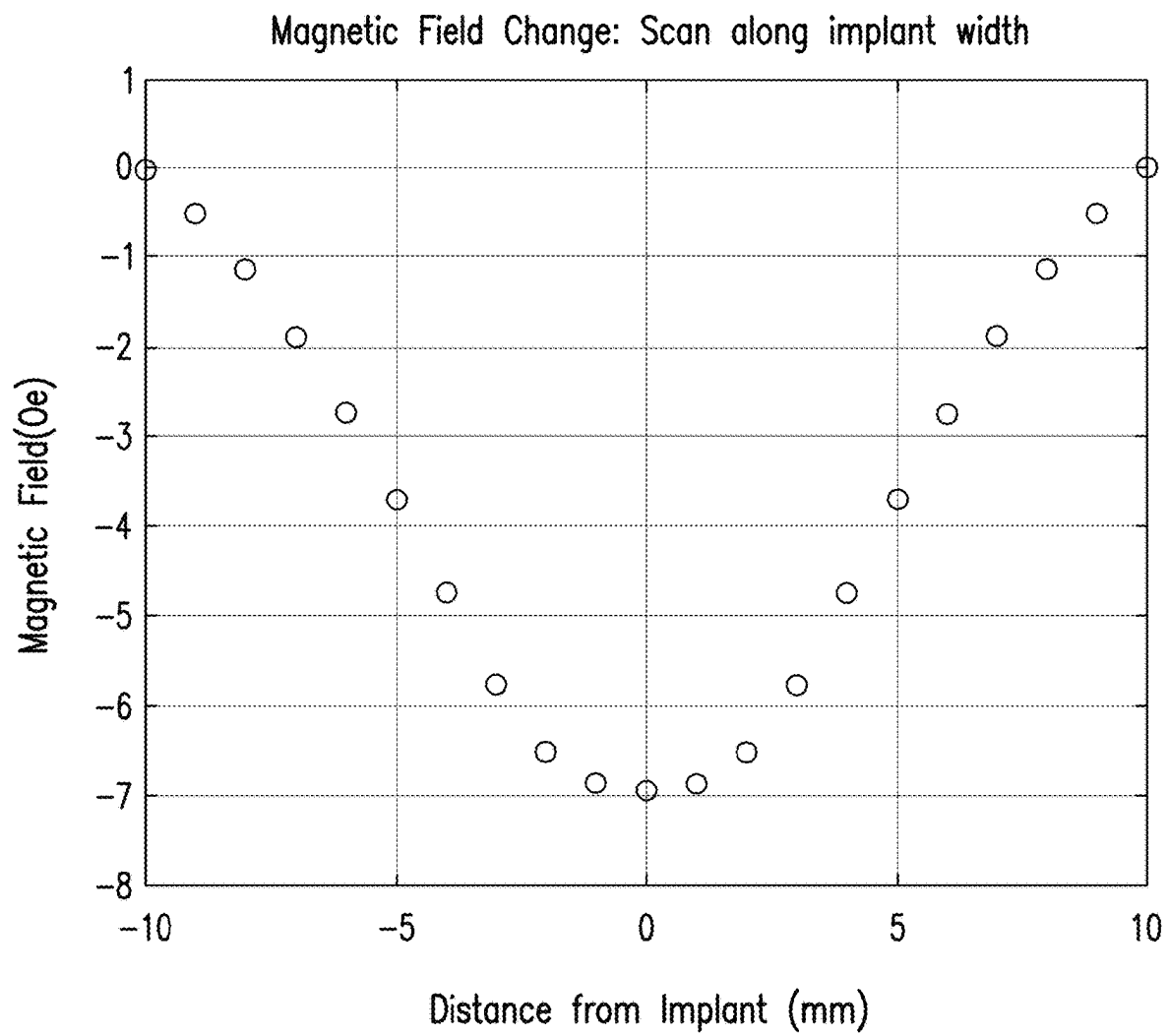


FIG. 14B

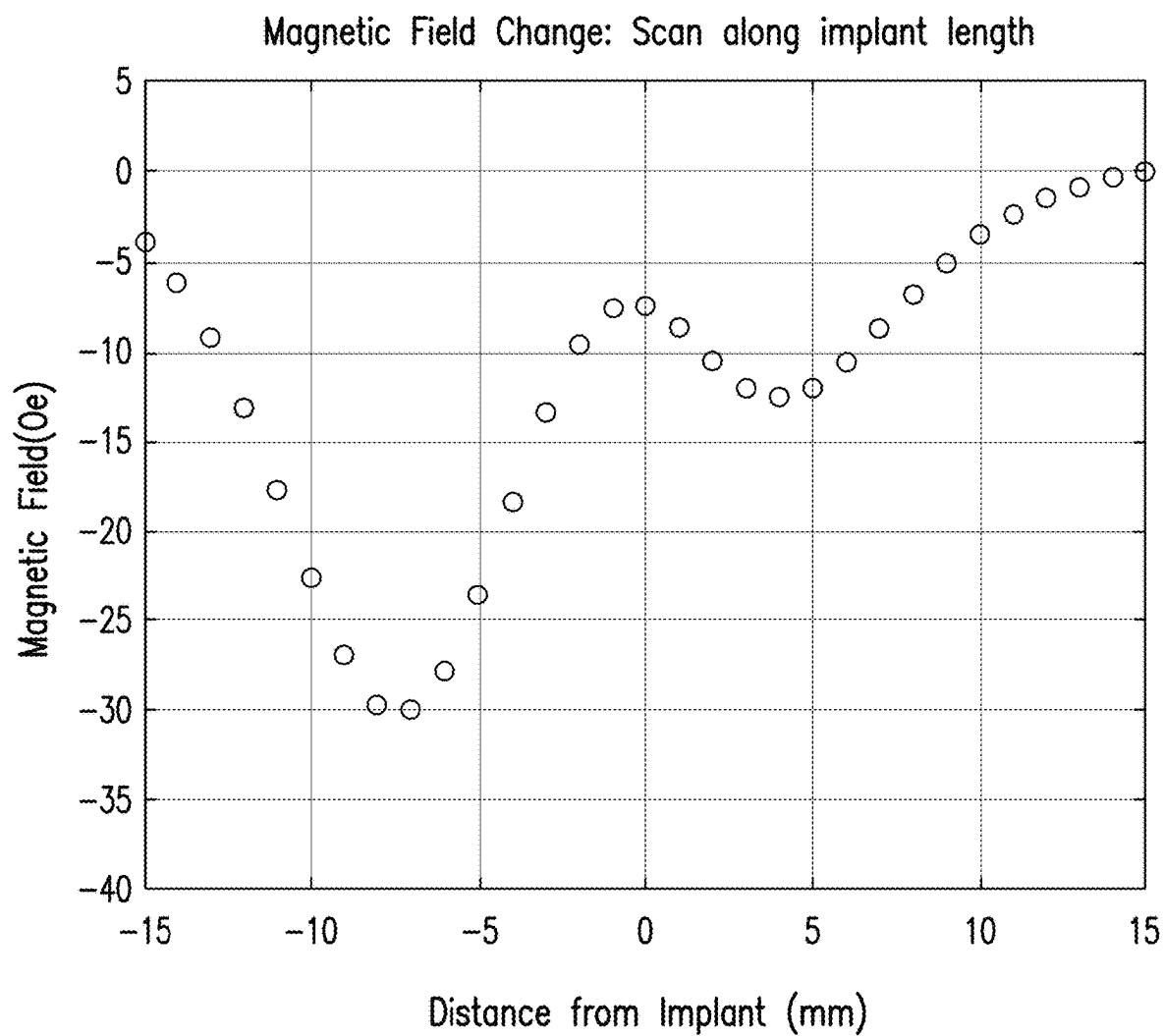
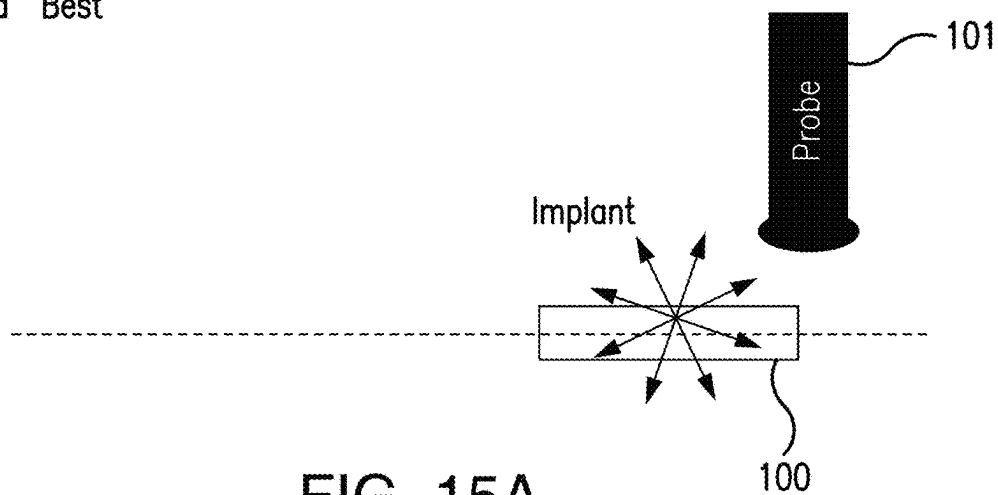
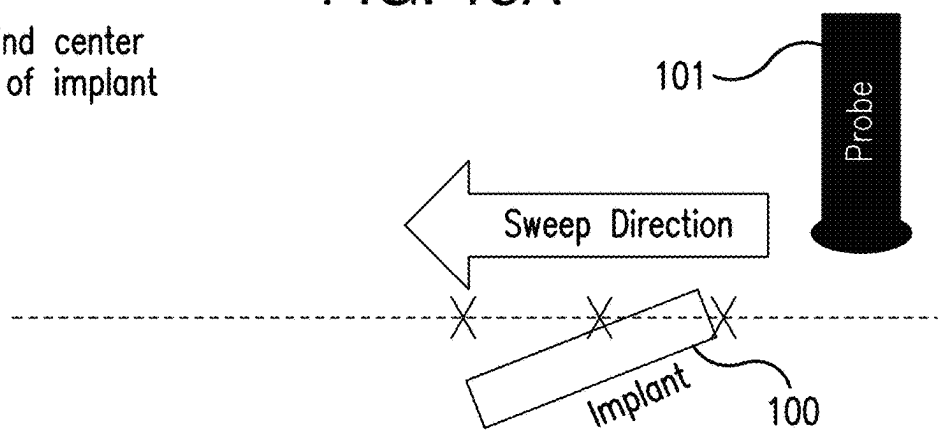


FIG. 14C

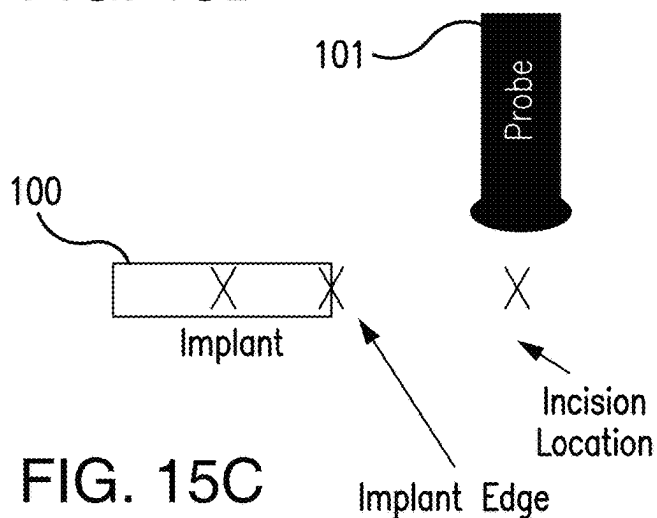
Step 1: Find "Best"
Orientation



Step 2: Find center
and edge of implant



Step 3: Read
Depth, Mark Entry



Step 1: Approximate location

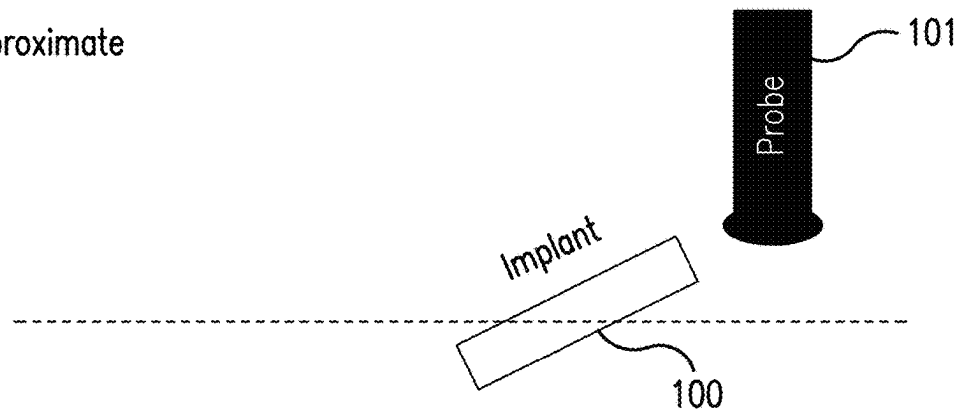


FIG. 16A

Step 2: Place Template Sticker and Sweep Course

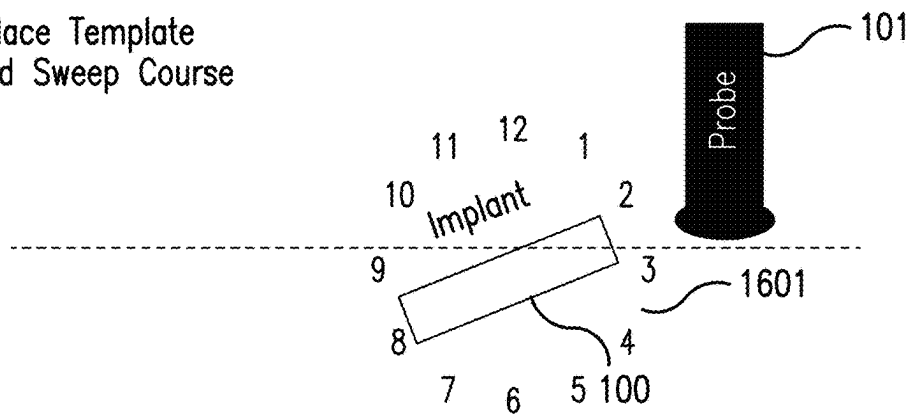


FIG. 16B

Step 3: Receive coordinates and orientation info

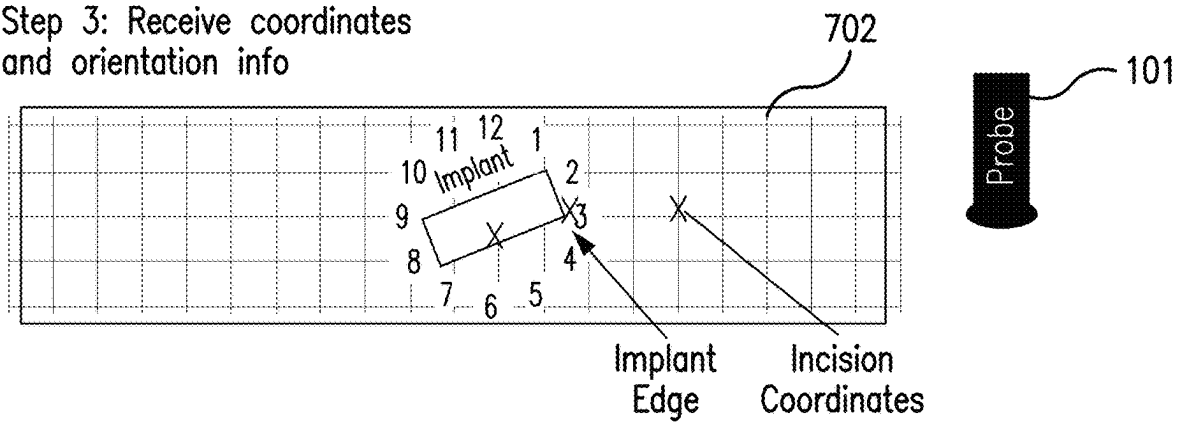


FIG. 16C

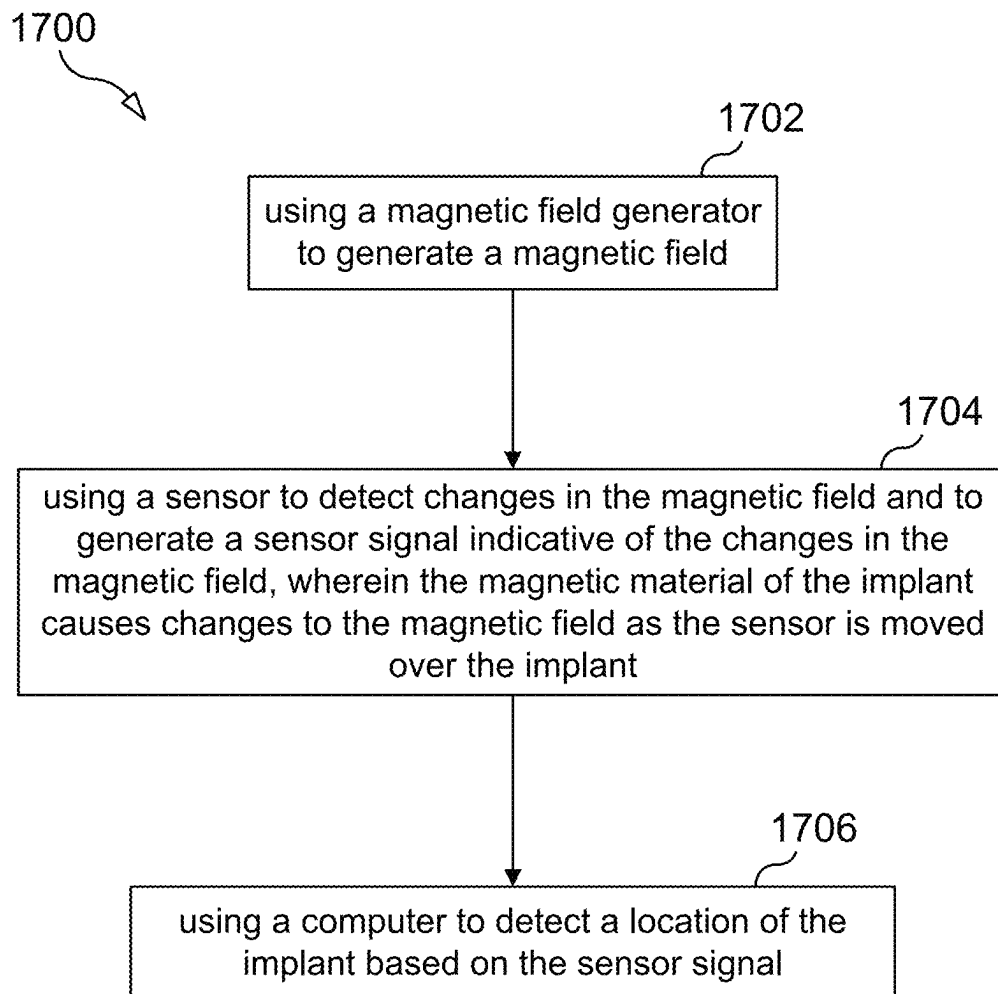


FIG. 17

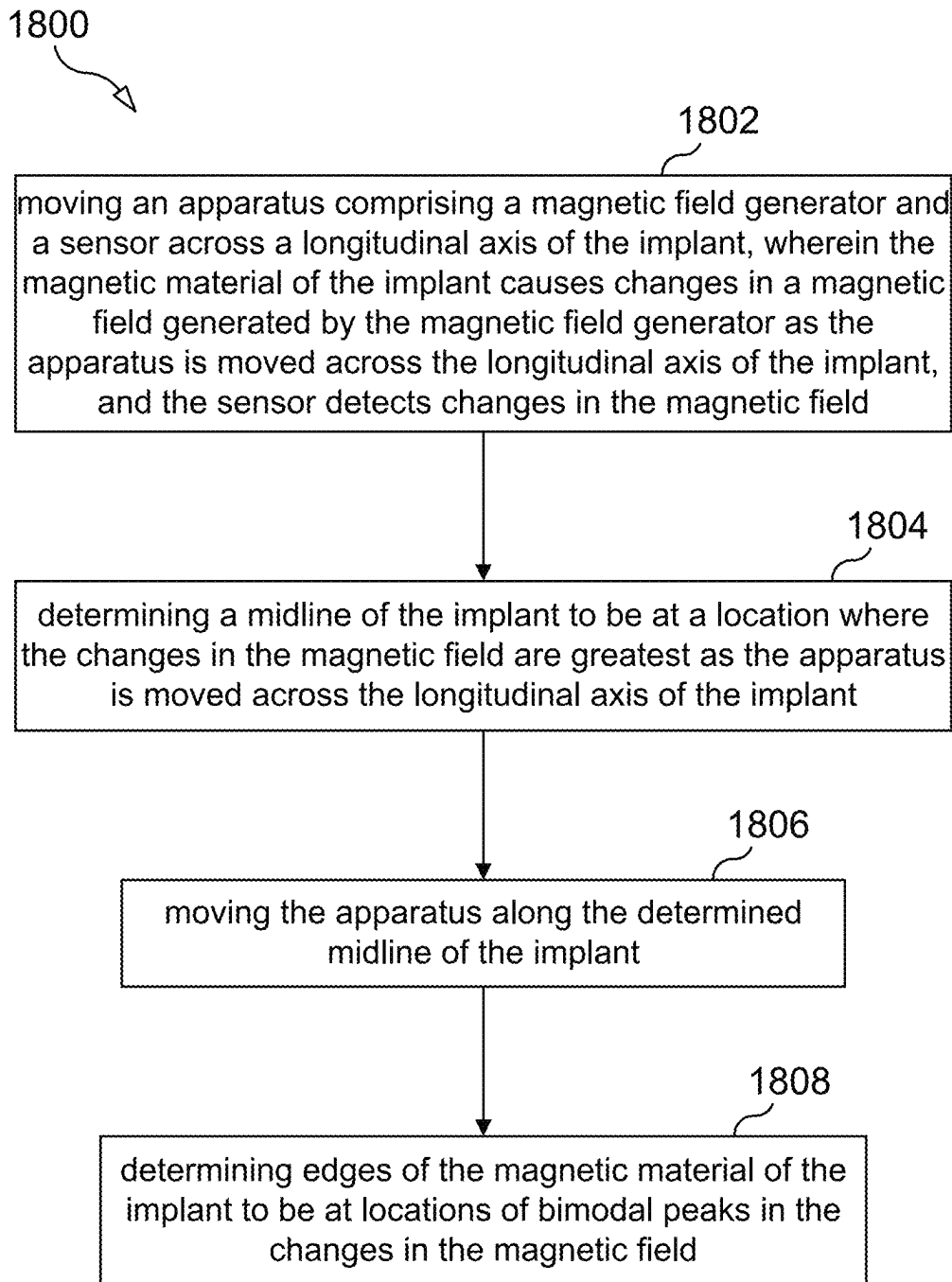


FIG. 18

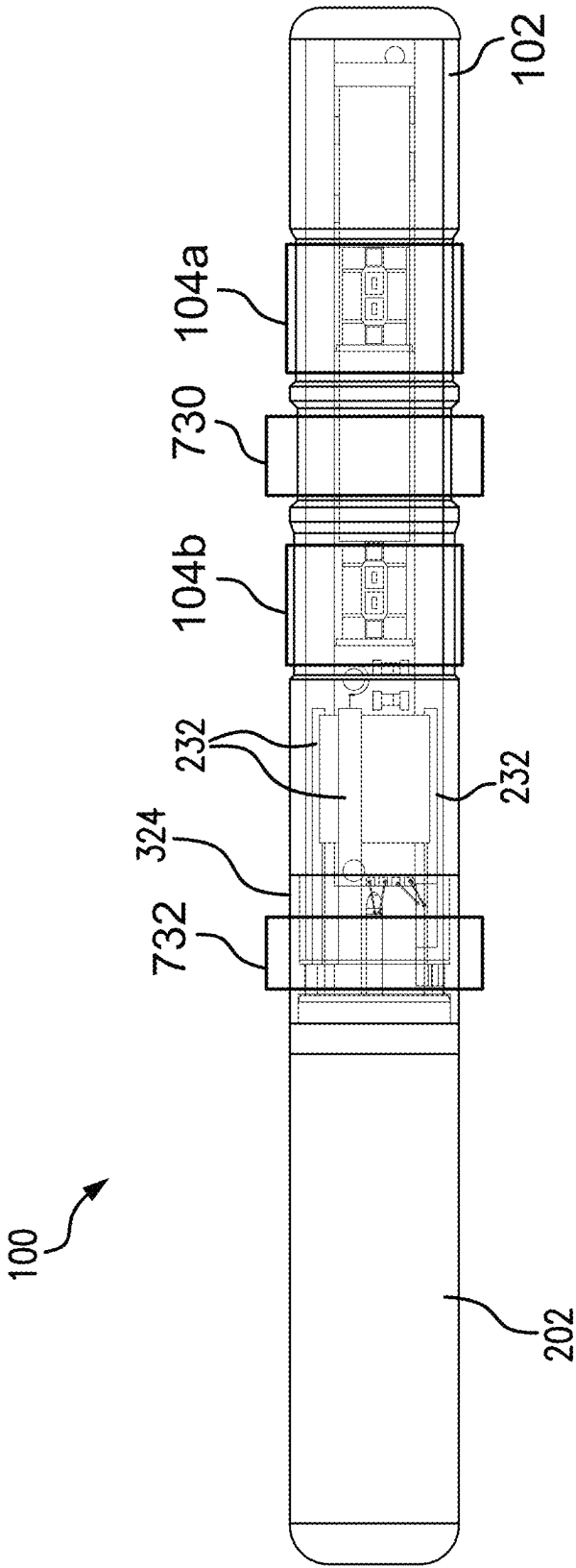


FIG. 19

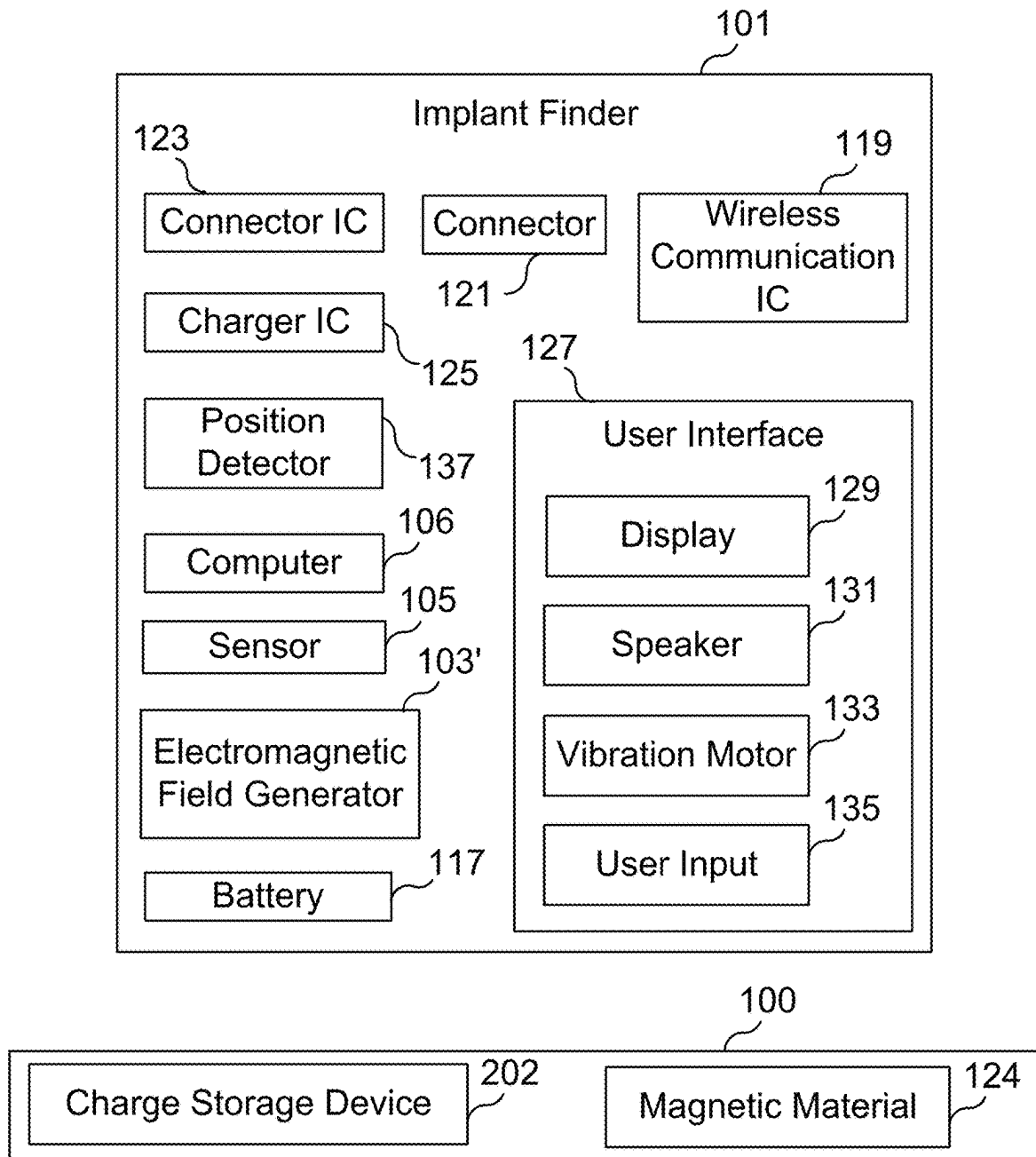


FIG. 20

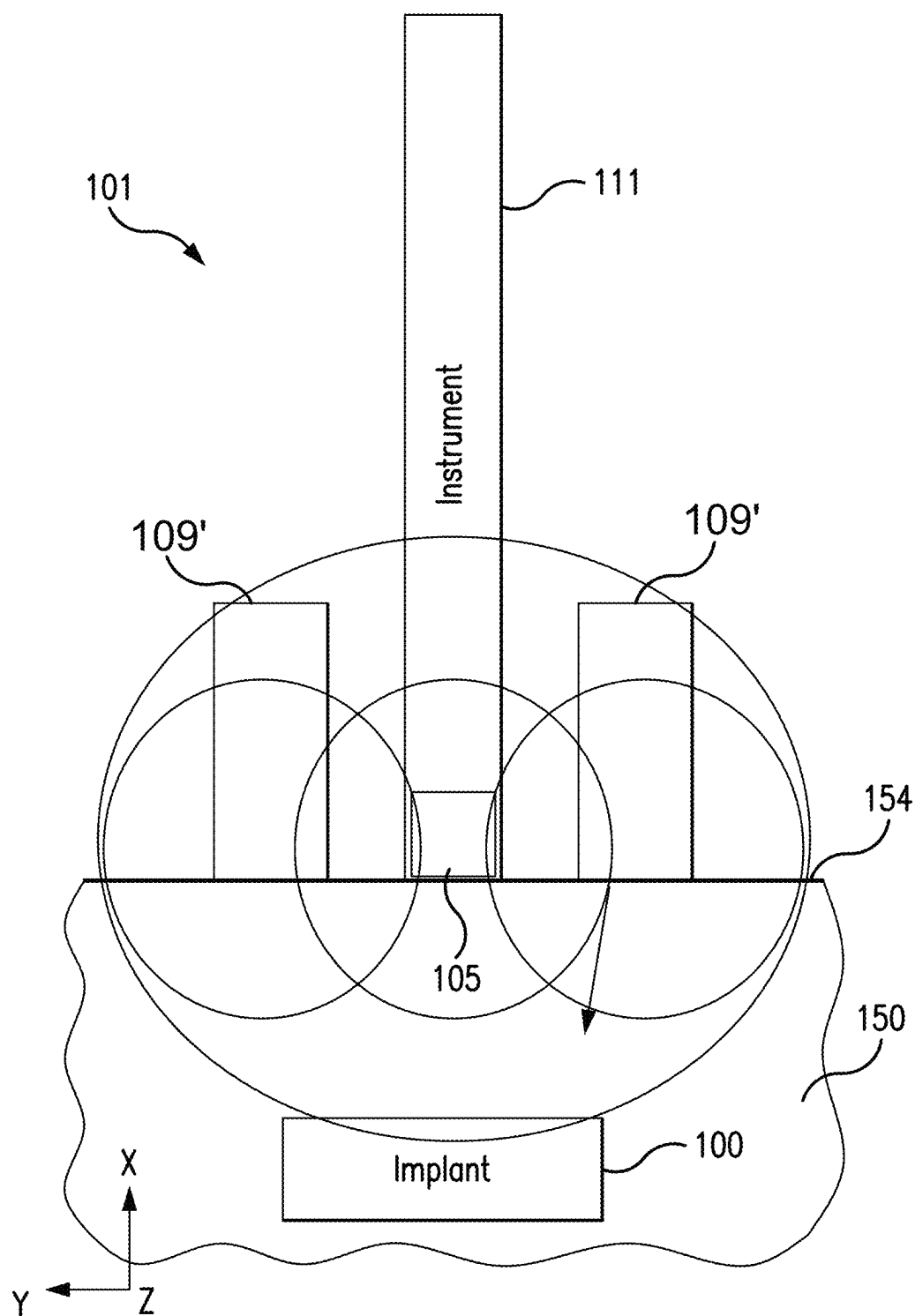


FIG. 21A

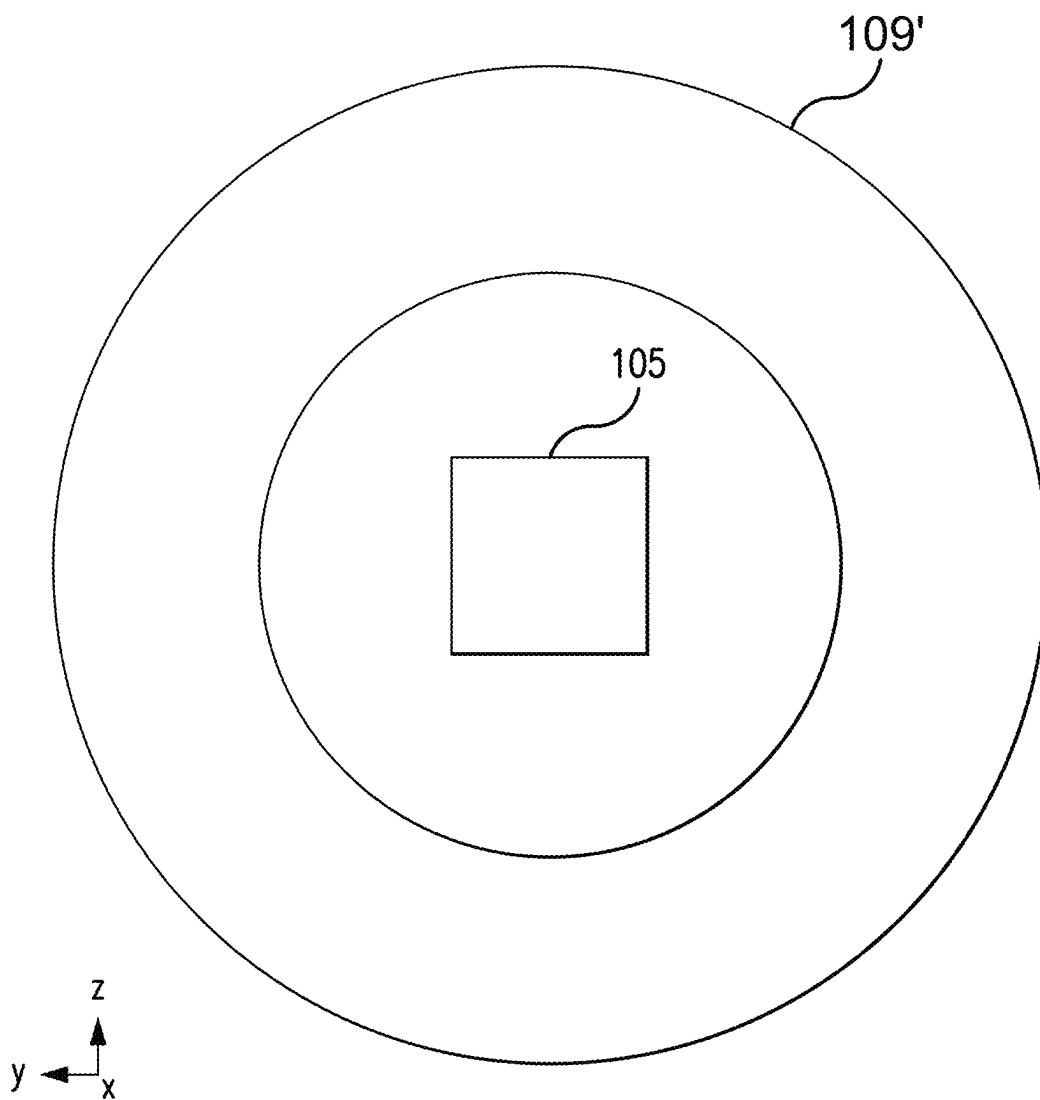


FIG. 21B

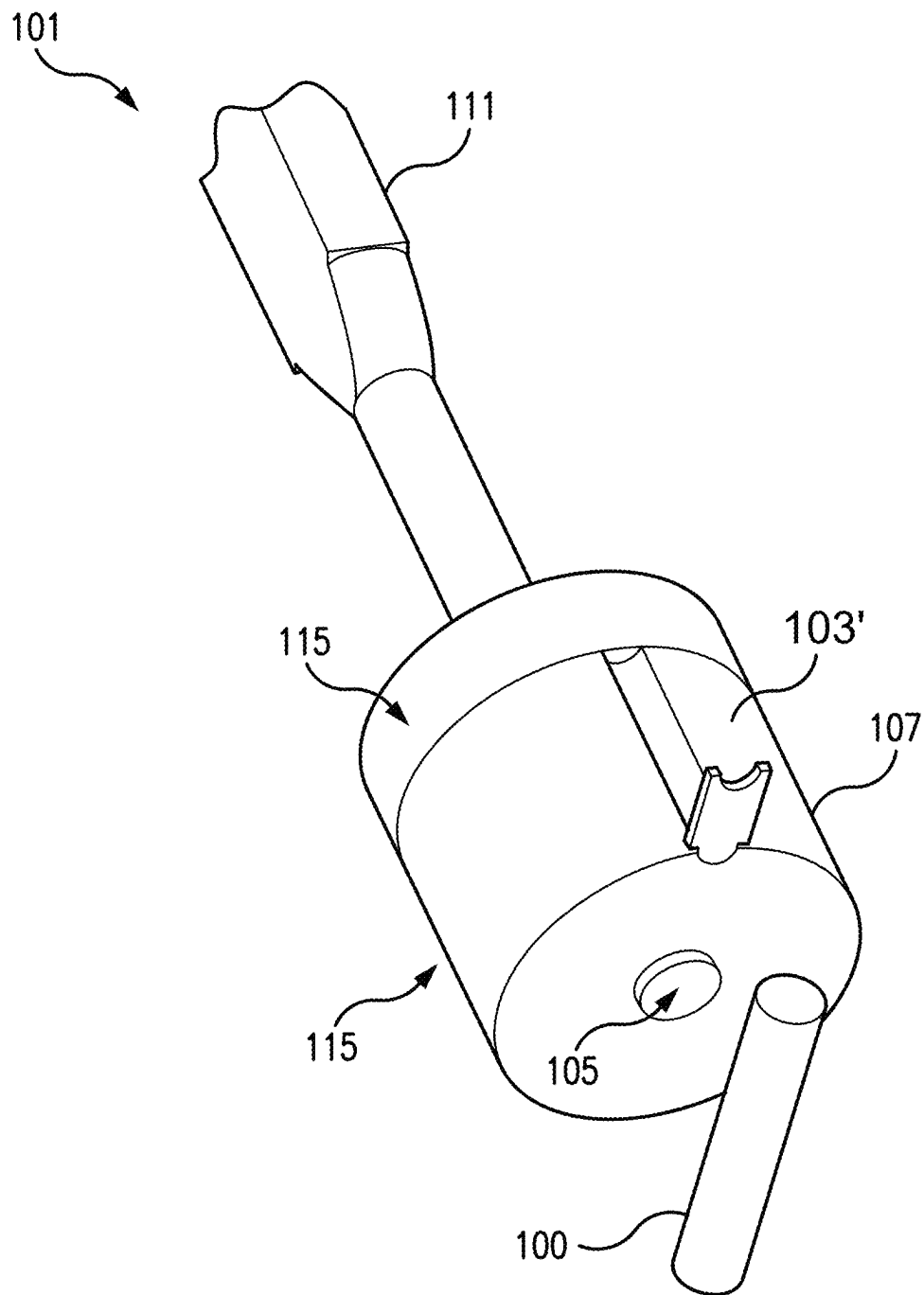


FIG. 21C

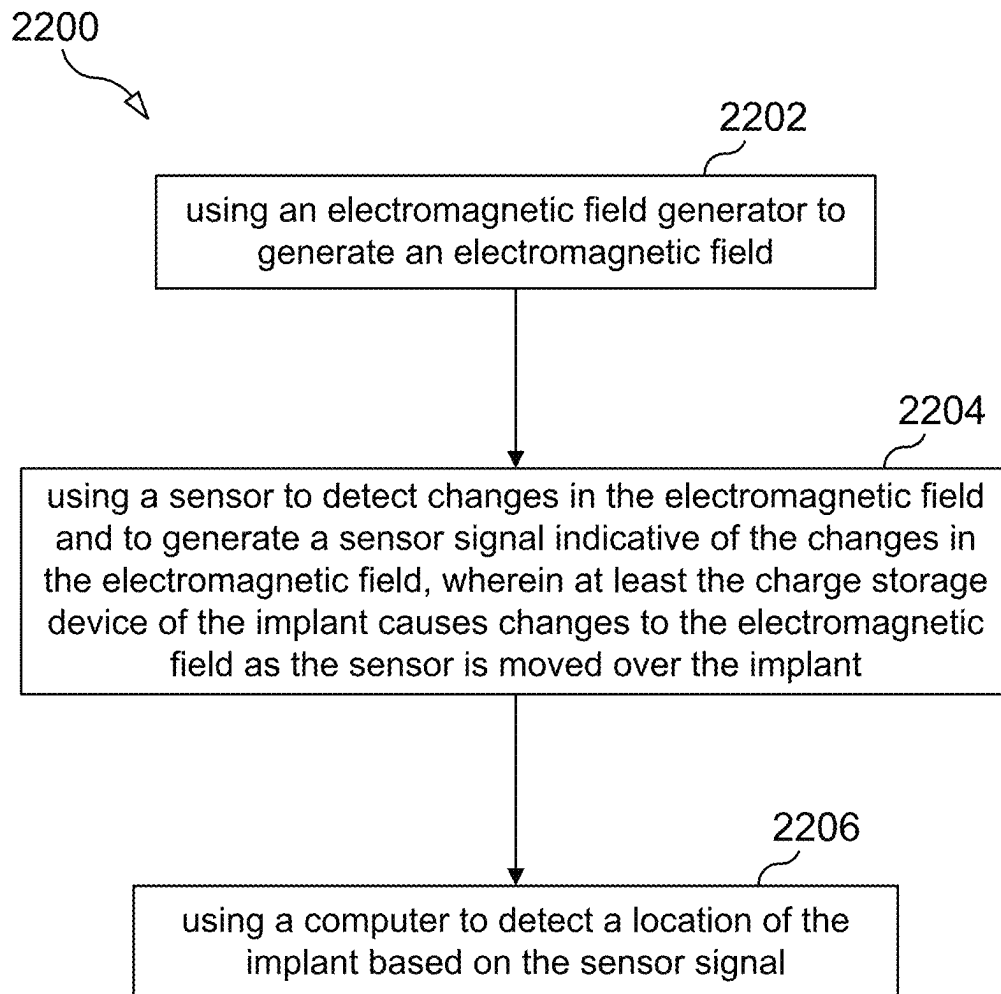


FIG. 22

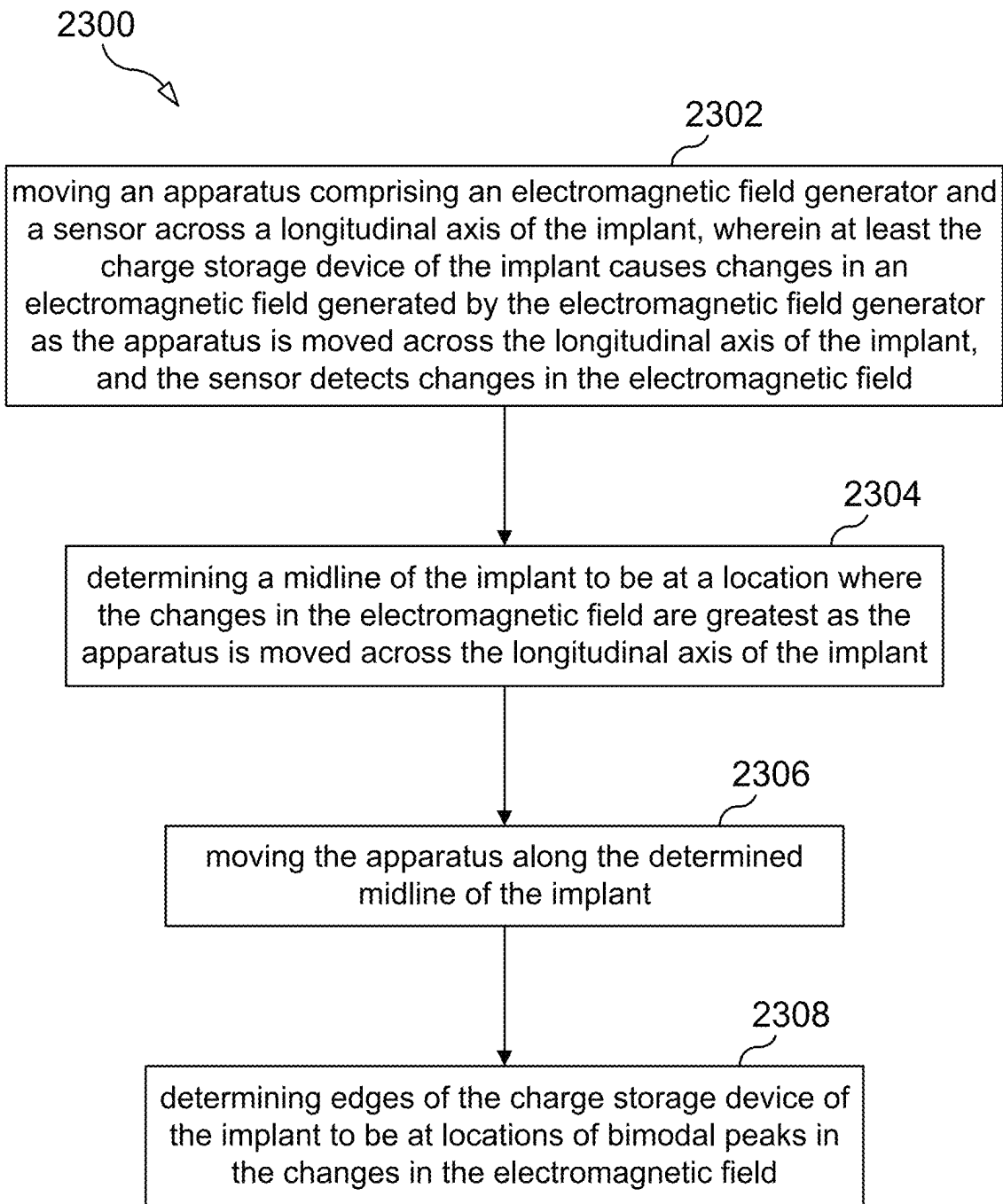


FIG. 23

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IMPLANT FINDER**CROSS-REFERENCE TO RELATED APPLICATION**

The present application claims the benefit of priority to U.S. Provisional Application Ser. No. 63/327,566, filed on Apr. 5, 2022, which is incorporated herein by reference in its entirety.

BACKGROUND**Field of Invention**

The present disclosure relates to apparatuses and method for locating an implant including magnetic material and/or a charge storage device in a living animal.

Discussion of the Background

Implants, such as sensors, may be implanted within a living animal (e.g., a human). An implant may, for example, detect the presence or amount of an analyte (e.g., glucose or oxygen) in a medium (e.g., blood or interstitial fluid) within the living animal. Some implants may be relatively small (e.g., 2-4 mm diameter capsule) and may be implanted in the subcutaneous tissue of a host. The implant may need to be located (e.g., for removal of the implant). Due to the small size of the implant, it may be difficult to locate and retrieve the implant from the subcutaneous tissue of the host.

SUMMARY

One aspect of the invention may provide an apparatus for locating an implant including magnetic material. The apparatus may include a magnetic field generator, a sensor, and a computer. The magnetic field generator may be configured to generate a magnetic field. The sensor may be configured to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field. The magnetic material of the implant may cause changes to the magnetic field as the sensor is moved over the implant. The computer may be configured to use the sensor signal to detect a location of the implant.

In some aspects, the magnetic field generator may include a cylindrical magnet. In some aspects, the cylindrical magnet may be hollow.

In some aspects, the magnetic field generator may include one or more magnets. In some aspects, the magnetic field generator may include two or more magnets. In some aspects, the magnetic field generator may further include a housing configured to hold the two or more magnets. In some aspects, the magnetic field generator may include four magnets. In some aspects, the magnetic field generator may include six magnets. In some aspects, the one or more magnets may include one or more permanent magnets. In some aspects, the one or more magnets may include one or more electromagnets.

In some aspects, the magnetic field generated by the magnetic field generator may be a substantially uniform magnetic field.

In some aspects, the magnetic field may be substantially symmetric about a longitudinal axis at the center of the magnetic field generator. In some aspects, the magnetic field may be non-uniform and/or asymmetric about a longitudinal axis at the center of the magnetic field generator. In some

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aspects, the sensor may be located along or offset from the longitudinal axis at the center of the magnetic field generator.

In some aspects, the computer may be configured to, in detecting the location of the implant, determine edges of the magnetic material of the implant based on locations of bimodal peaks in the changes in the magnetic field during movement of the sensor along a longitudinal axis of the implant. In some aspects, the computer may be configured to determine edges of the implant based on the determined edges of the magnetic material of the implant and one of more offsets between edges of the implant and edges of the magnetic material of the implant. In some aspects, the computer may be configured to determine a depth of the implant based on a magnitude of the change in the magnetic field at the bimodal peaks in the changes in the magnetic field during the movement of the sensor along the longitudinal axis of the implant. In some aspects, the computer may be configured to determine an orientation of the implant based on a difference between magnitudes of the change in the magnetic field at the bimodal peaks in the changes in the magnetic field during the movement of the sensor along the longitudinal axis of the implant.

In some aspects, the computer may be configured to calculate a derivative of the changes in the magnetic field and to use the calculated derivative to detect the location of the implant. In some aspects, the derivative of the changes in the magnetic field may be with respect to time (during the movement of the sensor along the longitudinal axis of the implant). In some alternative aspects, the derivative of the changes in the magnetic field may be with respect to the position of the sensor (during movement of the sensor along the longitudinal axis of the implant). In some aspects, the computer may be configured to determine edges of the magnetic material of the implant based on locations where the derivative of the changes in the magnetic field equals zero during movement of the sensor along a longitudinal axis of the implant. In some aspects, the computer may be configured to determine a depth of the implant based on magnitudes of the changes in the magnetic field at the locations where the derivative of the changes in the magnetic field equals zero during the movement of the sensor along the longitudinal axis of the implant. In some aspects, the computer may be configured to determine an orientation of the implant based on a difference between magnitudes of the changes in the magnetic field at the locations where the derivative of the changes in the magnetic field equals zero during the movement of the sensor along the longitudinal axis of the implant. In some aspects, the computer may be configured to determine a midline of the implant based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor across a longitudinal axis of the implant. In some aspects, the derivative of the changes in the magnetic field may be with respect to time or position (as the sensor is moved across the longitudinal axis of the implant).

In some aspects, the computer may be further configured to use the one or more sensor signals to detect an orientation of the implant.

In some aspects, the apparatus may further include a display, and the computer may be configured to cause the display to display an indication of the detected location of the implant. In some aspects, the display may be located above the sensor. In some aspects, the indication of the detected location of the implant may include an implant image, and a location of the implant image on a screen of the display relative to a point on the screen of the display may

correspond to the detected location of the implant relative to the sensor. In some aspects, the implant image may have an orientation that corresponds to a detected orientation of the implant.

In some aspects, the apparatus may further include an incision marking tool configured to identify an incision location for removing the implant.

In some aspects, a diameter of the magnetic field generator may be equal to the length of the implant.

In some aspects, the apparatus may further include a position detector configured to generate a location signal indicative of a location of the sensor on the skin surface. In some aspects, the position detector may include a motion detector configured to detect movement of the sensor and to generate a motion signal indicative of the detected movement of the sensor, and the location signal may include the motion signal. In some aspects, the computer may be configured to use the sensor signal and the location signal to generate a map of sensor signals at different locations of the sensor on the skin surface.

Another aspect of the invention may provide a method for locating an implant including magnetic material. The method may include using a magnetic field generator to generate a magnetic field. The method may include using a sensor to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field. The magnetic material of the implant may cause changes to the magnetic field as the sensor is moved over the implant. The method may include using a computer to detect a location of the implant based on the sensor signal.

In some aspects, detecting the location of the implant may include determining edges of the magnetic material of the implant based on locations of bimodal peaks in the changes in the magnetic field during movement of the sensor along a longitudinal axis of the implant.

Yet another aspect of the invention may provide a method for locating an implant including magnetic material. The method may include moving an apparatus including a magnetic field generator and a sensor across a longitudinal axis of the implant. The magnetic material of the implant may cause changes in a magnetic field generated by the magnetic field generator as the apparatus is moved across the longitudinal axis of the implant, and the sensor may detect changes in the magnetic field. The method may include determining a midline of the implant based on a location where the changes in the magnetic field are greatest as the apparatus is moved across the longitudinal axis of the implant. The method may include moving the apparatus along the determined midline of the implant. The method may include determining edges of the magnetic material of the implant based on locations of bimodal peaks in the changes in the magnetic field as the apparatus is moved along the determined midline of the implant.

In some aspects, the method may further include using an incision marking tool of the apparatus to mark an incision location for removing the implant.

Still another aspect of the invention may provide an apparatus for locating an implant including a charge storage device. The apparatus may include an electromagnetic field generator, a sensor, and a computer. The electromagnetic field generator may be configured to generate an electromagnetic field. The sensor may be configured to detect changes in the electromagnetic field and to generate a sensor signal indicative of the changes in the electromagnetic field. The charge storage device of the implant may cause changes to the electromagnetic field as the sensor is moved over the

implant. The computer may be configured to use the sensor signal to detect a location of the implant.

Yet another aspect of the invention may provide a method for locating an implant including a charge storage device. The method may include using an electromagnetic field generator to generate an electromagnetic field. The method may include using a sensor to detect changes in the electromagnetic field and to generate a sensor signal indicative of the changes in the electromagnetic field. The charge storage device of the implant may cause changes to the electromagnetic field as the sensor is moved over the implant. The method may include using a computer to detect a location of the implant based on the sensor signal.

Still another aspect of the invention may provide a method for locating an implant including a charge storage device. The method may include moving an apparatus including an electromagnetic field generator and a sensor across a longitudinal axis of the implant. The charge storage device of the implant may cause changes in an electromagnetic field generated by the electromagnetic field generator as the apparatus is moved across the longitudinal axis of the implant, and the sensor may detect changes in the electromagnetic field. The method may include determining a midline of the implant based on a location where the changes in the electromagnetic field are greatest as the apparatus is moved across the longitudinal axis of the implant. The method may include moving the apparatus along the determined midline of the implant. The method may include determining edges of the charge storage device of the implant based on locations of bimodal peaks in the changes in the electromagnetic field as the apparatus is moved along the determined midline of the implant.

These and other embodiments encompassed within the systems and methods are described in the detailed description of the invention below.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated herein and form part of the specification, illustrate various, non-limiting embodiments of the present invention. In the drawings, like reference numbers indicate identical or functionally similar elements.

FIG. 1 is a block diagram illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

FIG. 2A is a cross-sectional side view illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

FIG. 2B is a cross-sectional top view illustrating a non-limiting example of an implant finder embodying aspects of the present invention.

FIGS. 2C and 2D are two-dimensional contour plot slices in the XY and ZY planes, respectively, illustrating the magnitude of a magnetic field generated by a non-limiting example of a hollow, cylindrical magnetic field generator embodying aspects of the present invention.

FIG. 2E is a cross-sectional top view illustrating a non-limiting example of an implant finder embodying aspects of the present invention.

FIGS. 2F and 2G are two-dimensional contour plot slices in the XY and ZY planes, respectively, illustrating the magnitude and vector of a magnetic field generated by a non-limiting example of a magnetic field generator including six magnets embodying aspects of the present invention.

FIGS. 2H and 2I are two-dimensional contour plot slices in the XY and ZY planes, respectively, illustrating the

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magnitude and vector of a magnetic field generated by a non-limiting example of a magnetic field generator including four magnets embodying aspects of the present invention.

FIG. 3A is a cross-sectional side-view illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

FIGS. 3B and 3C are two-dimensional contour plot slices in the XY and ZY planes, respectively, illustrating the magnitude and vector of a magnetic field generated by a non-limiting example of a cylindrical magnetic field generator embodying aspects of the present invention.

FIGS. 4A and 4B illustrate a non-limiting example of a magnetometer including the sensor of an implant finder embodying aspects of the present invention.

FIGS. 5A and 5B are perspective views illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

FIG. 5C is a see-through perspective view illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

FIGS. 5D and 5E are perspective and top views illustrating a non-limiting example of an implant and magnetic field generator and sensor of an implant finder embodying aspects of the present invention.

FIG. 5F is a see-through perspective view illustrating a non-limiting example of an implant and magnetic field generator and sensor of an implant finder embodying aspects of the present invention.

FIG. 6 is a block diagram illustrating a non-limiting example of a computer of an implant finder embodying aspects of the present invention.

FIG. 7A is a cross-sectional side view illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

FIGS. 7B-7F illustrate non-limiting examples of a display screen of the user interface of an implant finder embodying aspects of the present invention.

FIGS. 8A and 8B are perspective views illustrating a non-limiting example of an implant embodying aspects of the present invention.

FIGS. 9A and 9B illustrate movement of the implant finder across a longitudinal axis of an implant that is not tilted relative to the skin surface and a magnitude of changes to a magnetic field during the movement, respectively, embodying aspects of the present invention.

FIGS. 9C and 9D illustrate movement of the implant finder along a longitudinal axis of the implant that is not tilted relative to the skin surface and a magnitude of changes to a magnetic field during the movement, respectively, embodying aspects of the present invention.

FIG. 9E illustrates an identified implant edge and marked incision location embodying aspects of the present invention.

FIGS. 10A and 10B illustrate movement of the implant finder across a longitudinal axis of a downward-tilted implant and a magnitude of changes to a magnetic field during the movement, respectively, embodying aspects of the present invention.

FIGS. 10C and 10D illustrate movement of the implant finder along a longitudinal axis of the downward-tilted implant and a magnitude of changes to a magnetic field during the movement, respectively, embodying aspects of the present invention.

FIG. 10E illustrates an identified implant edge and marked incision location embodying aspects of the present invention.

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FIGS. 11A and 11B illustrate movement of the implant finder across a longitudinal axis of an implant and a magnitude of changes to a magnetic field during the movement for different distances between the sensor and implant, respectively, embodying aspects of the present invention.

FIGS. 12A and 12B illustrate movement of the implant finder along a longitudinal axis of an implant and a magnitude of changes to a magnetic field during the movement for different distances between the sensor and implant, respectively, embodying aspects of the present invention.

FIGS. 13A and 13B illustrate movement of the implant finder in a diagonal direction relative to a longitudinal axis of an implant and a magnitude of changes to a magnetic field during the movement, respectively, embodying aspects of the present invention.

FIGS. 13C-13E illustrate X-, Y-, and Z-components, respectively, of the magnetic field change as the implant finder is moved in the diagonal direction relative to the longitudinal axis of the implant embodying aspects of the present invention.

FIGS. 14A-14C illustrate an implant twisted out-of-plane, changes to a magnetic field during movement of the implant finder across a longitudinal axis of the implant, and changes to a magnetic field during movement of the implant finder along a longitudinal axis of the implant, respectively, embodying aspects of the present invention.

FIGS. 15A-15C illustrate movement of the implant finder to identify an orientation of an implant, movement of the implant finder along a longitudinal axis of the implant, and an identified implant edge and marked incision location, respectively, embodying aspects of the present invention.

FIGS. 16A-16C illustrate identification of an approximate location of an implant, markings indicating a course for movement of an implant finder, and an identification of implant edge and incision coordinates, respectively, embodying aspects of the present invention.

FIG. 17 is a flowchart illustrating a non-limiting example of a process for locating an implant including magnetic material embodying aspects of the present invention.

FIG. 18 is a flowchart illustrating a non-limiting example of a process for locating an implant including magnetic material embodying aspects of the present invention.

FIG. 19 is a side view illustrating a non-limiting example of an implant embodying aspects of the present invention.

FIG. 20 is a block diagram illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

FIG. 21A is a cross-sectional side view illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

FIG. 21B is a cross-sectional top view illustrating a non-limiting example of an implant finder embodying aspects of the present invention.

FIG. 21C is a perspective view illustrating a non-limiting example of an implant and an implant finder embodying aspects of the present invention.

FIG. 22 is a flowchart illustrating a non-limiting example of a process for locating an implant including a charge storage device embodying aspects of the present invention.

FIG. 23 is a flowchart illustrating a non-limiting example of a process for locating an implant including a charge storage device embodying aspects of the present invention.

DETAILED DESCRIPTION

FIG. 1 is a block diagram illustrating an implant 100 and an implant finder 101 embodying aspects of the present invention.

invention. In some aspects, the implant finder **101** may be for locating the implant **100**, which may have been implanted in tissue below the surface of skin of a living animal. In some aspects, as shown in FIG. 1, the implant **100** may include magnetic material **124** (e.g., a magnetic core). In some aspects, the magnetic material **124** may include, for example and without limitation, a ferromagnetic material (e.g., iron) or a ferrimagnetic material (e.g., ferrite).

In some aspects, the implant finder **101** may include a magnetic field generator **103**, a sensor **105**, a computer **106**, a battery **117**, a wireless communication integrated circuit (IC) **119**, a connector **121**, a connector IC **123**, a charger IC **125**, a user interface **127**, and/or a position detector **137**. In some aspects, the user interface **127** of the implant finder **101** may include a display **129** (e.g., an optical display such as, for example, a light emitting diode (LED) display), a speaker **131**, a vibration motor **133**, and/or a user input **135**.

In some aspects, the magnetic field generator **103** may be configured to generate a magnetic field. In some aspects, the sensor **105** may be configured to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field. In some aspects, the magnetic material **124** of the implant **100** may cause changes to the magnetic field as the sensor **105** is moved over the implant **100**. In some aspects, the computer **106** may be configured to use the sensor signal to detect a location of the implant **100**.

In some aspects, as shown in FIGS. 2A, 2B, 2E, and 3A, the magnetic field generator **103** may include one or more magnets **109**. In some aspects, the one or more magnets **109** may be, for example and without limitation, permanent magnets and/or electromagnets (e.g., a coil of wire wound around a magnetic core made from a ferromagnetic or ferrimagnetic material). In some aspects, as shown in the FIGS. 5A-5C, the magnetic field generator **103** may include a housing **107**, and the housing **107** may be configured to hold the one or more magnets **109**. However, a housing **107** is not required, and, in some alternative aspects (e.g., some aspects in which the magnetic field generator **103** consists of only a single magnet **109**), the magnetic field generator **103** does not include a housing **107**.

In some aspects, as shown in FIGS. 2A, 2B, and 2E, the sensor **105** may be located on a longitudinal axis at the center of the magnetic field generator **103**. However, this is not required, and, in some alternative aspects, the sensor **105** is not located on a longitudinal axis at the center of the magnetic field generator **103**. For example, as shown in FIG. 3A, the sensor **105** may be located adjacent to the magnet **109** of the magnetic field generator **103** (as opposed to at the center of one or more magnets **109** of the magnetic field generator **103**).

In some aspects, as shown in FIG. 2B, the magnetic field generator **103** may include a hollow cylindrical magnet **109**. In some alternative aspects, as shown in FIG. 2E, the magnetic field generator **103** may include two or more magnetics **109** (e.g., four or six magnets). In some other alternative aspects, as shown in FIG. 3A, the magnetic field generator **103** may include a solid cylindrical magnet **109**.

In some aspects, the one or more magnets **109** of the magnetic field generator **103** may generate a substantially uniform magnetic field. In some aspects, the hollow cylindrical magnet **109** of the magnetic field generator **103** illustrated in FIG. 2B may generate a magnetic field having the magnitudes shown in the two-dimensional contour plot slices of FIGS. 2C and 2D. In some aspects, the six magnets **109** of the magnetic field generator **103** illustrated in FIG. 2E may generate a magnetic field having the magnitudes

shown in the two-dimensional contour plot slices of FIGS. 2F and 2G. In some aspects, the magnetic field generator **103** may include four magnets **109**, and the four magnets **109** may generate a magnetic field having the magnitudes shown in the two-dimensional contour plot slices of FIGS. 2H and 2I. In some aspects, the solid cylindrical magnet **109** of the magnetic field generator **103** illustrated in FIG. 3A may generate a magnetic field having the magnitudes shown in the two-dimensional contour plot slices of FIGS. 3B and 3C. In some aspects, as shown in FIGS. 2C, 2D, 2F-2I, 3B, and 3C, the magnetic field generated by the one or more magnetics **109** of the magnetic field generator **103** may be substantially symmetric about a longitudinal axis at the center of the magnetic field generator **103**. However, this is not required, and in some alternative aspects, the magnetic field may have a different profile (e.g., the magnetic field may be non-uniform and/or asymmetric about the longitudinal axis at the center of the magnetic field generator). In some aspects, as shown in FIGS. 2A, 2B, and 2E, the sensor **105** may be located on the longitudinal axis at the center of the magnetic field generator **103**.

In some aspects, the sensor **105** may be a magnetic field sensor. In some aspects, the sensor **105** may be a tunnel magnetoresistance (TMR) magnetic field sensor (e.g., a TMR full bridge magnetic field sensor) or a Hall sensor (e.g., a high field Hall sensor). In some aspects, as shown in FIGS. 4A and 4B, the sensor **105** may be part of a magnetometer **400** (e.g., an axial low-field magnetometer). In some aspects, the magnetometer **400** may include the sensor **105**, a probe **402** (e.g., a rigid probe), a cable **404** (e.g., a flexible cable), and data acquisition electronics **406**, and/or a connector **408** (e.g., a USB connector). In some aspects, the connector **408** of the magnetometer **400** may be connected to the computer **106** of the implant finder **101**.

In some aspects, as shown in FIGS. 5A-5F, the implant finder **101** may include the magnetic field generator **103** and the sensor **105**. In some aspects, the sensor **105** may be located on the longitudinal axis at the center of the magnetic field generator **103**. In some aspects, the magnetic field generator **103** may include a housing **107**. In some aspects, the housing **107** may include housing bodies **115** (e.g., polymer bodies). In some aspects, as shown in FIGS. 5C-5F, the magnetic field generator **103** may include one or more magnets **109**. In some aspects, the housing **107** may be configured to house the one or more magnets **109**. In some aspects, as shown in FIGS. 5C-5F, the magnetic field generator **103** may include six magnets **109**. However, in some alternative aspects, the magnetic field generator **103** may include a different number of magnets **109** (e.g., ten, nine, eight, seven, five, four, three, or two magnets or a single magnet, such as a hollow cylindrical magnet). In some aspects, as shown in FIG. 5E, a diameter or width of the magnetic field generator **103** may be equal to the length of the implant **100**. However, this is not required, and, in some alternative aspects, the magnetic field generator **103** may have a different size (e.g., a diameter or width that is less than or greater than the length of the implant **100**).

In some aspects, as shown in FIGS. 5A-5C, the implant finder **101** may include a handle **111**. In some aspects, as shown in FIGS. 5A-5C, the implant finder **101** may include an incision marking tool **113**. In some aspects, the incision marking tool **113** may be configured such that a user may use the incision marking tool **113** to mark an appropriate location on the skin surface **154** for an incision that would enable removal of the implant **100**. In some aspects, the user may use the incision marking tool **113** to mark the incision location when the implant finder **101** is located over the

implant **100**. In some aspects, the incision marking tool **113** may be configured to mark the skin surface **154** above a determined edge of the implant **100**, a determined center of the implant **100**, a determined edge of the magnetic material **124**, or a determined center of the magnetic material **124**. In some alternative aspects, the incision marking tool **113** may be configured to mark the skin surface **154** above a location offset from a determined edge of the implant **100**, a determined center of the implant **100**, a determined edge of the magnetic material **124**, and/or a determined center of the magnetic material **124**. In some alternative aspects, the incision marking tool **113** may be configured mark the skin surface at a longitudinal axis at the center of the magnetic field generator **103**.

In some aspects in which the implant finder **101** includes the connector **121**, the connector **121** may be, for example and without limitation, a Micro-Universal Serial Bus (USB) connector. In some aspects, the connector **121** may be configured to enable a wired connection to an external device, such as a personal computer or display device. In some aspects, the implant finder **101** may exchange data to and from the external device through the connector **121** and/or may receive power through the connector **121**. In some embodiments, the connector IC **123** may be, for example and without limitation, a USB-IC, which may control transmission and receipt of data through the connector **121**.

In some aspects in which the implant finder **101** includes the battery **117**, the battery **117** may provide operating power for the implant finder **101**. In some aspects, the battery **117** may be a rechargeable battery. In some aspects, the battery **117** may be, for example and without limitation, a lithium-polymer battery. In some aspects, the battery **117** may have a short recharge duration and/or may have a small size. In some aspects in which the display device **105** includes the charger IC **125**, the charger IC **125** may receive power via the connector **121** and charge the battery **117**.

In some aspects in which the implant finder **101** includes the wireless communication IC **119**, the wireless communication IC **119** may enable wireless communication with one or more external devices, such as, for example, one or more personal computers and/or one or more other display devices (e.g., a smartphone or tablet executing an application). In some aspects, the wireless communication IC **119** may employ one or more wireless communication standards to wirelessly transmit data. The wireless communication standard employed may be any suitable wireless communication standard, such as an IEEE 802.11 standard, an ANT standard, a Bluetooth standard, or a Bluetooth Low Energy (BLE) standard (e.g., BLE **4.0**). In some aspects, the wireless communication IC **119** may include an antenna (e.g., a Bluetooth antenna, a Wi-Fi antenna, and/or one or more cellular antennas). In some aspects, the antenna of the wireless communication IC **119** may be entirely contained within a housing (e.g., housing **107**) of the implant finder **101**. However, this is not required, and, in alternative embodiments, all or a portion of the antenna of the wireless communication IC **119** may be external to a housing of the implant finder **101**.

In some aspects in which the implant finder **101** includes the position detector **137**, the position detector **137** may be configured to generate a location signal indicative of a location of the sensor on the skin surface **154**. In some aspects, the position detector **137** may, for example and without limitation, include a motion detector configured to detect movement of the sensor **105** (and/or implant finder **101**) relative to the skin surface **154** and to generate a motion

signal indicative of the detected movement of the sensor **105**, and the location signal may include the motion signal. In some aspects, the motion detector may include a mechanical motion detector (e.g., that makes use of a motion of a ball inside the mechanical motion detector) and/or an optical motion detector (e.g., that uses light to detect movement of the sensor **105**).

In some aspects in which the implant finder **101** includes the user interface **127**, the user interface **127** may include one or more of a display **129** and a user input **135**. In some aspects, the display **129** may include a liquid crystal display (LCD) and/or a light emitting diode (LED) display. In some aspects, the user input **135** may include one or more buttons, a keyboard, a keypad, and/or a touchscreen. In some aspects, the user interface **127** may include one or more of a speaker **131** (e.g., a beeper) and a vibration motor **133**, which may be activated, for example, in the event that a condition is met (e.g., detection of an edge of the implant **100** and/or a detection of a midline of the implant **100**).

In some aspects in which the implant finder **101** includes the computer **106**, the computer **106** may control the overall operation of the implant finder **101**. For example, the implant finder **101** may control the wireless communication IC **119**, the connector IC **123**, the charger IC **125**, the position detector **137**, the magnetic field generator **103**, the sensor **105**, and/or the user interface **127**. In some aspects, the computer **106** may receive and/or process data from the sensor **105**, the position detector **137**, and/or user input **135** of the user interface **127**. For example, in some aspects, the computer **106** may be configured to use the sensor signal generated by the sensor **105** to detect a location of the implant **100**. In some aspects, the computer **106** may be configured to use the sensor signal generated by the sensor **105** and the location signal generated by the position detector **137** to generate a map of sensor signals at different locations of the sensor **105** on the skin surface **154**. In some aspects, the computer **106** may additionally or alternatively control the display **129**, speaker **131**, and/or vibration motor **135** to provide information about the detected location of the implant **100**. For example, in some aspects, the computer **106** may cause the display **129** to display a notification (e.g., one or more optical displays produced by one or more LEDs), the speaker **131** to beep, and/or the vibration motor **135** to vibrate when the computer **106** detects an edge of the implant **100** and/or a midline of the implant **100**. In some aspects, the computer **106** may additionally or alternatively be configured to cause the display **129** to display an indication of the detected location of the implant **100**.

FIG. **6** is a block diagram of the computer **106** of the implant finder **101** according to some aspects. In some aspects, the computer **106** can be adapted to perform any of methods, processes, or steps disclosed herein. As shown in FIG. **6**, the computer **106** may include processing circuitry (PC) **302**, which may include one or more processors (P) **355** (e.g., one or more general purpose microprocessors and/or one or more other processors, such as an application specific integrated circuit (ASIC), field-programmable gate arrays (FPGAs), and the like).

In some aspects, as shown in FIG. **6**, the computer **106** may include one or more storage units (a.k.a., "data storage systems") **308** which may be co-located or geographically distributed and which may include one or more non-volatile storage devices and/or one or more volatile storage devices. In some aspects where the PC **302** includes a programmable processor, the one or more storage units may include a computer program product (CPP) **341**. In some aspects, the CPP **341** may include a computer readable medium (CRM)

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342 storing a computer program (CP) 343 comprising computer readable instructions (CRI) 344. In some aspects, the CRM 342 may be a non-transitory computer readable medium, such as, magnetic media (e.g., a hard disk), optical media, memory devices (e.g., random access memory, flash memory), and the like. In some aspects, the CRI 344 of the computer program 343 is adapted such that when executed by the PC 302, the CRI 341 may cause the computer 106 to perform steps described herein (e.g., steps described herein with reference to the flow charts). In other aspects, the computer 106 may be adapted to perform steps described herein without the need for code. That is, for example, the PC 302 may consist merely of one or more ASICs. Hence, the features of the aspects described herein may be implemented in hardware and/or software.

In some aspects, as shown in FIG. 7A, the display 129 of the user interface 127 of the implant finder 101 may be located above the sensor 105. In some aspects, the magnetic field generator 103 may be configured to hold the display 129 in place at the top of the implant finder 101. In some aspects, as shown in FIG. 7A, the housing 107 of the magnetic field generator 103 may be configured to house the one or more magnets 109 of the magnetic field generator 103, the computer 106, and/or the display 129. However, this is not required, and, in some alternative aspects, the display 129 may be located above the magnetic field generator 103.

In some aspects, as shown in FIGS. 7B-7F, the display 129 of the user interface 127 may include a display screen 702. In some aspects (e.g., in some aspects in which the display 129 is located above the sensor 105 and the computer 106 is configured to cause the display 129 to display an indication of the detected location of the implant 100), the indication of the detected location of the implant 100 may include an implant image 704, and a location of the implant image 704 on the screen 702 of the display 129 relative to a point on (e.g., the center of) the screen 702 of the display 129 may correspond to the detected location of the implant 100 relative to the sensor 105. In some aspects, as shown in FIGS. 7B-7E, the implant image 704 may have an orientation that corresponds to an orientation of the implant 100 detected by the computer 106 using sensor signals generated by the sensor 105.

In some aspects, as shown in FIG. 7C, depending on the location of the implant 100 relative to the sensor 105, the implant image 704 may be a partial implant image. In some aspects, as shown in FIGS. 7B-7F, a mark 706 may identify the point on the screen 702 that corresponds to the location of the sensor 105. In some aspects (e.g., some aspects in which the screen 702 of the display 129 is larger than and/or extends beyond the magnetic field generator 103), as shown in FIG. 7E, the computer 106 may cause the display 129 to display a magnetic field generator image 708 indicative of the location of the outer perimeter of the magnetic field generator 103. In some aspects, the location of the implant image 704 relative to the magnetic field generator image 708 may correspond to the detected location of the implant 100 relative to the location of magnetic field generator 103. In some aspects, as shown in FIG. 7F, the computer 106 may cause the display 129 to display a target region 710, and the implant finder 101 may be properly positioned over the implant 100 such that the incision location can be marked (e.g., using the incision marking tool 113) when the implant image 704 is located within the target region 710.

FIGS. 8A and 8B illustrate a non-limiting example of an implant 100 that may be located by the implant finder 101. In some aspects, the implant 100 may be a small, fully

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subcutaneously implantable sensor, e.g., that measures the amount or concentration of an analyte (e.g., glucose) in a medium (e.g., interstitial fluid) of a living animal (e.g., a living human). In some alternative aspects, the implant 100 is not a sensor and may instead be a different type of implantable device, such as, for example and without limitation, a tag, an insulin pump, a pacemaker, or an electrical/heat therapy device.

In some aspects, as illustrated in FIG. 8A, the implant 100 may be implanted in the tissue 150 (e.g., subcutaneous tissue) of the living animal, where the implant 100 may rest in a pocket 152 in the tissue 150 below the skin surface 154, and the pocket 152 may surround the implant 100. In some aspects, the pocket 152 may be created by a tissue dissector tool before implantation of the implant 100 or by the implantation process.

In some aspects, as illustrated in FIG. 8A, the implant 100 may include a housing 102 and one or more analyte indicators 104 coated, diffused, adhered, embedded, or grown on or in at least a portion of the exterior surface of the housing 102. In some aspects, the one or more analyte indicators 104 (e.g., polymer grafts or hydrogels) of the implant 100 may include one or more indicator molecules (e.g., fluorescent indicator molecules). In some aspects, the indicator molecules may produce (e.g., exhibit) one or more detectable properties (e.g., optical properties) that vary in accordance with the amount or concentration of the analyte in proximity to an analyte indicator 104. In some aspects, the indicator molecules may emit an amount of emission light (e.g., fluorescent light) that varies in accordance with the amount or concentration of the analyte in proximity to the analyte indicator 104.

In some embodiments, the implant 100 may include a substrate 110 (e.g., a printed circuit board (PCB) or flexible PCB), one or more light sources 111 mounted on or fabricated in the substrate 110, and one or more photodetectors 112, 114, and 116 (e.g., photodiodes, phototransistors, photoresistors, or other photosensitive elements) mounted on or fabricated in the substrate 110. In some embodiments, the light source 111 may be configured to emit an excitation light over an excitation wavelength range that interacts with the one or more indicator molecules in the analyte indicator 104. In some aspects, one or more of the photodetectors 112, 114, and 116 may output a signal indicative of an amount of light received by the photodetector. In some examples, the signal output by the one or more internal photodetectors 112, 114, and 116 may be indicative of an amount or concentration of an analyte in a medium in proximity to the analyte indicator 104.

In some aspects, as shown in FIGS. 8A and 8B, the implant 100 may include an inductor 120, which may be, for example, a ferrite based micro-antenna. In some aspects, the inductor 120 may include a conductor 122 in the form of a coil and a magnetic material 124 in form of a magnetic core 124. In some aspects, the core 124 may be, for example and without limitation, a ferrite core. In some aspects, the inductor 120 may be connected to circuitry (e.g., an application specification integrated circuit (ASIC)) of the implantable device 100. In some aspects, the inductor 120 may communicate with an external device (not shown) by, passive telemetry (e.g., near field communication), such that power and/or data is transferred between the implant 100 and the external device.

In some aspects, the implant finder 101 may be used for locating an implant 100 (e.g., an implant 100 that has been implanted in tissue 150 below a skin surface 154 of a living animal). In some aspects, a first process for locating the

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implant 100 may include a first step of determining a midline of the implant 100. In some aspects, determining the midline of the implant 100 may include, as shown in FIGS. 9A, 10A, and 11A, moving the implant finder 101 across (e.g., transverse to) a longitudinal axis of the implant 100. FIGS. 9A and 10A illustrate the movement of the implant finder 101 across the longitudinal axis of an implant 100 that is not tilted relative to the skin surface and across the longitudinal axis of a downward-tilted implant 100, respectively. In some aspects, the orientation of the implant 100 may be assumed (e.g., the longitudinal axis of an implant 100 implanted in an arm of a living animal may be assumed to be approximately along the long axis of the arm), and the movement of the implant finder 101 across the longitudinal axis of the implant 100 may be transverse to the assumed orientation of the implant 100. In some aspects, the orientation of the implant 100 may be determined using records regarding the implantation of the implant 100, and the movement of the implant finder 101 across the longitudinal axis of the implant 100 may be transverse to the orientation of the implant 100 determined using the implantation records.

In some aspects, locating the implant 100 may include using implantation information for the implant 100. In some aspects, the implantation information may include implantation location information identifying the location at which the implant 100 was implanted, and the implantation location information may be used to identify and/or limit an initial search area for the implant 100. In some aspects, the implantation information may additionally or alternatively include implantation depth information identifying the depth at which the implant was implanted, and the implantation depth information may be used to estimate an expected signal range. In some aspects in which there is an offset between the center of the magnetic material 124 and the center of the implant 100, the implantation information may additionally or alternatively include offset direction information identifying a direction of the offset of the center of the magnetic material 124 relative to the center of the implant 100, which may be used for identifying edges of the implant 100 and/or an incision location.

In some aspects, the magnetic material 124 of the implant 100 may cause changes in a magnetic field generated by the magnetic field generator 103 of the implant finder 101 as the implant finder 101 is moved across the longitudinal axis of the implant 100, and the sensor 105 of the implant finder 101 may detect changes in the magnetic field. FIG. 9B illustrates changes in the magnetic field caused by the magnetic material 124 of the implant 100 that is not tilted relative to the skin surface as detected by the sensor 105 during movement across the longitudinal axis of the implant 100 according to some aspects. FIGS. 10B and 14B illustrate changes in the magnetic field caused by the magnetic material 124 of an out-of-plane or downward-tilted implant 100 as detected by the sensor 105 during movement across the longitudinal axis of the implant 100 according to some aspects. In some aspects, as shown in FIGS. 9B, 10B, and 14B, the change in the magnetic field as the implant finder 101 is moved across the longitudinal axis of the implant 100 may be greatest at the location of the midline of the implant 100.

In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may be configured to determine the midline of the implant 100 based on a location where the changes in the magnetic field are greatest as the implant finder 101 is moved across the longitudinal axis of the implant 100. In some aspects, the computer 106 of the implant finder 101 may be configured to determine a deriva-

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tive of the detected changes in the magnetic field, and the computer 106 may be configured to determine the midline of the implant 100 (and the location where the changes in the magnetic field are greatest) based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor 105 across a longitudinal axis of the implant 100. In some aspects, the derivative of the changes in the magnetic field may be with respect to time (as the sensor 105 is moved across the longitudinal axis of the implant 100). In some alternative aspects, the derivative of the changes in the magnetic field may be with respect to the position of the sensor (as the sensor 105 is moved across the longitudinal axis of the implant 100).

In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may cause the user interface 127 to indicate when the implant finder 101 (e.g., the computer 106) has detected the midline of the implant 100 (e.g., when the computer 106 determines the derivative of the changes in the magnetic field equals zero during movement of the sensor 105 across the longitudinal axis of the implant 100). In some aspects, the user interface 127 may indicate detection of the midline using one or more of the display 129 (e.g., by displaying a visual indicator of midline detection), speaker 131 (e.g., by emitting an audible sound such as, for example and without limitation, a beep), and vibration motor 133 (e.g., by vibrating).

FIG. 11B illustrates detected magnetic field changes as the implant finder 101 is moved across the longitudinal axis of the implant 100 for implants 100 at different heights of the sensor 105 of the implant finder 101 relative to the implant 100. As shown in FIG. 11B, the magnitudes of the detected magnetic field changes increase as the height decreases. Accordingly, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may be configured to use the magnitude of the detected magnetic field change at the midline (e.g., the greatest detected magnetic field change during movement of the sensor 105 across the longitudinal axis of the implant 100) to determine the depth of the implant 100 in the tissue 150.

In some aspects, the first process for locating the implant 100 may include a second step of determining one or more edges of the magnetic material 124 of the implant 100 and/or a center of the magnetic material 124 of the implant 101. In some aspects, determining the edge(s) and/or center of magnetic material 124 of the implant 100 may include, as shown in FIGS. 9C, 10C, 12A, and 14A, moving the implant finder 101 along a longitudinal axis of the implant 100. FIG. 9C illustrated the movement of the implant finder 101 along the longitudinal axis of an implant 100 that is not tilted relative to the skin surface. FIGS. 10C and 14A illustrate the movement of the implant finder 101 along the longitudinal axis of a downward-tilted or out-of-plane implant 100. In some aspects, the determined midline may be used to move the implant finder 101 along the longitudinal axis of the implant 100. For example, in some aspects, moving the implant finder 101 along the longitudinal axis of the implant 100 may include moving implant finder 101 on the skin surface 154 along the determined midline of the implant 100. In some aspects, the movement along the longitudinal axis of the implant 100 may be parallel to the assumed orientation of the implant 100. In some aspects, the movement along the longitudinal axis of the implant 100 may be parallel to the orientation of the implant 100 determined using implantation records.

In some aspects, the magnetic material 124 of the implant 100 may cause changes in a magnetic field generated by the magnetic field generator 103 of the implant finder 101 as the

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implant finder 101 is moved along the longitudinal axis of the implant 100, and the sensor 105 of the implant finder 101 may detect changes in the magnetic field. FIG. 9D illustrates changes in the magnetic field caused by the magnetic material 124 of an implant 100 that is not tilted relative to the skin surface as detected by the sensor 105 during movement along the longitudinal axis of the implant 100 according to some aspects. FIGS. 10D and 14C illustrate changes in the magnetic field caused by the magnetic material 124 of an out-of-plane or downward-tilted implant 100 as detected by the sensor 105 during movement along the longitudinal axis of the implant 100 according to some aspects. In some aspects, as shown in FIGS. 9D, 10D, and 14C, the detected changes in the magnetic field as the implant finder 101 is moved along the longitudinal axis of the implant 100 may include bimodal peaks (e.g., local maxima) at the locations of the edges of the magnetic material 124 of the implant 100. In some aspects, as shown in FIGS. 9D, 10D, and 14C, the changes in the magnetic field may include a local minimum between the bimodal peaks at a location of the center of the magnetic material 124 of the implant 100.

In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may be configured to determine edges of the magnetic material 124 of the implant 100 based on locations of the bimodal peaks in the changes in the magnetic field. In some aspects, the computer 106 of the implant finder 101 may be configured to determine a derivative of the detected changes in the magnetic field, and the computer 106 may be configured to determine the edges of the magnetic material 124 of the implant 100 (and the locations of the bimodal peaks in the magnetic field changes) based on a locations where the derivative of the changes in the magnetic field equals zero during movement of the sensor 105 along a longitudinal axis of the implant 100. In some aspects, the derivative of the changes in the magnetic field may be with respect to time (as the sensor 105 is moved along the longitudinal axis of the implant 100). In some alternative aspects, the derivative of the changes in the magnetic field may be with respect to the position of the sensor (as the sensor 105 is moved along the longitudinal axis of the implant 100). In some aspects, the edges of the magnetic material 124 of the implant 100 may be offset from the edges of the implant 100. For example, the edges of the magnetic material 124 of the implant 100 are offset from the edges of the implant 100. In some aspects in which the edges of the magnetic material 124 are offset from the edges of the implant 100, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may be configured to determine one or more edges of the implant 100 using the one or more determined edges of the magnetic material 124 and the offset(s).

In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may be configured to determine a center of the magnetic material 124 of the implant 100 based on the location of a local minimum in the magnetic field changes between the bimodal peaks in the magnetic field changes. In some aspects, the computer 106 of the implant finder 101 may be configured to determine a derivative of the detected changes in the magnetic field, and the computer 106 may be configured to determine the center of the magnetic material 124 of the implant 100 (and the locations of local minimum between the bimodal peaks in the magnetic field changes) based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor 105 along a longitudinal axis of the implant 100. In some aspects, the center of the

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magnetic material 124 of the implant 100 may be offset from the center of the implant 100 (i.e., the center of the magnetic material 124 may be different than the center of the implant 100). For example, in the implant 100 illustrated in FIG. 8A, the center of the magnetic material 124 of the implant 100 is offset from the center of the implant 100 (e.g., by 2.5 mm). In some aspects in which the magnetic material 124 is offset from the center of the implant 100, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may be configured to determine a center of the implant 100 using the determined center of the magnetic material 124 and the offset. In some aspects, the direction of the offset of the center of the magnetic material 124 relative to the center of the implant 100 may be assumed (e.g., the center of the magnetic material 124 may be assumed to be higher along the long axis of the arm than the center of the implant 100). In some aspects, the direction of the offset of the center of the magnetic material 124 relative to the center of the implant 100 may be additionally or alternatively confirmed or determined (e.g., using implantation records and/or detected changes in the magnetic field caused by metal and/or circuitry on and/or in the substrate 110, which may extend from one side of the magnetic material 124 of the implant 100 as shown in FIGS. 8A and 8B).

In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may cause the user interface 127 to indicate when the implant finder 101 (e.g., the computer 106) has detected the edges and/or center of the magnetic material 124 of the implant 100 (e.g., when the computer 106 determines the derivative of the changes in the magnetic field equals zero during movement of the sensor 105 along the longitudinal axis of the implant 100). In some aspects, the user interface 127 may indicate detection of the edges and/or center using one or more of the display 129 (e.g., by displaying a visual indicator of edges and/or center detection), speaker 131 (e.g., by emitting an audible sound such as, for example and without limitation, a beep), and vibration motor 133 (e.g., by vibrating).

FIG. 12B illustrates detected magnetic field changes as the implant finder 101 is moved along the longitudinal axis of the implant 100 for implants 100 at different heights of the sensor 105 of the implant finder 101 relative to the implant 100. As shown in FIG. 12B, the magnitudes of the detected magnetic field change at the bimodal peaks increase as the height decreases. Accordingly, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may be configured to use the magnitude of the detected magnetic field change at one or more of the bimodal peaks (e.g., the local maxima in the detected magnetic field change during movement of the sensor 105 along the longitudinal axis of the implant 100) to determine the depth of the implant 100 in the tissue 150. In some aspects, the implant finder 101 may additionally or alternatively use the magnitude of the magnetic field change at the midline of the implant 101 to determine the depth of the implant 100 in the tissue 150.

In some aspects, as shown in FIG. 9D, the magnitudes of the magnetic field change at the bimodal peaks may be equal (or approximately equal such as, for example and without limitation, within 2 or 3 oersted (Oe)) for an implant 100 that is not tilted relative to the skin surface and has a longitudinal axis that runs parallel to the skin surface 154. In some aspects, as shown in FIGS. 10D and 14C, the magnitudes of the magnetic field change at the bimodal peaks may be different for an implant 100 having a longitudinal axis that is not parallel to the skin surface 154 (e.g., for a downward-tilted implant 100). In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101)

may determine an orientation of the implant 100 using the magnitudes of the magnetic field change at the bimodal peaks. In some aspects, determining the orientation may include calculating a difference between the magnitudes of the magnetic field change at the bimodal peaks and calculating an angle of the implant 100 relative to the skin surface 154 based on the calculated difference. In some aspects, determining the orientation of the implant 100 may additionally or alternatively include using the magnitudes of the magnetic field change at the bimodal peaks to calculate depths of the edges of the magnetic material 124 (and/or of the edges of the implant 100).

In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may be configured to cause the user interface 127 (e.g., the display 129 of the user interface 127) to provide information indicative of a location and/or orientation of the implant 100. In some aspects, the information indicative of the location and/or orientation of the implant 100 may include information indicative of a midline of the implant 100, one or more edges of the magnetic material 124, one or more edges of the implant 100, a center of the magnetic material 124, a center of the implant 100, a depth of the implant 100 in the tissue 150, an orientation of the implant 100, and/or an incision location for removal of the implant 100. In some aspects, the information indicative of the orientation of the implant 100 may include an indication of the angle of the implant 100, depths of the edges of the magnetic material 124, and/or depths of the edges of the implant 100. In some aspects, the information indicative of the location and/or orientation of the implant 100 may include the implant image 704 (e.g., as shown in FIGS. 7B-7F). In some aspects, the information indicative of the location and/or orientation of the implant 100 may include one or more plots of the magnetic field changes during the movement/sweeps of the implant finder 101 (e.g., the plots shown in FIGS. 9B, 9D, 10B, 10D, 13B-13E, 14B, 14C). In some aspects, the plots of may show one or more peaks in the magnetic field change (e.g., depending on whether the movement is across, along, or diagonal to the longitudinal axis of the implant 100).

In some aspects, as shown in FIGS. 9E and 10E, the first process for locating the implant 100 may include a third step of marking a location for an incision to remove the implant 100. In some aspects, the incision location may be marked using the incision marking tool 113 of the implant finder 101 (e.g., when the sensor 105 of the implant finder 101 is positioned over the center of the implant 100, over the center of the magnetic material 101 of the implant 100, and/or with the edges of the magnetic field generator 103 positioned over the edges of the magnetic material 124 of the implant 100).

In some aspects, as shown in FIG. 13A, the implant finder 101 may be moved in a diagonal direction (e.g., at a 45 degree angle or at a smaller angle such as, for example and without limitation, a 10, 15, 20, 25, or 30 degree angle) relative to a longitudinal axis of the implant 100 (e.g., when a user is attempting to locate the implant 100 and/or by accident when the user is attempting to move the implant finder 101 along the longitudinal axis). FIG. 13B illustrates detected magnetic field changes as the implant finder 101 is moved diagonal to the longitudinal axis of the implant 100. FIGS. 13C-13E illustrate X-, Y-, and Z-components of the magnetic field change as the implant finder 101 is moved diagonal to the longitudinal axis of the implant 100. In some aspects, the location of the greatest magnitude of the magnetic field change may be used to determine the midline of the implant 100. In some aspects, the movement/scan direc-

tion of the implant finder 101 may be changed until the orientation of the implant 100 (e.g., the longitudinal axis of the implant 100) is found.

In some aspects, the implant finder 101 (e.g., a computer 106 of the implant finder 101) may be configured to control the user interface 127 (e.g., the display 129) to output information indicative of the sensor signal generated by the sensor 105 (and therefore indicative of the detected changes in the magnetic field caused by the magnetic material 124 of the implant 100). In some aspects, a user (e.g., clinician) may use the information indicative of the sensor signal live to map out an approximate location of the implant 100.

In some aspects, a second process may be used to locate the implant 100. In some aspects, the second process may be used if the first process (described above with respect to FIGS. 9A-10E) does not expose the double/bimodal peak pattern present when the implant finder 101 is moved along the longitudinal axis of the implant 100. In some aspects, the second process may include a first step of finding a "best" orientation of the longitudinal axis of the implant 100. In some aspects, as shown in FIG. 15A, the first step may include moving the implant finder 101 at different angles (e.g., at incremental 15 degree angles) until double/bimodal peak pattern is identified.

In some aspects, as shown in FIG. 15B, the second process may include a second step identifying a center of the magnetic material 124 of the implant 100. In some aspects, the second step may include moving the implant finder 101 along the longitudinal axis identified in the first step. In some aspects, as shown in FIGS. 9D, 10D, and 14C, the detected changes in the magnetic field as the implant finder 101 is moved along the longitudinal axis of the implant 100 may include bimodal peaks (e.g., local maxima) at the locations of the edges of the magnetic material 124 of the implant 100. In some aspects, the center of the magnetic material 124 may be identified by finding a local minimum in the magnetic field change between the bimodal peaks in the magnetic field changes. In some aspects, the computer 106 of the implant finder 101 may be configured to determine a derivative of the detected changes in the magnetic field, and the computer 106 may be configured to determine the center of the magnetic material 124 of the implant 100 (and the locations of local minimum between the bimodal peaks in the magnetic field changes) based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor 105 along a longitudinal axis of the implant 100. In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may be configured to cause the user interface 127 to provide an indication of the center of the magnetic material 124. In some aspects, the indication of the center of the magnetic material 124 may include a plot of the magnetic field changes during the sweep and/or a visual (e.g., flashing light), aural (e.g., beep), or vibratory indication when a derivative of the magnitude of the magnetic field change is zero between the bimodal peaks).

In some aspects, the second process may include a third step in which the implant finder 101 (e.g., the computer 106 of the implant finder 101) calculates the depth and/or orientation (e.g., downward angulation) of the implant 100. In some aspects, the implant finder 101 may calculate the depth and/or orientation of the implant 100 using the magnitudes of the magnetic field change at the bimodal peaks and/or a difference between the magnitudes of the magnetic field change at the bimodal peaks. In some aspects, the implant finder 101 may calculate the depth and/or orientation of the implant 100 in response to a user input (e.g.,

button press) received via the user input 135 of the user interface 127. In some aspects, the implant finder 101 may cause the user interface 127 (e.g., the display 129 of the user interface 127) to provide the calculated depth and/or orientation of the implant 100 to a user (e.g., a clinician).

In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may calculate the depth and/or orientation (e.g., downward angulation) of the implant 100 using a data library to map the sensor signal generated by the sensor 105 during one or more movements/sweeps of the implant finder 101 to a depth and/or orientation. In some aspects, the library may include sensor signal curves characteristic of particular orientations of the implant 100. In some aspects, the characteristic curves may have been identified through experimentation. In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may compare the sensor signal generated by the sensor 105 during one or more movements/sweeps of the implant finder 101 to the characteristic curves. In some aspects, the implant finder 101 may determine that the implant 100 has approximately the implant orientation (e.g., in-plane and/or out-of-plane angulation) associated with the characteristic curve to which the generated sensor signal is most similar. In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may calculate the depth (or an estimated depth range) using the magnitude of the sensor signal generated during the movement/sweep (e.g., at one or more peaks thereof).

In some aspects, if the implant finder 101 determines that the sensor signal generated by the sensor 105 during movement of the implant finder 101 is indicative of in-plane angulation (e.g., indicative that the movement/sweep direction is diagonal with respect to the longitudinal axis of the implant 100), the third step of the second process may include the implant finder 101 (e.g., the computer 106 of the implant finder 101) determining suggested modification to the movement/sweep direction. In some aspects, the suggested modification may be a particular angle with respect to set direction (e.g., the long axis of the arm) or an angle change (e.g., +5° or -15° with respect to the current/most-recent movement direction. In some aspects, the implant finder 101 may cause the user interface 127 (e.g., the display 129 of the user interface 127) to provide the suggested modification to the movement/sweep direction to a user. In some aspects, the movement/sweep direction may be adjusted until the generated sensor signal looks as close as the baseline as possible. In some aspects, if one peak is still present in the generated sensor signal, the one peak is most likely due to out-of-plane angulation, which cannot be corrected by adjusting sweep direction.

In some aspects, as shown in FIG. 15C, the third step of the second process may include marking an implant edge (e.g., using fittings on the outer edge of the magnetic field generator 103) and/or marking an incision location (e.g., using the incision marking tool 113).

In some aspects, a third process may be used to locate the implant 100. In some aspects, the third process may include a first step of finding an approximate location of the longitudinal axis of the implant 100. In some aspects, as shown in FIG. 16A, the first step may include moving the implant finder 101 along the skin surface 154 and searching for any signal. In some aspects, the implant 100 may be located in the region where the signal is found.

In some aspects, as shown in FIG. 16B, the third process may include a second step in which a template 1601 (e.g., a template sticker) is placed over the region where the signal was found. In some aspects, as shown in FIG. 16B, the

template 1601 may include markings (e.g., numbers) that act as way-points identifying a course of movement/sweeping for the implant finder 101. For example, in some aspects, the template 1601 may identify a course, for example and without limitation, from waypoints 9 to 3, 4 to 10, 11 to 5, 6 to 12, 1 to 7, and 8 to 2. In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may cause the user interface 127 (e.g., the display 129) to provide instructions for the course of movement/sweeping of the implant finder 101 using the template 1601. In some aspects, the second step of the third process may include moving the implant finder 101 along the course identified by the template 1601. In some aspects, the second step of the third process may end with movement of the implant finder 101 to a final position on top of the template 1601.

In some aspects, the third process may include a third step in which the implant finder 101 (e.g., the computer 106 of the implant finder 101) calculates the depth and/or orientation (e.g., in-plane angulation and/or out-of-plane angulation) of the implant 100. In some aspects, the implant finder 101 may calculate the depth and/or orientation of the implant 100 using the sensor signal generated by the sensor 105, which is indicative of magnetic field changes caused by the magnetic material 124 of the implant 100, as the implant finder 101 is moved along the course identified by the template 1601. In some aspects, the implant finder 101 (e.g., the computer 106 of the implant finder 101) may calculate the depth and/or orientation of the implant 100 using the data library to map the sensor signal generated by the sensor 105 as the implant finder 101 is moved along the course identified by the template 1601 to a depth and/or orientation. In some aspects, the implant finder 101 may calculate the depth and/or orientation of the implant 100 in response to a user input (e.g., button press) received via the user input 135 of the user interface 127. In some aspects, the implant finder 101 may cause the user interface 127 (e.g., the display 129 of the user interface 127) to provide the calculated depth and/or orientation of the implant 100 to a user (e.g., a clinician).

In some aspects, as shown in FIG. 16C, the implant finder 101 may display an implant image 704 and/or a template image 1603 on the screen 702 of the display 129. In some aspects, as shown in FIG. 16C, the location of the implant image 704 relative to the template image 1603 on the screen 704 of the display 129 may correspond to a detected location of the implant 100 relative to the template 1601. In some aspects, as shown in FIG. 16C, the implant image 704 may have an orientation that corresponds to the calculated orientation of the implant 100. In some aspects, the implant image 704 may show in-plane angulation and/or out-of-plane (e.g., downward) angulation of the calculated orientation. In some aspects, the implant image 704 may additionally or alternatively show the calculated depth.

In some aspects, the third step of the third process may additionally or alternatively include the implant finder 101 (e.g., the computer 106 of the implant finder 101) calculating one or more edges of the implant 100, a center of the implant 100, and/or an incision location. In some aspects, the implant finder 101 may calculate the one or more implant edges, the implant center, and/or the incision location using the sensor signal generated by the sensor 105, which is indicative of magnetic field changes caused by the magnetic material 124 of the implant 100, as the implant finder 101 is moved along the course identified by the template 1601. In some aspects, the implant finder 101 may calculate the one or more implant edges and/or the incision location using the calculated depth and/or orientation of the implant 100. In

some aspects, the implant finder **101** may cause the user interface **127** (e.g., the display **129** of the user interface **127**) to provide the calculated one or more implant edges, implant center, and/or incision location to a user (e.g., a clinician). In some aspects, as shown in FIG. **16C**, the implant finder **101** may display the calculated one or more implant edges, implant center, and/or incision location on the screen **702** of the display **129** as coordinates.

FIG. **17** is a flow chart illustrating a process **1700** for locating an implant **100** including magnetic material **124** according to some aspects. In some aspects, the process **1700** may include a step **1702** of using a magnetic field generator **103** to generate a magnetic field. In some aspects, the magnetic field generator **103** may include one or more magnets **109**. In some aspects, the magnetic field generator **103** may include a cylindrical magnet **109** (e.g., a hollow cylindrical magnet). In some alternative aspects, the magnetic field generator **103** may include two or more magnets **109** (e.g., four or six magnets). In some aspects, the magnetic field generator **103** may include a housing **107** configured to hold the two or more magnets **109**. In some aspects, the magnetic field generated by the magnetic field generator **103** may be a substantially uniform magnetic field.

In some aspects, the process **1700** may include a step **1704** of using a sensor **105** to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field. In some aspects, the magnetic material **124** of the implant **100** may cause changes to the magnetic field as the sensor **105** is moved over the implant **100**. In some aspects, the magnetic field may be substantially symmetric about a longitudinal axis at the center of the magnetic field generator **103**. In some aspects, the sensor **105** may be located along or offset from the longitudinal axis at the center of the magnetic field generator **103**.

In some aspects, the process **1700** may include a step **1706** of using a computer **106** to detect a location of the implant **100** based on the sensor signal. In some aspects, the step **1706** may include the computer **106** causing the user interface **127** to provide an output indicative of the sensor signal. In some aspects, the user (e.g., a clinician) may use the output sensor signal to detect the location of the implant **100**. In some aspects, the step **1706** may include the computer **106** causing the user interface **127** to provide indications of when a derivative of the changes in the magnetic field equals zero during movement of the sensor **105** across a longitudinal axis of the implant **100**. In some aspects, the indications of the derivative of the changes in the magnetic field equaling zero during movement of the sensor **105** across the longitudinal axis of the implant **100** may be indicative of the location of the implant **100**.

In some aspects, the step **1706** may additionally or alternatively include the computer **106** determining edges of the magnetic material **124** of the implant **100** based on locations of bimodal peaks in the changes in the magnetic field during movement of the sensor **105** along a longitudinal axis of the implant **100**. In some aspects, the computer **106** may determine edges of the implant **100** based on the determined edges of the magnetic material **124** of the implant **100** and one of more offsets between edges of the implant **100** and edges of the magnetic material **124** of the implant **100**. In some aspects, the computer **106** may be configured to calculate a derivative of the changes in the magnetic field and to use the calculated derivative to detect the location of the implant **100**. In some aspects, the computer **106** may be configured to determine edges of the magnetic material **124** of the implant **100** based on locations where the derivative of the changes in the magnetic field

equals zero during movement of the sensor **105** along a longitudinal axis of the implant **100**.

In some aspects, the step **1706** may additionally or alternatively include the computer **106** determining a midline of the implant **100**. In some aspects, the computer **106** may be configured to determine the midline based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor **105** across a longitudinal axis of the implant **100**.

In some aspects, the process **1700** may include a step in which the computer **106** determines a depth of the implant **100** based on a magnitude of the change in the magnetic field at the bimodal peaks in the changes in the magnetic field during the movement of the sensor **105** along the longitudinal axis of the implant **100**. In some aspects, the computer **106** may determine a depth of the implant **100** based on magnitudes of the changes in the magnetic field at the locations where the derivative of the changes in the magnetic field equals zero during the movement of the sensor **105** along the longitudinal axis of the implant **100**.

In some aspects, the process **1700** may additionally or alternatively include a step in which the computer **106** determines an orientation of the implant **100**. In some aspects, the computer **106** may determine the orientation based on a difference between magnitudes of the change in the magnetic field at the bimodal peaks in the changes in the magnetic field during the movement of the sensor **105** along the longitudinal axis of the implant **100**. In some aspects, the computer **106** may determine an orientation of the implant **100** based on a difference between magnitudes of the changes in the magnetic field at the locations where the derivative of the changes in the magnetic field equals zero during the movement of the sensor **105** along the longitudinal axis of the implant **100**.

In some aspects, the process **1700** may additionally or alternatively include computer **106** causing a display **129** to display an indication of the detected location of the implant **100**. In some aspects, the indication of the detected location of the implant **100** may include an implant image **704**, and a location of the implant image **704** on a screen **702** of the display **129** relative to a point **706** on the screen **702** of the display **129** may correspond to the detected location of the implant **100** relative to the sensor **105**. In some aspects, the implant image **704** may have an orientation that corresponds to a detected orientation of the implant **100**.

In some aspects in which the implant finder **101** includes the position detector **137** configured to generate a location signal indicative of a location of the sensor **105**, the process **1700** may additionally or alternatively include a step in which the computer **106** uses the sensor signal and the location signal (e.g., a motion signal generated by a motion detector of the position detector **137** that is indicative of movement of the sensor **105**) to generate a map of sensor signals at different locations of the sensor **105**. In some aspects, generating a map of sensor signals at different locations of the sensor may include, for example and without limitation, measuring the sensor signal at each of two or more different locations of the sensor **105** and storing the measured sensor signals along with identifications of the locations at which the sensor signals were measured. In some aspects, generating a map of sensor signals at different locations of the sensor **105** may include generating a visualization of the measured sensor signals at the different locations of the sensor **105**.

In some aspects, the process **1700** may include an optional step of removing the implant **100**. In some aspects, removing the implant **100** may include making an incision at an

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identified edge of the implant **100**. In some aspects, removing the implant **100** may include grabbing (e.g., using forceps) the implant **100** and pulling the implant **100** out of the body through the incision.

FIG. **18** is a flow chart illustrating a process **1800** for locating an implant **100** including magnetic material **124** according to some aspects. In some aspects, the process **1800** may include a step **1802** of moving an apparatus (e.g., implant finder **101**) including a magnetic field generator **103** and a sensor **105** across a longitudinal axis of the implant **100**. In some aspects, the magnetic material **124** of the implant **100** may cause changes in a magnetic field generated by the magnetic field generator **103** as the apparatus is moved across the longitudinal axis of the implant **100**, and the sensor **105** may detect changes in the magnetic field. In some aspects, the process **1800** may include a step **1804** of determining a midline of the implant **100** based on a location where the changes in the magnetic field are greatest as the apparatus is moved across the longitudinal axis of the implant **100**. In some aspects, the process **1800** may include a step **1806** of moving the apparatus along the determined midline of the implant **100**. In some aspects, the process **1800** may include a step **1808** of determining edges of the magnetic material **124** of the implant **100** based on locations of bimodal peaks in the changes in the magnetic field. In some aspects, the process **1800** may include an optional step of using an incision marking tool **113** of the apparatus to mark an incision location for removing the implant **100**. In some aspects, the process **1800** may include an optional step of removing the implant **100**. In some aspects, removing the implant **100** may include making an incision at the incision location. In some aspects, removing the implant **100** may include grabbing (e.g., using forceps) the implant **100** and pulling the implant **100** out of the body through the incision.

In some aspects, as shown in FIG. **19**, the implant **100** may include more than one analyte indicator **104** (e.g., first analyte indicator **104a** and second analyte indicator **104b**). In some aspects, as shown in FIG. **19**, the implant **100** may include a charge storage device **202** (e.g., a battery). In some aspects, the charge storage device **202** may be a battery with a metal housing (e.g., a titanium housing). In some aspects, the charge storage device **202** may be attached to the housing **102**. In some aspects, a coupler **324** may attach the housing **102** and the charge storage device **202**. In some aspects, the coupler **324** may be between the housing **102** and the charge storage device **202**. In some aspects, as shown in FIG. **19**, the coupler **324** may include one or more supports **232** (e.g., reinforcement rods, bars, or beams), which may be attached to and/or integral with the coupler **324**. In some aspects, the implant **100** may include first and second electrically conductive connectors, which connect positive and negative terminals, of the charge storage device **202** to the circuitry of the implant **100**. In some aspects, as shown in FIG. **19**, the circuitry of the implant **100** may extend away from the charge storage device **202** along the longitudinal axis of the charge storage device **202**. In some aspects, the circuitry of the implant **100** may include the inductor **120**, which may include the conductor **122** and the magnetic material **124** in form of a magnetic core. In some aspects, as shown in FIG. **19**, the charge storage device **202** may include a first edge at one end of the implant **100** and a second edge adjacent to the coupler **324** and/or housing **102**. In some aspects, the second edge of the charge storage device **202** may be located in a central region of the implant **100**.

In some aspects, the implant **100** may include one or more drug eluting polymer matrices **730** and **732**. In some aspects,

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the one or more drug eluting polymer matrices **730** and **732** may be in or on portions of the exterior surface of the housing **102** of the implant **100**. In some aspects, one or more therapeutic agents may be dispersed within the one or more drug eluting polymer matrices **730** and **732**. In some aspects, the one or more therapeutic agents may reduce or stop the migration of neutrophils from entering the space in which the implant **100** has been implanted and, thus, reduce or stop the production of hydrogen peroxide and fibrotic encapsulation. Accordingly, in some aspects, the one or more therapeutic agents may reduce deterioration of the one or more analyte indicators **104** (e.g., first analyte indicator **104a** and second analyte indicator **104b**). In some aspects, the one or more therapeutic agents, which may be dispersed within the one or more drug eluting polymer matrices **730** and **732**, may include one or more anti-inflammatory drugs, such as, for example, non-steroidal anti-inflammatory drug (e.g., acetylsalicylic acid (aspirin) and/or isobutylphenylpropanoic acid (ibuprofen)). In some aspects, the one or more therapeutic agents dispersed within the one or more drug eluting polymer matrices **730** and **732** may include one or more glucocorticoids. In some aspects, the one or more therapeutic agents may include one or more of dexamethasone, triamcinolone, betamethasone, methylprednisolone, beclometasone, fludrocortisone, derivatives thereof, and analogs thereof. In some aspects, the one or more therapeutic agents may reduce the production of hydrogen peroxide by neutrophils and macrophages.

In some aspects (e.g., some aspects in which the implant **100** includes a charge storage device **202**), as shown in FIG. **20**, the magnetic field generator **103** of the implant finder **101** may be an electromagnetic field generator **103'** configured to generate an electromagnetic field. In some aspects, the electromagnetic field may include low frequency waves (e.g., very low frequency (VLF) waves). In some aspects, the low frequency waves may have a frequency, for example and without limitation, within a range between 3 kHz and 30 kHz. In some aspects, the low frequency waves may have a frequency, for example and without limitation, within a range between 5 kHz and 15 kHz. In some aspects, the sensor **105** of the implant finder **101** may be configured to detect changes in the electromagnetic field and to generate a sensor signal indicative of the changes in the electromagnetic field. In some aspects, at least the charge storage device **202** of the implant **100** may cause changes to the electromagnetic field as the sensor **105** is moved over the implant **100**. In some aspects, the computer **106** of the implant finder **101** may be configured to use the sensor signal to detect a location of the implant **100**.

In some electromagnetic field aspects, as shown in FIGS. **21A** and **21B**, the electromagnetic field generator **103'** may include one or more electromagnets **109'**. In some aspects, the one or more electromagnets **109'** may be, for example and without limitation, a coil of wire wound around a magnetic core made from a ferromagnetic or ferrimagnetic material. In some aspects, the electromagnetic field generator **103'** may supply an alternating current to the one or more electromagnets **109'**. The alternating current may have, for example and without limitation, a frequency within a range of between 3 kHz and 30 kHz or, more particularly, within a range between 5 kHz and 15 kHz. In some aspects, as shown in FIG. **21C**, the electromagnetic field generator **103'** may include a housing **107**, and the housing **107** may be configured to hold the one or more electromagnets **109'**. However, a housing **107** is not required, and, in some alternative aspects (e.g., some aspects in which the magnetic field generator **103'** consists of only a single electromagnet

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109'), the electromagnetic field generator 103' may not include a housing 107. In some aspects, as shown in FIGS. 21A and 21B, the sensor 105 may be located on a longitudinal axis at the center of the magnetic field generator 103'. However, this is not required, and, in some alternative aspects, the sensor 105 is not located on a longitudinal axis at the center of the electromagnetic field generator 103'. For example, the sensor 105 may be located adjacent to the electromagnet 109' of the electromagnetic field generator 103' (as opposed to at the center of one or more electromagnets 109' of the electromagnetic field generator 103'). See FIG. 3A.

FIG. 22 is a flow chart illustrating a process 2200 for locating an implant 100 including a charge storage device 202 according to some aspects. In some aspects, the process 2200 may include a step 2202 of using an electromagnetic field generator 103' to generate an electromagnetic field. In some aspects, the electromagnetic field generator 103' may include one or more electromagnets 109'. In some aspects, the electromagnetic field generator 103' may include a cylindrical electromagnet 109' (e.g., a hollow cylindrical magnet). In some alternative aspects, the electromagnetic field generator 103' may include two or more electromagnets 109' (e.g., four or six electromagnets). In some aspects, the electromagnetic field generator 103' may include a housing 107 configured to hold the one or more electromagnets 109'.

In some aspects, the process 2200 may include a step 2204 of using a sensor 105 to detect changes in the electromagnetic field and to generate a sensor signal indicative of the changes in the electromagnetic field. In some aspects, at least the charge storage device 202 of the implant 100 may cause changes to the electromagnetic field as the sensor 105 is moved over the implant 100. In some aspects, the electromagnetic field may be substantially symmetric about a longitudinal axis at the center of the electromagnetic field generator 103'. In some aspects, the sensor 105 may be located along or offset from the longitudinal axis at the center of the electromagnetic field generator 103'.

In some aspects, the process 2200 may include a step 2206 of using a computer 106 to detect a location of the implant 100 based on the sensor signal. In some aspects, the step 2206 may include the computer 106 causing the user interface 127 to provide an output indicative of the sensor signal. In some aspects, the user (e.g., a clinician) may use the output sensor signal to detect the location of the implant 100. In some aspects, the step 2206 may include the computer 106 causing the user interface 127 to provide indications of when a derivative of the changes in the electromagnetic field equals zero during movement of the sensor 105 across a longitudinal axis of the implant 100. In some aspects, the indications of the derivative of the changes in the electromagnetic field equaling zero during movement of the sensor 105 across the longitudinal axis of the implant 100 may be indicative of the location of the implant 100.

In some aspects, the step 2206 may additionally or alternatively include the computer 106 determining edges of the charge storage device 202 of the implant 100 based on locations of bimodal peaks in the changes in the electromagnetic field during movement of the sensor 105 along a longitudinal axis of the implant 100. In some aspects, the computer 106 may determine one or more edges of the implant 100 based on the determined edges of the charge storage device 202 of the implant 100. In some aspects, the computer 106 may be configured to calculate a derivative of the changes in the electromagnetic field and to use the calculated derivative to detect the location of the implant 100. In some aspects, the computer 106 may be configured

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to determine edges of the charge storage device 202 of the implant 100 based on locations where the derivative of the changes in the magnetic field equals zero during movement of the sensor 105 along a longitudinal axis of the implant 100.

In some aspects, the magnitude of the bimodal peak in the changes in the electromagnetic field at the first edge of the charge storage device 202, which is at an edge of the implant 100, may be greater than the magnitude of the bimodal peak in the changes in the electromagnetic field at the second edge of the charge storage device 202, which is adjacent to the coupler 324 and/or housing 102. In some aspects, the magnetic material 124 in the housing 102 of the implant 100 may cause the magnitude of the bimodal peak in the changes in the electromagnetic field at the second edge of the charge storage device 202 to be lower than the magnitude of the bimodal peak in the changes in the electromagnetic field at the first edge of the charge storage device 202. In some aspects, the computer 106 may be configured to determine which of the bimodal peaks in the changes in the electromagnetic field has a greater magnitude. In some aspects, the computer 106 may be configured to determine the first edge of charge storage device 202 to be at the location of the bimodal peak having the greater magnitude. In some aspects, the computer 106 may determine one edge of the implant 100 to be at the first edge of the charge storage device 202. In some aspects, the incision location for removing the implant 100 may be at or near the location of the bimodal peak in the changes in the electromagnetic field having the greater magnitude.

In some aspects, the computer 106 may be configured to additionally or alternatively determine a second edge of the implant 100. In some aspects, the computer 106 may determine the second edge of the implant 100 using the location of the second edge of the charge storage device 202 (e.g., the location of the bimodal peak in the changes in the electromagnetic field having the lower magnitude) and an offset between the second edge of the charge storage device 202 and an end of the housing 102. In some aspects, the second edge of the charge storage device 202 may be the edge of the charge storage device 202 that is adjacent to the coupler 324 and/or housing 102.

In some aspects, the step 2206 may additionally or alternatively include the computer 106 determining a midline of the implant 100. In some aspects, the computer 106 may be configured to determine the midline based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor 105 across a longitudinal axis of the implant 100.

In some aspects, the process 2200 may include a step in which the computer 106 determines a depth of the implant 100 based on a magnitude of the change in the electromagnetic field at the bimodal peaks in the changes in the electromagnetic field during the movement of the sensor 105 along the longitudinal axis of the implant 100. In some aspects, the computer 106 may determine a depth of the implant 100 based on magnitudes of the changes in the electromagnetic field at the locations where the derivative of the changes in the electromagnetic field equals zero during the movement of the sensor 105 along the longitudinal axis of the implant 100.

In some aspects, the process 2200 may additionally or alternatively include a step in which the computer 106 determines an orientation of the implant 100. In some aspects, the computer 106 may determine the orientation based on a difference between magnitudes of the change in the electromagnetic field at the bimodal peaks in the changes

in the electromagnetic field during the movement of the sensor **105** along the longitudinal axis of the implant **100** (e.g., after taking into account the expected difference between the magnitudes at the bimodal peaks due to the second edge of the charge storage device **202** being closer to the magnetic material **124** in the housing **102**). In some aspects, the computer **106** may determine an orientation of the implant **100** based on a difference between magnitudes of the changes in the electromagnetic field at the locations where the derivative of the changes in the electromagnetic field equals zero during the movement of the sensor **105** along the longitudinal axis of the implant **100** (e.g., after taking into account the reduction in the magnitude of the bimodal peak in the changes in the electromagnetic field at the second edge of the charge storage device **202**, which may be caused by the proximity of the second edge of the charge storage device **202** to the magnetic material **124**).

In some aspects, the process **2200** may additionally or alternatively include computer **106** causing a display **129** to display an indication of the detected location of the implant **100**. See, e.g., FIGS. **1** and **7A-7F**. In some aspects, the indication of the detected location of the implant **100** may include an implant image **704**, and a location of the implant image **704** on a screen **702** of the display **129** relative to a point **706** on the screen **702** of the display **129** may correspond to the detected location of the implant **100** relative to the sensor **105**. In some aspects, the implant image **704** may have an orientation that corresponds to a detected orientation of the implant **100**.

In some aspects in which the implant finder **101** includes the position detector **137** configured to generate a location signal indicative of a location of the sensor **105**, the process **2200** may additionally or alternatively include a step in which the computer **106** uses the sensor signal and the location signal (e.g., a motion signal generated by a motion detector of the position detector **137** that is indicative of movement of the sensor **105**) to generate a map of sensor signals at different locations of the sensor **105**. In some aspects, generating a map of sensor signals at different locations of the sensor may include, for example and without limitation, measuring the sensor signal at each of two or more different locations of the sensor **105** and storing the measured sensor signals along with identifications of the locations at which the sensor signals were measured. In some aspects, generating a map of sensor signals at different locations of the sensor **105** may include generating a visualization of the measured sensor signals at the different locations of the sensor **105**.

In some aspects, the process **2200** may include an optional step of removing the implant **100**. In some aspects, removing the implant **100** may include making an incision at an identified edge of the charge storage device **202**. In some aspects, removing the implant **100** may include grabbing (e.g., using forceps) the charge storage device **202** of the implant **100**, which may be stronger than the housing **102** of the implant **100**, and pulling the implant **100** out of the body through the incision.

FIG. **23** is a flow chart illustrating a process **2300** for locating an implant **100** including a charge storage device **202** according to some aspects. In some aspects, the process **2300** may include a step **2302** of moving an apparatus (e.g., implant finder **101**) including an electromagnetic field generator **103'** and a sensor **105** across a longitudinal axis of the implant **100**. In some aspects, at least the charge storage device **202** of the implant **100** may cause changes in an electromagnetic field generated by the electromagnetic field generator **103'** as the apparatus is moved across the longitudinal axis of the implant **100**, and the sensor **105** may detect changes in the electromagnetic field. In some aspects, the process **2300** may include a step **2304** of determining a midline of the implant **100** based on a location where the changes in the electromagnetic field are greatest as the apparatus is moved across the longitudinal axis of the implant **100**. In some aspects, the process **2300** may include a step **2306** of moving the apparatus along the determined midline of the implant **100**. In some aspects, the process **2300** may include a step **2308** of determining edges of the charge storage device **202** of the implant **100** based on locations of bimodal peaks in the changes in the electromagnetic field. In some aspects, the process **2300** may include an optional step of using an incision marking tool **113** of the apparatus to mark an incision location for removing the implant **100**. In some aspects, the incision location may be at or near the location of the bimodal peak in the changes in the electromagnetic field that has the greater magnitude. In some aspects, the process **2300** may include an optional step of removing the implant **100**. In some aspects, removing the implant **100** may include making an incision at the incision location. In some aspects, removing the implant **100** may include grabbing (e.g., using forceps) the charge storage device **202** of the implant **100**, which may be stronger than the housing **102** of the implant **100**, and pulling the implant **100** out of the body through the incision.

While the subject matter of this disclosure has been described and shown in considerable detail with reference to certain illustrative embodiments, including various combinations and sub-combinations of features, those skilled in the art will readily appreciate other embodiments and variations and modifications thereof as encompassed within the scope of the present disclosure. Moreover, the descriptions of such embodiments, combinations, and sub-combinations is not intended to convey that the claimed subject matter requires features or combinations of features other than those expressly recited in the claims. Accordingly, the scope of this disclosure is intended to include all modifications and variations encompassed within the spirit and scope of the following appended claims.

What is claimed is:

1. An apparatus for locating an implant comprising magnetic material in a living animal, the apparatus comprising:
 - a magnetic field generator configured to generate a magnetic field;
 - a sensor configured to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field, wherein the magnetic material of the implant causes changes to the magnetic field as the sensor is moved over the implant; and
 - a computer configured to use the sensor signal to detect a location of the implant, wherein, in detecting the location of the implant, the computer is configured to determine edges of the magnetic material of the implant based on locations of bimodal peaks in the changes in the magnetic field during movement of the sensor along a longitudinal axis of the implant.
2. The apparatus of claim 1, wherein the magnetic field generator comprises a cylindrical magnet.
3. The apparatus of claim 2, wherein the cylindrical magnet is hollow.
4. The apparatus of claim 1, wherein the magnetic field generator comprises two or more magnets.
5. The apparatus of claim 4, wherein the magnetic field generator further comprises a housing configured to hold the two or more magnets.

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6. The apparatus of claim 4, wherein the magnetic field generator comprises four magnets.

7. The apparatus of claim 4, wherein the magnetic field generator comprises six magnets.

8. The apparatus of claim 1, wherein the magnetic field generator comprises one or more permanent magnets.

9. The apparatus of claim 1, wherein the magnetic field generator comprises one or more electromagnets.

10. The apparatus of claim 1, wherein the magnetic field generated by the magnetic field generator is a substantially uniform magnetic field.

11. The apparatus of claim 1, wherein the magnetic field is substantially symmetric about a longitudinal axis at a center of the magnetic field generator.

12. The apparatus of claim 11, wherein the sensor is located along or offset from the longitudinal axis at the center of the magnetic field generator.

13. The apparatus of claim 1, wherein the magnetic field is non-uniform and/or asymmetric about a longitudinal axis at a center of the magnetic field generator.

14. The apparatus of claim 1, wherein the computer is configured to determine edges of the implant based on the determined edges of the magnetic material of the implant and one of more off sets between edges of the implant and edges of the magnetic material of the implant.

15. The apparatus of claim 1, wherein the computer is configured to determine a depth of the implant based on a magnitude of the change in the magnetic field at the bimodal peaks in the changes in the magnetic field during the movement of the sensor along the longitudinal axis of the implant.

16. The apparatus of claim 1, wherein the computer is configured to determine an orientation of the implant based on a difference between magnitudes of the change in the magnetic field at the bimodal peaks in the changes in the magnetic field during the movement of the sensor along the longitudinal axis of the implant.

17. The apparatus of claim 1, wherein the computer is further configured to use one or more sensor signals to detect an orientation of the implant.

18. The apparatus of claim 1, further comprising a display, and the computer is configured to cause the display to display an indication of the detected location of the implant.

19. The apparatus of claim 18, wherein the display is located above the sensor.

20. The apparatus of claim 18, wherein the indication of the detected location of the implant comprises an implant image, and a location of the implant image on a screen of the display relative to a point on the screen of the display corresponds to the detected location of the implant relative to the sensor.

21. The apparatus of claim 20, wherein the implant image has an orientation that corresponds to a detected orientation of the implant.

22. The apparatus of claim 1, further comprising an incision marking tool configured to identify an incision location for removing the implant.

23. The apparatus of claim 1, further comprising a position detector configured to generate a location signal indicative of a location of the sensor on a skin surface.

24. The apparatus of claim 23, wherein the position detector comprises a motion detector configured to detect movement of the sensor and to generate a motion signal indicative of the detected movement of the sensor, wherein the location signal comprises the motion signal.

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25. The apparatus of claim 23, wherein the computer is configured to use the sensor signal and the location signal to generate a map of sensor signals at different locations of the sensor on the skin surface.

26. An apparatus for locating an implant comprising magnetic material in a living animal, the apparatus comprising:

a magnetic field generator configured to generate a magnetic field;

a sensor configured to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field, wherein the magnetic material of the implant causes changes to the magnetic field as the sensor is moved over the implant; and

a computer configured to use the sensor signal to calculate a derivative of the changes in the magnetic field and to use the calculated derivative to detect a location of the implant, wherein the computer is configured to determine edges of the magnetic material of the implant based on locations where the derivative of the changes in the magnetic field equals zero during movement of the sensor along a longitudinal axis of the implant.

27. The apparatus of claim 26, wherein the computer is configured to determine a depth of the implant based on magnitudes of the changes in the magnetic field at the locations where the derivative of the changes in the magnetic field equals zero during the movement of the sensor along the longitudinal axis of the implant.

28. The apparatus of claim 26, wherein the computer is configured to determine an orientation of the implant based on a difference between magnitudes of the changes in the magnetic field at the locations where the derivative of the changes in the magnetic field equals zero during the movement of the sensor along the longitudinal axis of the implant.

29. An apparatus for locating an implant comprising magnetic material in a living animal, the apparatus comprising:

a magnetic field generator configured to generate a magnetic field;

a sensor configured to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field, wherein the magnetic material of the implant causes changes to the magnetic field as the sensor is moved over the implant; and

a computer configured to use the sensor signal to calculate a derivative of the changes in the magnetic field and to use the calculated derivative to detect a location of the implant, wherein the computer is configured to determine a midline of the implant based on a location where the derivative of the changes in the magnetic field equals zero during movement of the sensor across a longitudinal axis of the implant.

30. A method for locating an implant comprising magnetic material in a living animal, the method comprising: using a magnetic field generator to generate a magnetic field;

using a sensor to detect changes in the magnetic field and to generate a sensor signal indicative of the changes in the magnetic field, wherein the magnetic material of the implant causes changes to the magnetic field as the sensor is moved over the implant; and

using a computer to detect a location of the implant based on the sensor signal, wherein detecting the location of the implant comprises determining edges of the magnetic material of the implant based on locations of

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bimodal peaks in the changes in the magnetic field during movement of the sensor along a longitudinal axis of the implant.

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